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JEE MAIN · PHYSICS · CHAPTER-WISE PYQS

Current Electricity

JEE Main Physics · Previous Year Questions with Answer Key

408	191	2002 – 2026	34	English
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Numerical	97 questions
MCQ (Single Correct Answer)	277 questions
Out of syllabus	34 questions
Answer key	at the end

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Numerical 97 questions

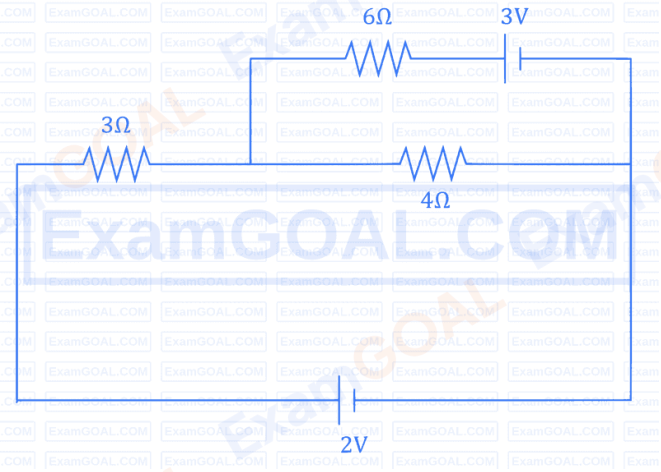
Q1. JEE Main 2026 (Online) 6th April Evening Shift

Two cells of emfs 1 V and 2 V and internal resistance 2Ω and 1Ω, respectively connected in parallel, gave a current of 1 A through an external resistance. If the polarity of one cell is reversed, then value of current through the external resistance will be $\frac{\alpha}{5}$ A. The value of α is ____.

Open solution

Q2. JEE Main 2026 (Online) 6th April Morning Shift

Refer to the circuit diagram given below. The heat generated across the 6Ω resistance in 100 second is $\frac{\alpha}{100}$ J. The value of α is ____ . (Nearest integer)



Open solution

Q3. JEE Main 2026 (Online) 5th April Evening Shift

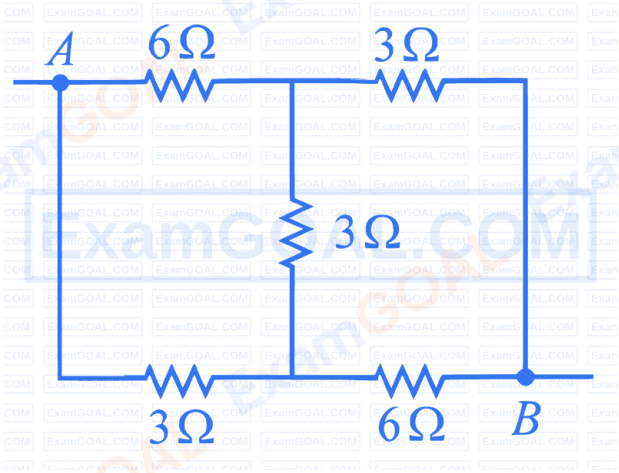
When an external resistance of 5Ω is connected across terminals of a cell, a current of 0.25 A flows through it. When the 5Ω resistor is replaced by a 2Ω resistor, a current of 0.5 A flows through it. The internal resistance of the cell is ____ Ω.

Open solution

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Q4. JEE Main 2026 (Online) 28th January Morning Shift

The equivalent resistance between the points A and B in the following circuit is $\frac{x}{5}\Omega$. The value of x is ____ .



Open solution

Q5. JEE Main 2026 (Online) 24th January Evening Shift

In a meter bridge experiment to determine the value of unknown resistance, first the resistances 2Ω and 3Ω are connected in the left and right gaps of the bridge and the null point is obtained at a distance l cm from the left. Now when an unknown resistance $x\Omega$ is connected in parallel to 3Ω resistance, the null point is shifted by 10 cm to the right of wire. The value of unknown resistance x is

____ Ω.

Open solution

Q6. JEE Main 2026 (Online) 22nd January Evening Shift

A cylindrical conductor of length 2 m and area of cross-section $0.2\ \text{mm}^2$ carries an electric current of 1.6 A when its ends are connected to a 2 V battery. Mobility of electrons in the conductor is $\alpha \times 10^{-3}\ \text{m}^2/\text{V.s}$. The value of α is :

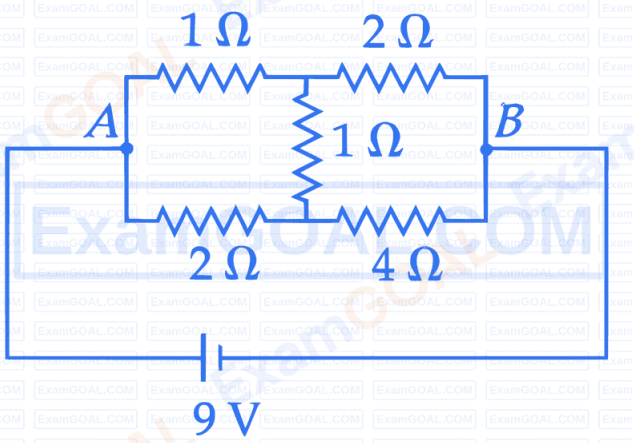
(electron concentration $= 5 \times 10^{28}/\text{m}^3$ and electron charge $= 1.6 \times 10^{-19}\text{C}$)

Open solution

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Q7. JEE Main 2026 (Online) 21st January Morning Shift

The heat generated in 1 minute between points A and B in the given circuit, when a battery of 9 V with internal resistance of 1Ω is connected across these points is ____ J.



Open solution

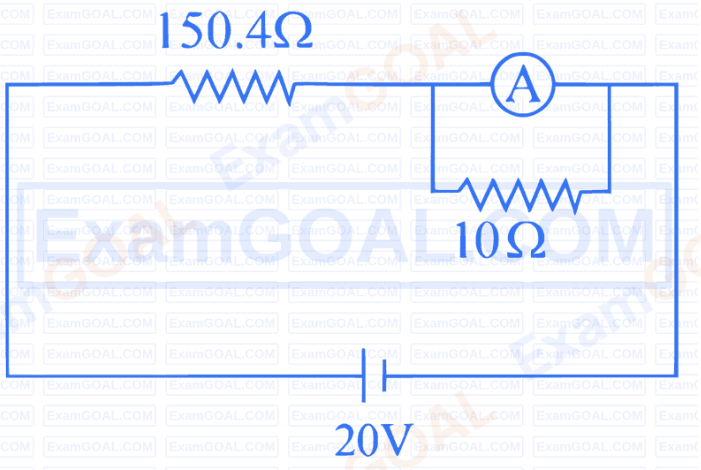
Q8. JEE Main 2025 (Online) 3rd April Evening Shift

Two cells of emfs 1 V and 2 V and internal resistances 2Ω and 1Ω, respectively, are connected in series with an external resistance of 6Ω. The total current in the circuit is I_1. Now the same two cells in parallel configuration are connected to same external resistance. In this case, the total current drawn is I_2. The value of $\left(\frac{\mathrm{I}_1}{\mathrm{I}_2}\right)$ is $\frac{x}{3}$. The value of x is_____.

Open solution

Q9. JEE Main 2025 (Online) 3rd April Morning Shift

In the figure shown below, a resistance of 150.4Ω is connected in series to an ammeter A of resistance 240Ω. A shunt resistance of 10Ω is connected in parallel with the ammeter. The reading of the ammeter is_____mA .

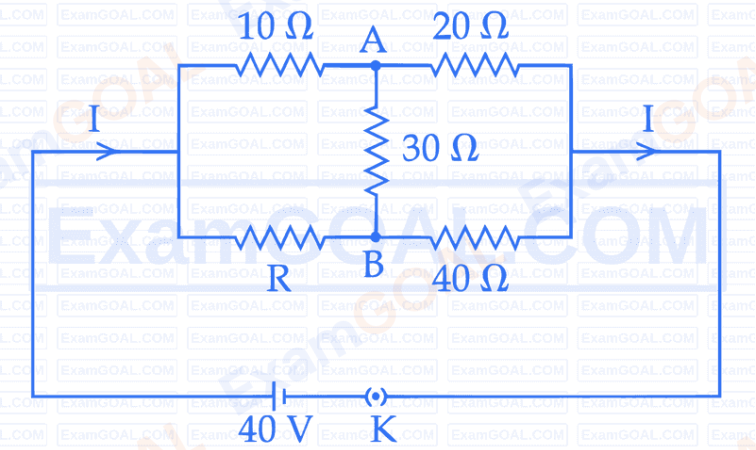


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Q10. JEE Main 2025 (Online) 28th January Evening Shift

The value of current I in the electrical circuit as given below, when potential at A is equal to the potential at B, will be _____ A.



Open solution

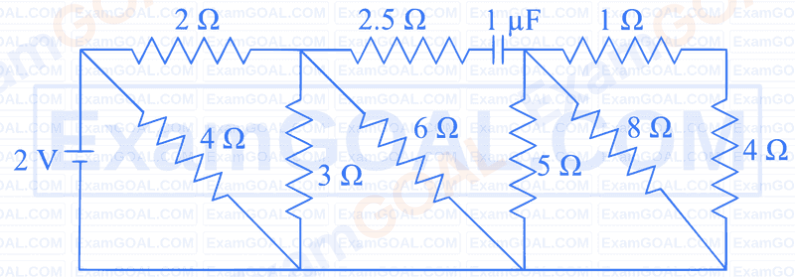
Q11. JEE Main 2025 (Online) 24th January Morning Shift

A wire of resistance 9Ω is bent to form an equilateral triangle. Then the equivalent resistance across any two vertices will be _____ ohm.

Open solution

Q12. JEE Main 2025 (Online) 22nd January Evening Shift

The net current flowing in the given circuit is _____ A.

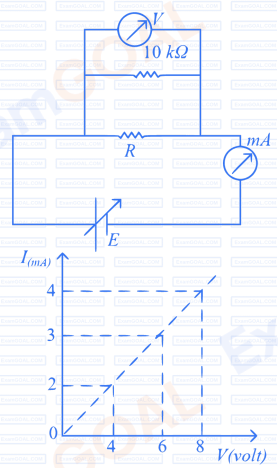


Open solution

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Q13. JEE Main 2024 (Online) 9th April Evening Shift

To determine the resistance (R) of a wire, a circuit is designed below. The V-I characteristic curve for this circuit is plotted for the voltmeter and the ammeter readings as shown in figure. The value of R is _____ Ω.



Open solution

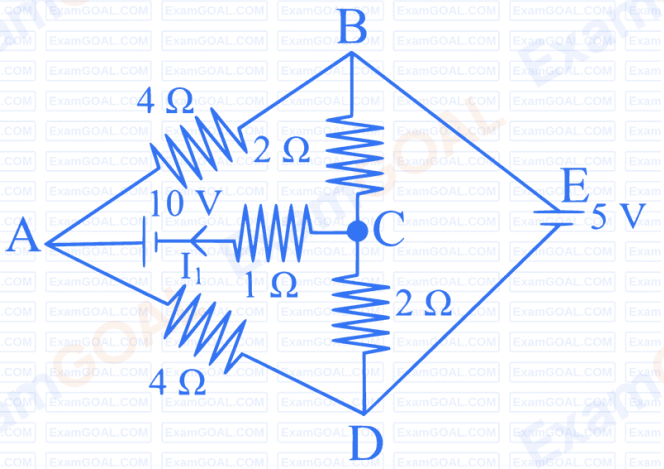
Q14. JEE Main 2024 (Online) 9th April Evening Shift

At room temperature $(27^{\circ}\mathrm{C})$, the resistance of a heating element is 50Ω. The temperature coefficient of the material is $2.4 \times 10^{-4}{}^{\circ}\mathrm{C}^{-1}$. The temperature of the element, when its resistance is 62Ω, is _____ $^{\circ}\mathrm{C}$.

Open solution

Q15. JEE Main 2024 (Online) 9th April Morning Shift

The current flowing through the 1Ω resistor is $\frac{n}{10}$ A. The value of n is _____.

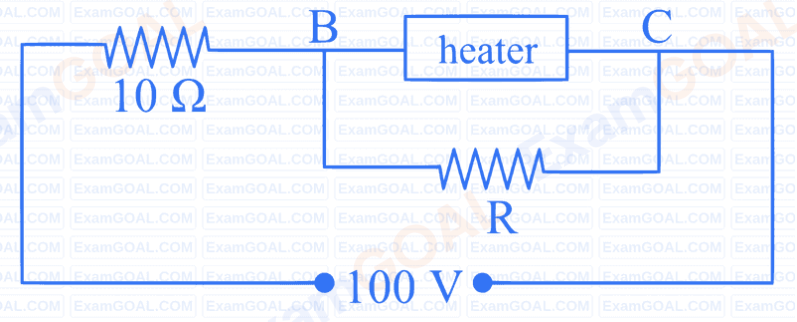


Open solution

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Q16. JEE Main 2024 (Online) 8th April Evening Shift

A heater is designed to operate with a power of $1000\ \mathrm{W}$ in a $100\ \mathrm{V}$ line. It is connected in combination with a resistance of 10Ω and a resistance R, to a $100\ \mathrm{V}$ mains as shown in figure. For the heater to operate at $62.5\ \mathrm{W}$, the value of R should be _____ Ω.



Open solution

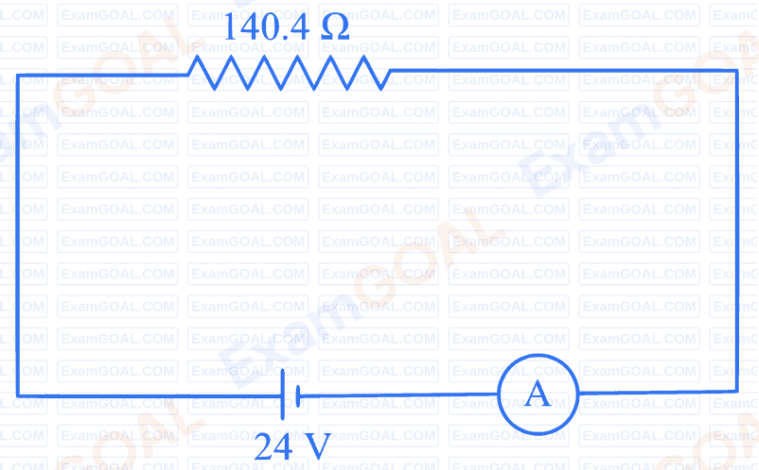
Q17. JEE Main 2024 (Online) 8th April Morning Shift

Resistance of a wire at $0^{\circ}\mathrm{C}, 100^{\circ}\mathrm{C}$ and $t^{\circ}\mathrm{C}$ is found to be $10\Omega, 10.2\Omega$ and 10.95Ω respectively. The temperature t in Kelvin scale is _____.

Open solution

Q18. JEE Main 2024 (Online) 6th April Evening Shift

In the given figure an ammeter A consists of a 240Ω coil connected in parallel to a 10Ω shunt. The reading of the ammeter is _____ mA.



Open solution

Q19. JEE Main 2024 (Online) 6th April Morning Shift

A wire of resistance R and radius r is stretched till its radius became $r/2$. If new resistance of the stretched wire is $x\ R$, then value of x is _____.

Open solution

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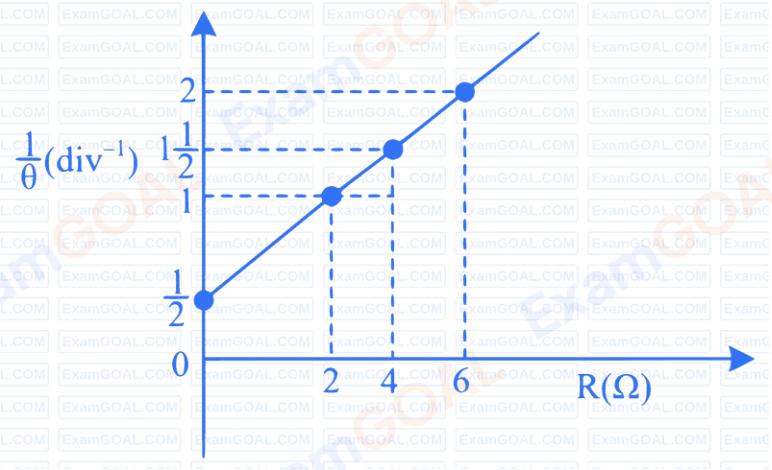
Q20. JEE Main 2024 (Online) 5th April Evening Shift

A wire of resistance 20Ω is divided into 10 equal parts, resulting pairs. A combination of two parts are connected in parallel and so on. Now resulting pairs of parallel combination are connected in series. The equivalent resistance of final combination is _____ Ω.

Open solution

Q21. JEE Main 2024 (Online) 5th April Morning Shift

In the experiment to determine the galvanometer resistance by half-deflection method, the plot of $1/\theta$ vs the resistance (R) of the resistance box is shown in the figure. The figure of merit of the galvanometer is _____ $\times 10^{-1}$ A/ division. [The source has emf 2 V]



Open solution

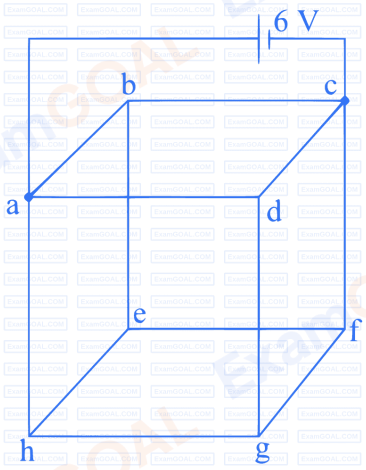
Q22. JEE Main 2024 (Online) 4th April Evening Shift

Two wires A and B are made up of the same material and have the same mass. Wire A has radius of 2.0 mm and wire B has radius of 4.0 mm. The resistance of wire B is 2Ω. The resistance of wire A is _____ Ω.

Open solution

Q23. JEE Main 2024 (Online) 4th April Morning Shift

Twelve wires each having resistance 2Ω are joined to form a cube. A battery of 6 V emf is joined across point a and c. The voltage difference between e and f is _____ V.



Open solution

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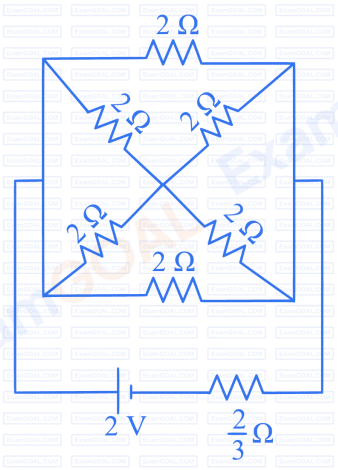
Q24. JEE Main 2024 (Online) 1st February Morning Shift

The current in a conductor is expressed as $I = 3t^2 + 4t^3$, where I is in Ampere and t is in second. The amount of electric charge that flows through a section of the conductor during $t = 1\,\text{s}$ to $t = 2\,\text{s}$ is _____ C.

Open solution

Q25. JEE Main 2024 (Online) 31st January Evening Shift

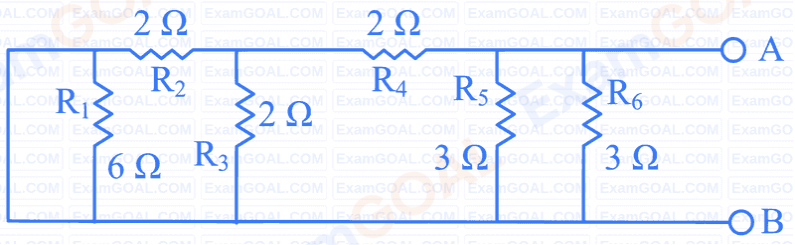
In the following circuit, the battery has an emf of $2\,\text{V}$ and an internal resistance of $\frac{2}{3}\Omega$. The power consumption in the entire circuit is _____ W.



Open solution

Q26. JEE Main 2024 (Online) 31st January Morning Shift

Equivalent resistance of the following network is _____ Ω.



Open solution

Q27. JEE Main 2024 (Online) 30th January Evening Shift

Two resistances of 100Ω and 200Ω are connected in series with a battery of $4\,\text{V}$ and negligible internal resistance. A voltmeter is used to measure voltage across 100Ω resistance, which gives reading as $1\,\text{V}$. The resistance of voltmeter must be _____ Ω.

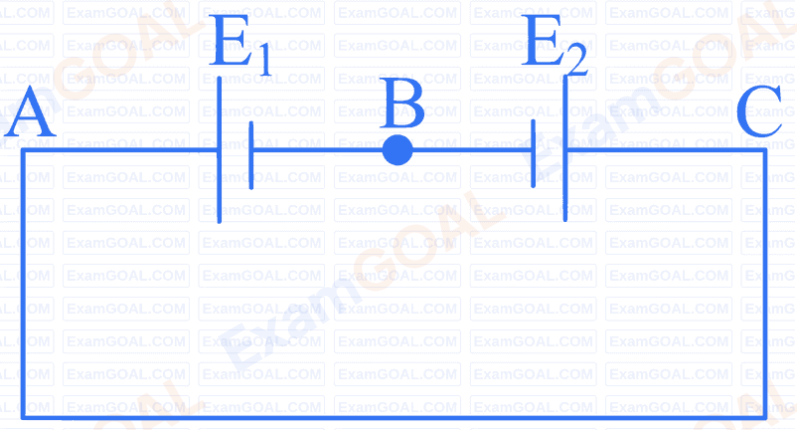
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Q28. JEE Main 2024 (Online) 30th January Morning Shift

Two cells are connected in opposition as shown. Cell E_1 is of $8\ V$ emf and 2Ω internal resistance; the cell E_2 is of $2\ V$ emf and 4Ω internal resistance. The terminal potential difference of cell E_2 is _____ V.

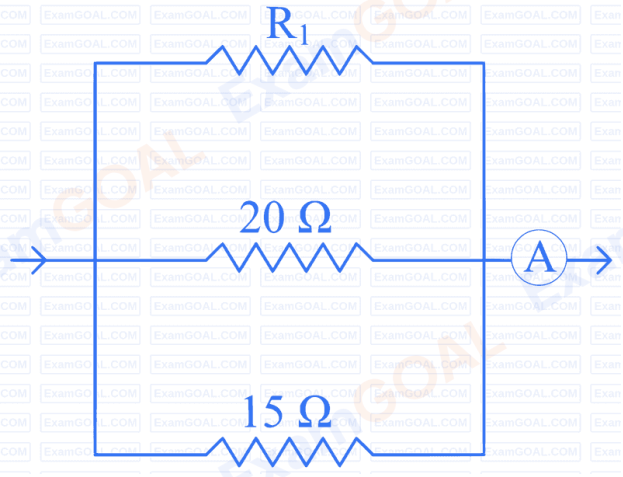
Open solution



Q29. JEE Main 2024 (Online) 29th January Evening Shift

In the given circuit, the current flowing through the resistance 20Ω is $0.3\ A$, while the ammeter reads $0.9\ A$. The value of R_1 is _____ Ω.

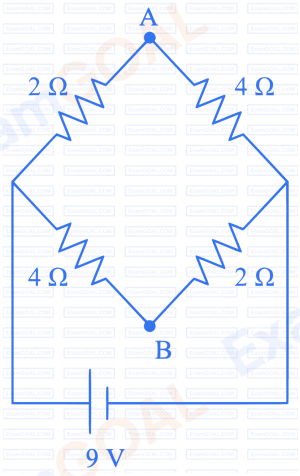
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Q30. JEE Main 2023 (Online) 15th April Morning Shift

A network of four resistances is connected to $9\ \mathrm{V}$ battery, as shown in figure. The magnitude of voltage difference between the points A and B is _____ V.



Open solution

Q31. JEE Main 2023 (Online) 13th April Morning Shift

When a resistance of $5\ \Omega$ is shunted with a moving coil galvanometer, it shows a full scale deflection for a current of $250\ \mathrm{mA}$, however when $1050\ \Omega$ resistance is connected with it in series, it gives full scale deflection for 25 volt. The resistance of galvanometer is _____ Ω.

Open solution

Q32. JEE Main 2023 (Online) 13th April Morning Shift

A potential V_0 is applied across a uniform wire of resistance R. The power dissipation is P_1. The wire is then cut into two equal halves and a potential of V_0 is applied across the length of each half. The total power dissipation across two wires is P_2. The ratio $P_2 : \mathrm{P}_1$ is $\sqrt{x} : 1$. The value of x is _____.

Open solution

Q33. JEE Main 2023 (Online) 12th April Morning Shift

The current flowing through a conductor connected across a source is $2\ \mathrm{A}$ and $1.2\ \mathrm{A}$ at $0^{\circ}\mathrm{C}$ and $100^{\circ}\mathrm{C}$ respectively. The current flowing through the conductor at $50^{\circ}\mathrm{C}$ will be _____ $\times 10^2\ \mathrm{mA}$.

Open solution

Q34. JEE Main 2023 (Online) 11th April Evening Shift

Two identical cells each of emf $1.5\ \mathrm{V}$ are connected in series across a $10\ \Omega$ resistance. An ideal voltmeter connected across $10\ \Omega$ resistance reads $1.5\ \mathrm{V}$. The internal resistance of each cell is _____ Ω.

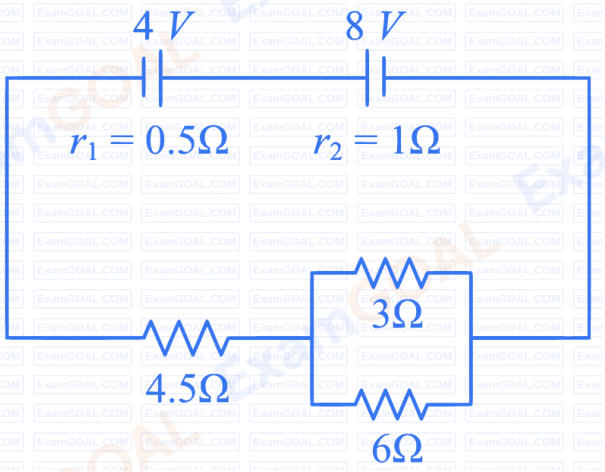
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Q35. JEE Main 2023 (Online) 11th April Morning Shift

In the circuit diagram shown in figure given below, the current flowing through resistance $3\ \Omega$ is $\frac{x}{3}A$.

The value of x is _____



Open solution

Q36. JEE Main 2023 (Online) 10th April Evening Shift

A rectangular parallelopiped is measured as $1\ \text{cm} \times 1\ \text{cm} \times 100\ \text{cm}$. If its specific resistance is $3 \times 10^{-7}\ \Omega\text{m}$, then the resistance between its two opposite rectangular faces will be _____ $\times 10^{-7}\ \Omega$.

Open solution

Q37. JEE Main 2023 (Online) 10th April Morning Shift

10 resistors each of resistance $10\ \Omega$ can be connected in such as to get maximum and minimum equivalent resistance. The ratio of maximum and minimum equivalent resistance will be _____.

Open solution

Q38. JEE Main 2023 (Online) 8th April Evening Shift

The number density of free electrons in copper is nearly $8 \times 10^{28}\ \text{m}^{-3}$. A copper wire has its area of cross section $= 2 \times 10^{-6}\ \text{m}^2$ and is carrying a current of $3.2\ \text{A}$. The drift speed of the electrons is _____ $\times 10^{-6}\text{ms}^{-1}$

Open solution

Q39. JEE Main 2023 (Online) 8th April Morning Shift

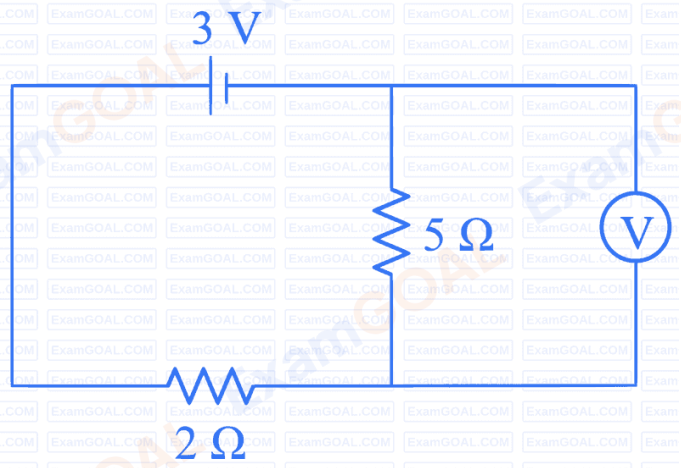
A current of $2\ \text{A}$ flows through a wire of cross-sectional area $25.0\ \text{mm}^2$. The number of free electrons in a cubic meter are 2.0×10^{28}. The drift velocity of the electrons is _____ $\times 10^{-6}\ \text{ms}^{-1}$ (given, charge on electron $= 1.6 \times 10^{-19}\text{C}$).

Open solution

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Q40. JEE Main 2023 (Online) 6th April Evening Shift

As shown in the figure, the voltmeter reads $2\ \text{V}$ across $5\ \Omega$ resistor. The resistance of the voltmeter is _____ Ω.



Open solution

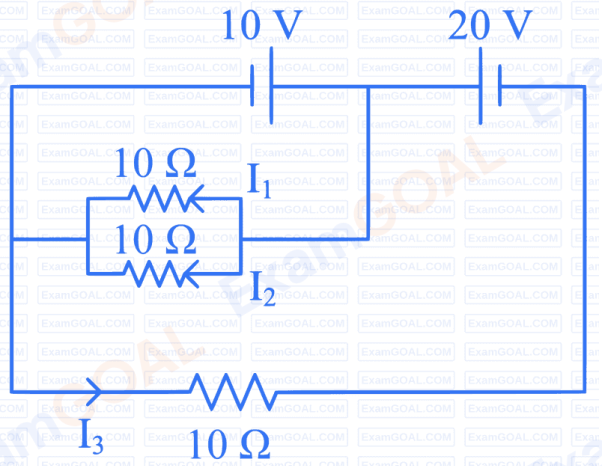
Q41. JEE Main 2023 (Online) 6th April Morning Shift

The length of a metallic wire is increased by 20% and its area of cross section is reduced by 4%. The percentage change in resistance of the metallic wire is _____.

Open solution

Q42. JEE Main 2023 (Online) 1st February Evening Shift

In the given circuit, the value of $\left|\frac{I_1+I_3}{I_2}\right|$ is _____



Open solution

Q43. JEE Main 2023 (Online) 31st January Morning Shift

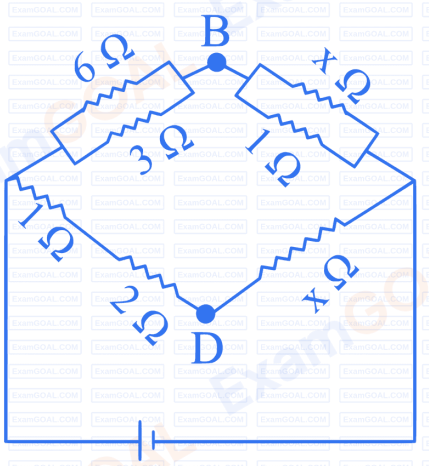
Two identical cells, when connected either in parallel or in series gives same current in an external resistance $5\ \Omega$. The internal resistance of each cell will be _____ Ω.

Open solution

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Q44. JEE Main 2023 (Online) 30th January Evening Shift

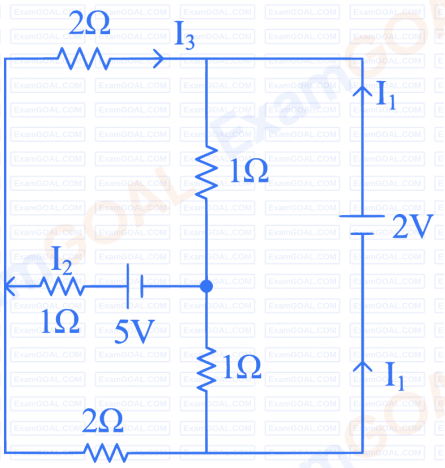
If the potential difference between B and D is zero, the value of x is $\frac{1}{n}\Omega$. The value of n is _____.



Open solution

Q45. JEE Main 2023 (Online) 30th January Morning Shift

In the following circuit, the magnitude of current I_1, is _____ A.



Open solution

Q46. JEE Main 2023 (Online) 29th January Evening Shift

When two resistance R_1 and R_2 connected in series and introduced into the left gap of a meter bridge and a resistance of 10 Ω is introduced into the right gap, a null point is found at 60 cm from left side. When R_1 and R_2 are connected in parallel and introduced into the left gap, a resistance of 3 Ω is introduced into the right gap to get null point at 40 cm from left end. The product of $\mathrm{R}_1\,\mathrm{R}_2$ is _____Ω^2

Open solution

Q47. JEE Main 2023 (Online) 29th January Morning Shift

In a metre bridge experiment the balance point is obtained if the gaps are closed by 2Ω and 3Ω. A shunt of X Ω is added to 3Ω resistor to shift the balancing point by 22.5 cm. The value of X is _____.

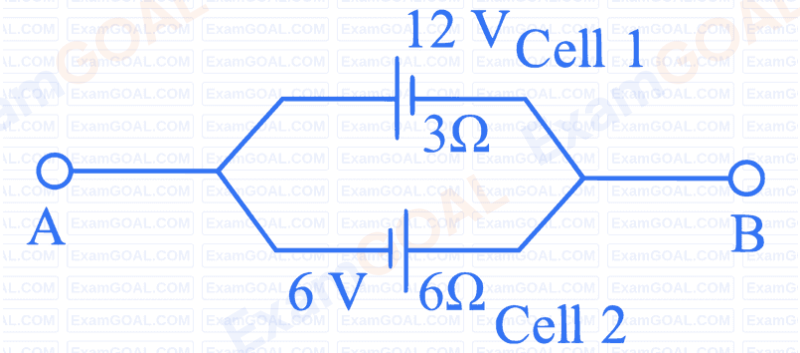
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Q48. JEE Main 2023 (Online) 25th January Evening Shift

Two cells are connected between points A and B as shown. Cell 1 has emf of 12 V and internal resistance of 3Ω. Cell 2 has emf of 6V and internal resistance of 6Ω. An external resistor R of 4Ω is connected across A and B. The current flowing through R will be _____ A.

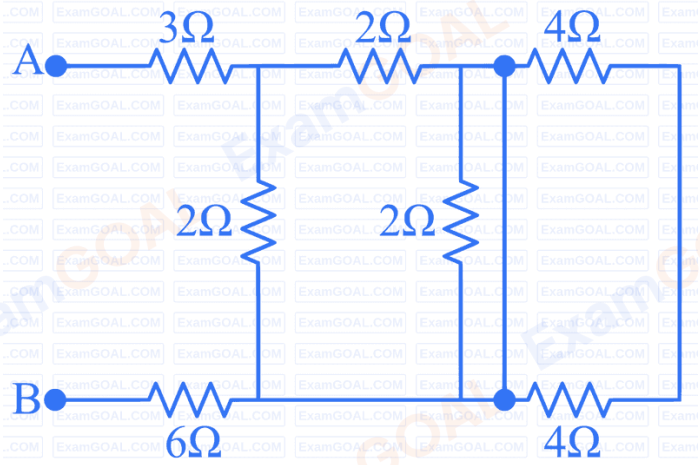
Open solution



Q49. JEE Main 2023 (Online) 25th January Morning Shift

In the given circuit, the equivalent resistance between the terminal A and B is _____ Ω.

Open solution



Q50. JEE Main 2023 (Online) 24th January Evening Shift

If a copper wire is stretched to increase its length by 20%. The percentage increase in resistance of the wire is _____%.

Open solution

Q51. JEE Main 2023 (Online) 24th January Morning Shift

A hollow cylindrical conductor has length of 3.14 m, while its inner and outer diameters are 4 mm and 8 mm respectively. The resistance of the conductor is $n \times 10^{-3}\Omega$. If the resistivity of the material is $2.4 \times 10^{-8}\Omega\text{m}$, the value of n is _____.

Open solution

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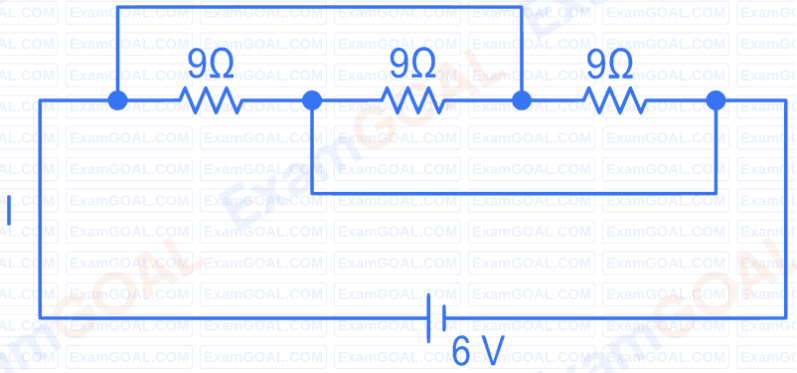
Q52. JEE Main 2022 (Online) 30th July Evening Shift

A cell sends a current through a resistance $4\ \Omega$ for time t and sends current through another resistance $16\ \Omega$ for the same time t. If the same amount of heat is developed in both the resistances, then the internal resistance of the cell is _____ Ω

Open solution

Q53. JEE Main 2022 (Online) 29th July Morning Shift

The current I flowing through the given circuit will be _____A.



Open solution

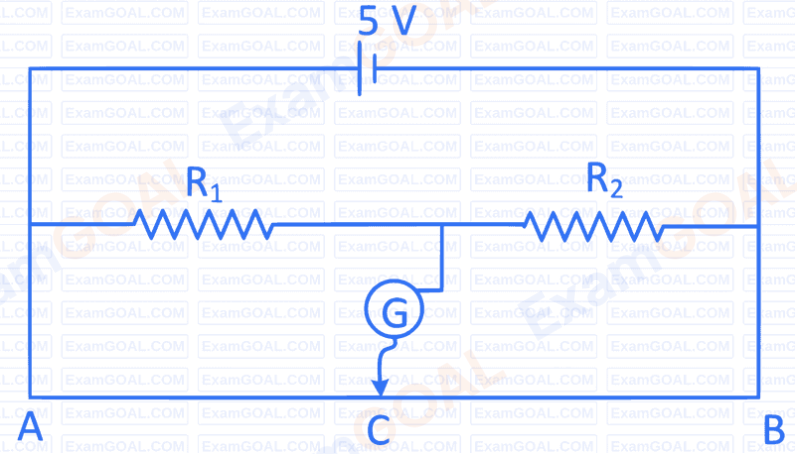
Q54. JEE Main 2022 (Online) 28th July Evening Shift

An electrical bulb rated 220 V, 100 W, is connected in series with another bulb rated 220 V, 60 W. If the voltage across combination is 220 V, the power consumed by the 100 W bulb will be about _____ W.

Open solution

Q55. JEE Main 2022 (Online) 27th July Evening Shift

In the given figure of meter bridge experiment, the balancing length AC corresponding to null deflection of the galvanometer is $40\ \text{cm}$. The balancing length, if the radius of the wire AB is doubled, will be _____ cm.



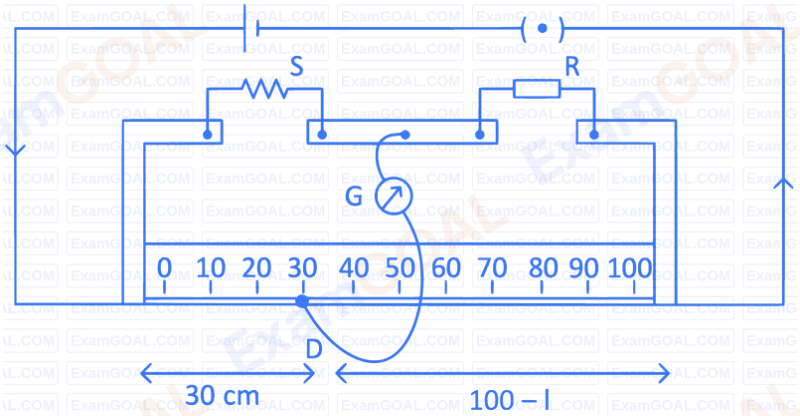
Open solution

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Q56. JEE Main 2022 (Online) 27th July Morning Shift

Open solution

In a meter bridge experiment, for measuring unknown resistance 'S', the null point is obtained at a distance $30\ \text{cm}$ from the left side as shown at point D. If R is $5.6\ \text{k}\Omega$, then the value of unknown resistance 'S' will be _____ Ω.



Q57. JEE Main 2022 (Online) 27th July Morning Shift

Open solution

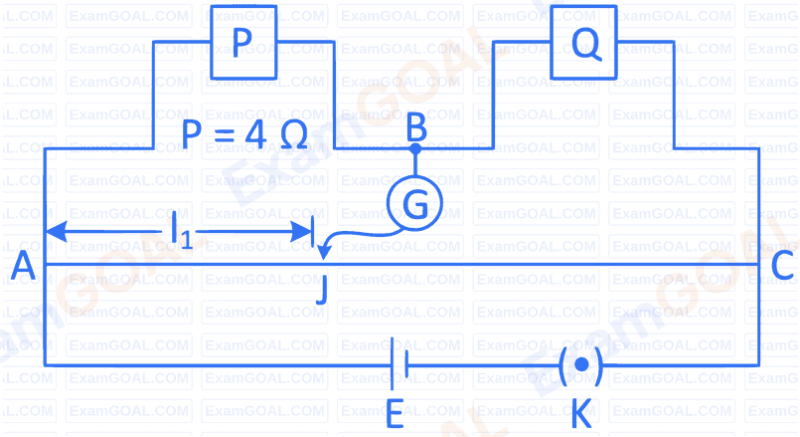
A $1\ \text{m}$ long copper wire carries a current of $1\ \text{A}$. If the cross section of the wire is $2.0\ \text{mm}^2$ and the resistivity of copper is $1.7 \times 10^{-8}\ \Omega\text{m}$, the force experienced by moving electron in the wire is _____ $\times 10^{-23}\ \text{N}$.

(charge on electron $= 1.6 \times 10^{-19}\ \text{C}$)

Q58. JEE Main 2022 (Online) 26th July Morning Shift

Open solution

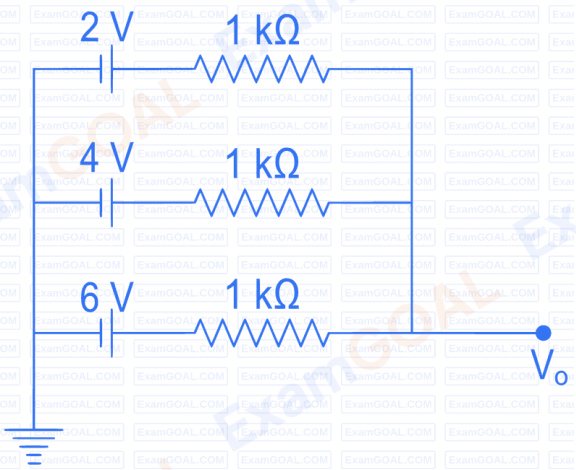
Resistances are connected in a meter bridge circuit as shown in the figure. The balancing length l_1 is $40\ \text{cm}$. Now an unknown resistance x is connected in series with P and new balancing length is found to be $80\ \text{cm}$ measured from the same end. Then the value of x will be _____ Ω.



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Q59. JEE Main 2022 (Online) 25th July Morning Shift

In the given figure, the value of V_o will be _____ V.



Open solution

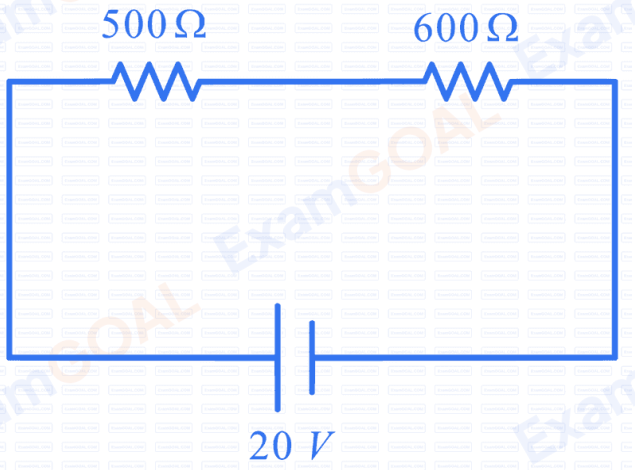
Q60. JEE Main 2022 (Online) 25th July Morning Shift

Eight copper wire of length l and diameter d are joined in parallel to form a single composite conductor of resistance R. If a single copper wire of length $2l$ have the same resistance (R) then its diameter will be _____ d.

Open solution

Q61. JEE Main 2022 (Online) 29th June Evening Shift

Two resistors are connected in series across a battery as shown in figure. If a voltmeter of resistance 2000 Ω is used to measure the potential difference across 500 Ω resistor, the reading of the voltmeter will be _____ V.



Open solution

Q62. JEE Main 2022 (Online) 29th June Morning Shift

Two coils require 20 minutes and 60 minutes respectively to produce same amount of heat energy when connected separately to the same source. If they are connected in parallel arrangement to the same source; the time required to produce same amount of heat by the combination of coils, will be _____ min.

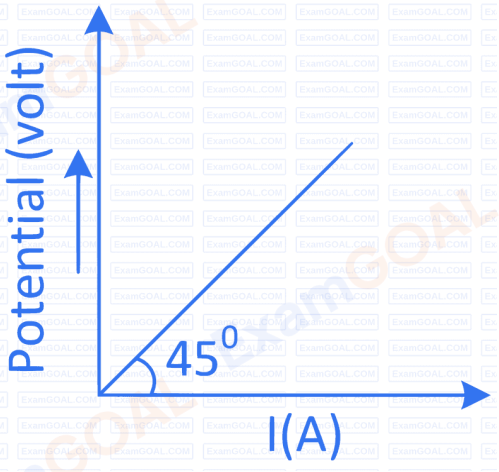
Open solution

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Q63. JEE Main 2022 (Online) 29th June Morning Shift

Open solution

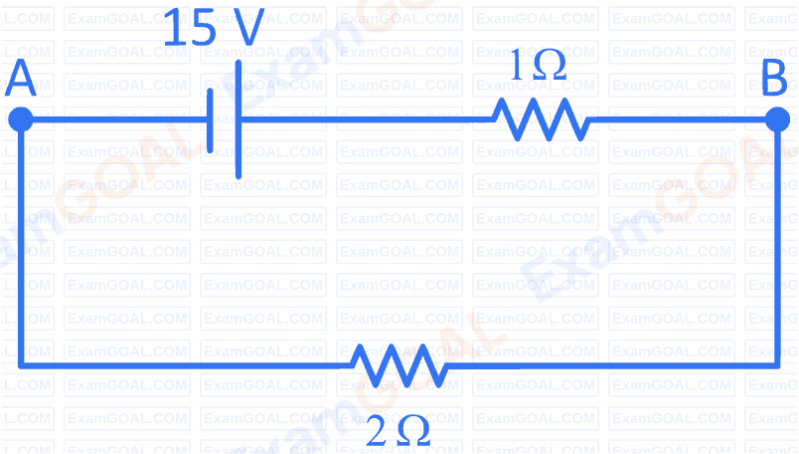
The variation of applied potential and current flowing through a given wire is shown in figure. The length of wire is 31.4 cm. The diameter of wire is measured as 2.4 cm. The resistivity of the given wire is measured as $x \times 10^{-3}\ \Omega$ cm. The value of x is _____. [Take π = 3.14]



Q64. JEE Main 2022 (Online) 29th June Morning Shift

Open solution

For the network shown below, the value of $V_B - V_A$ is _____ V.

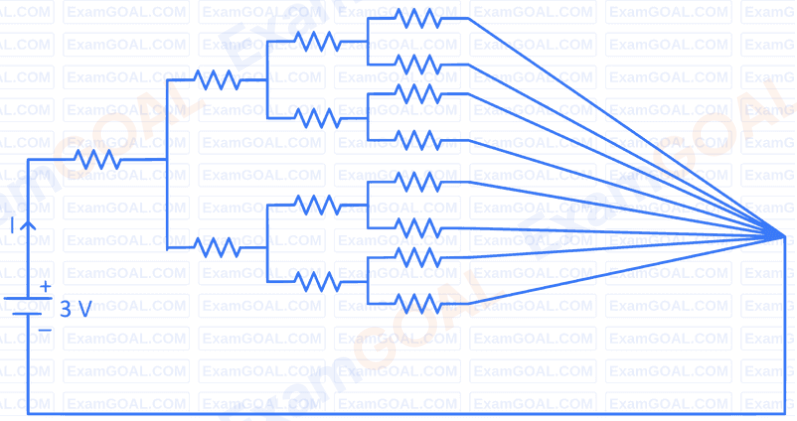


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Q65. JEE Main 2022 (Online) 28th June Evening Shift

All resistances in figure are 1 Ω each. The value of current 'I' is $\frac{a}{5}$ A. The value of a is _____.

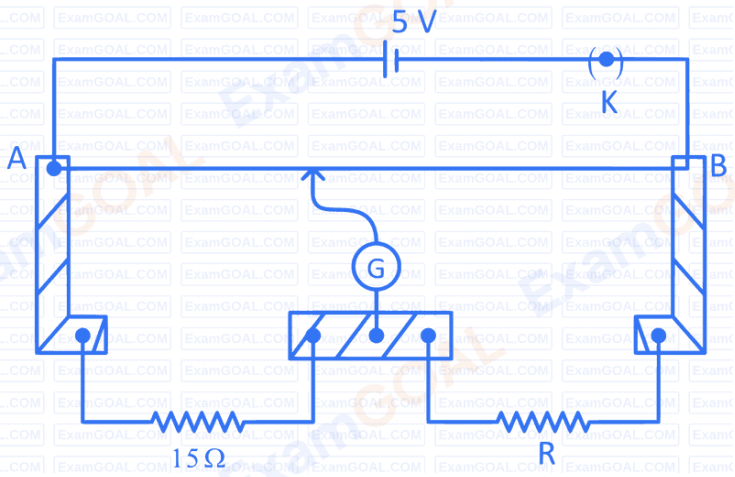
Open solution



Q66. JEE Main 2022 (Online) 28th June Morning Shift

A meter bridge setup is shown in the figure. It is used to determine an unknown resistance R using a given resistor of 15 Ω. The galvanometer (G) shows null deflection when tapping key is at 43 cm mark from end A. If the end correction for end A is 2 cm, then the determined value of R will be _____ Ω.

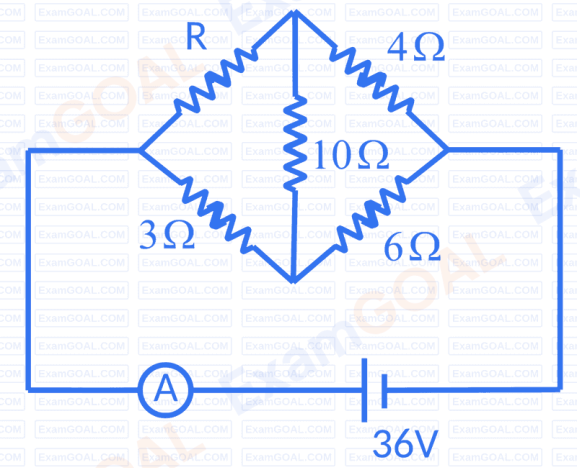
Open solution



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Q67. JEE Main 2022 (Online) 28th June Morning Shift

Current measured by the ammeter (A) in the reported circuit when no current flows through 10 Ω resistance, will be _____ A.



Open solution

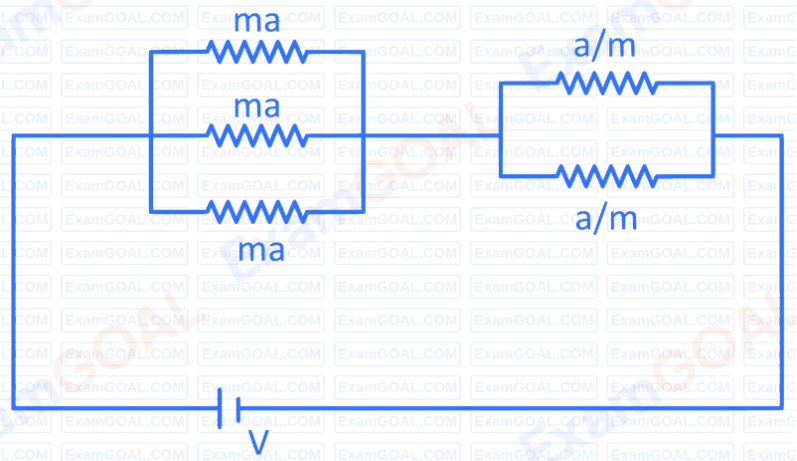
Q68. JEE Main 2022 (Online) 27th June Evening Shift

The current density in a cylindrical wire of radius r = 4.0 mm is 1.0×10^6 A/m^2. The current through the outer portion of the wire between radial distances $\frac{r}{2}$ and r is $x\pi$ A; where x is _____.

Open solution

Q69. JEE Main 2022 (Online) 27th June Evening Shift

In the given circuit 'a' is an arbitrary constant. The value of m for which the equivalent circuit resistance is minimum, will be $\sqrt{\frac{x}{2}}$. The value of x is _____.



Open solution

Q70. JEE Main 2022 (Online) 27th June Morning Shift

The current density in a cylindrical wire of radius 4 mm is 4×10^6 Am^{-2}. The current through the outer portion of the wire between radial distances $\frac{R}{2}$ and R is _____ π A.

Open solution

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Q71. JEE Main 2022 (Online) 25th June Evening Shift

The length of a given cylindrical wire is increased to double of its original length. The percentage increase in the resistance of the wire will be _____ %.

Open solution

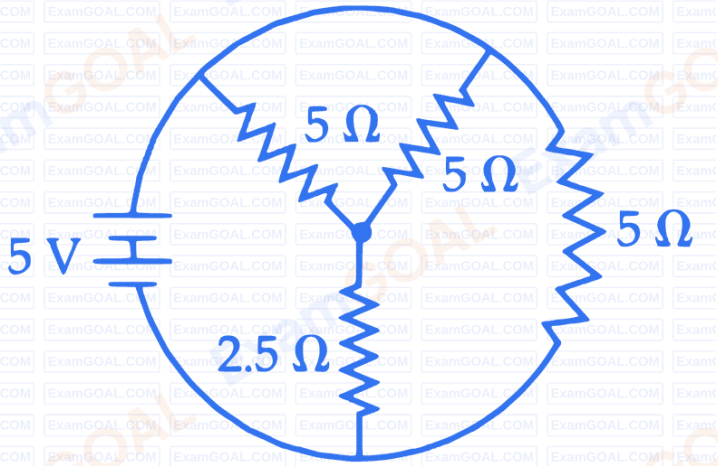
Q72. JEE Main 2022 (Online) 25th June Morning Shift

A resistor develops 300 J of thermal energy in 15 s, when a current of 2 A is passed through it. If the current increases to 3 A, the energy developed in 10 s is _____ J.

Open solution

Q73. JEE Main 2022 (Online) 25th June Morning Shift

The total current supplied to the circuit as shown in figure by the 5 V battery is _____ A.



Open solution

Q74. JEE Main 2021 (Online) 1st September Evening Shift

A uniform heating wire of resistance 36Ω is connected across a potential difference of 240 V. The wire is then cut into half and potential difference of 240V is applied across each half separately. The ratio of power dissipation in first case to the total power dissipation in the second case would be 1 : x, where x is _____

Open solution

Q75. JEE Main 2021 (Online) 31st August Evening Shift

A resistor dissipates 192 J of energy in 1s when a current of 4A is passed through it. Now, when the current is doubled, the amount of thermal energy dissipated in 5s in _____ J.

Open solution

Q76. JEE Main 2021 (Online) 31st August Morning Shift

A square shaped wire with resistance of each side 3Ω is bent to form a complete circle. The resistance between two diametrically opposite points of the circle in unit of Ω will be _____.

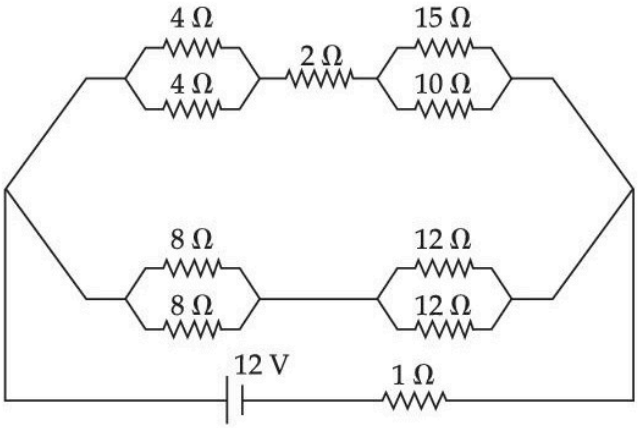
Open solution

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Q77. JEE Main 2021 (Online) 31st August Morning Shift

The voltage drop across 15Ω resistance in the given figure will be _____ V.

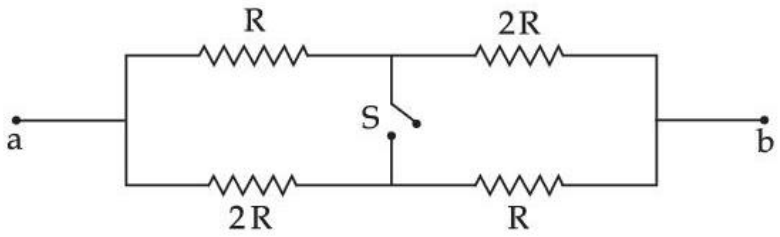
Open solution



Q78. JEE Main 2021 (Online) 27th August Evening Shift

The ratio of the equivalent resistance of the network (shown in figure) between the points a and b when switch is open and switch is closed is x : 8. The value of x is _____.

Open solution



Q79. JEE Main 2021 (Online) 27th August Morning Shift

First, a set of n equal resistors of 10 Ω each are connected in series to a battery of emf 20V and internal resistance 10Ω. A current I is observed to flow. Then, the n resistors are connected in parallel to the same battery. It is observed that the current is increased 20 times, then the value of n is ..... .

Open solution

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Q80. JEE Main 2021 (Online) 27th July Evening Shift

For the circuit shown, the value of current at time t = 3.2 s will be _____ A.

Open solution

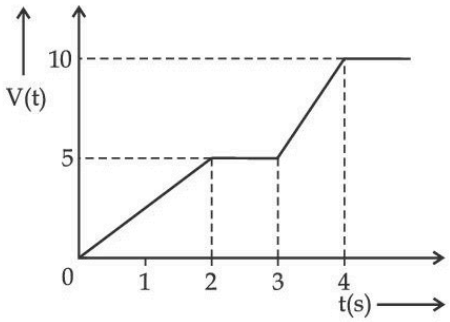


Figure 1

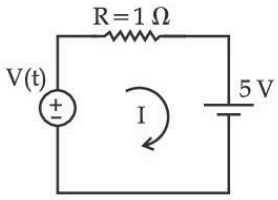


Figure 2

[Voltage distribution V(t) is shown by Fig. (1) and the circuit is shown in Fig. (2)]

Q81. JEE Main 2021 (Online) 25th July Evening Shift

A 16 Ω wire is bent to form a square loop. A 9V supply having internal resistance of 1Ω is connected across one of its sides. The potential drop across the diagonals of the square loop is _____ $\times 10^{-1}$ V

Open solution

Q82. JEE Main 2021 (Online) 25th July Morning Shift

An electric bulb rated as 200 W at 100 V is used in a circuit having 200 V supply. The resistance 'R' that must be put in series with the bulb so that the bulb delivers the same power is _____ Ω.

Open solution

Q83. JEE Main 2021 (Online) 22th July Evening Shift

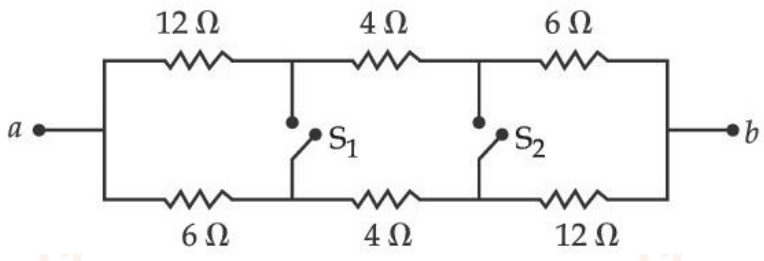
In an electric circuit, a cell of certain emf provides a potential difference of 1.25 V across a load resistance of 5Ω. However, it provides a potential difference of 1 V across a load resistance of 2Ω. The emf of the cell is given by $\frac{x}{10}V$. Then the value of x is _____.

Open solution

Q84. JEE Main 2021 (Online) 20th July Evening Shift

In the given figure switches S_1 and S_2 are in open condition. The resistance across ab when the switches S_1 and S_2 are closed is _____ Ω.

Open solution



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Q85. JEE Main 2021 (Online) 18th March Evening Shift

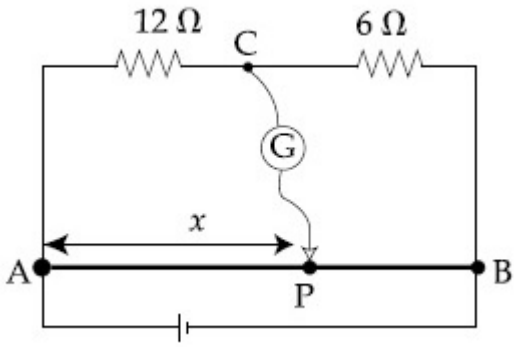
Two wires of same length and thickness having specific resistances 6Ω cm and 3Ω cm respectively are connected in parallel. The effective resistivity is $\rho\Omega$ cm. The value of ρ, to the nearest integer, is _____.

Open solution

Q86. JEE Main 2021 (Online) 18th March Evening Shift

Consider a 72 cm long wire AB as shown in the figure. The galvanometer jockey is placed at P on AB at a distance x cm from A. The galvanometer shows zero deflection.

Open solution

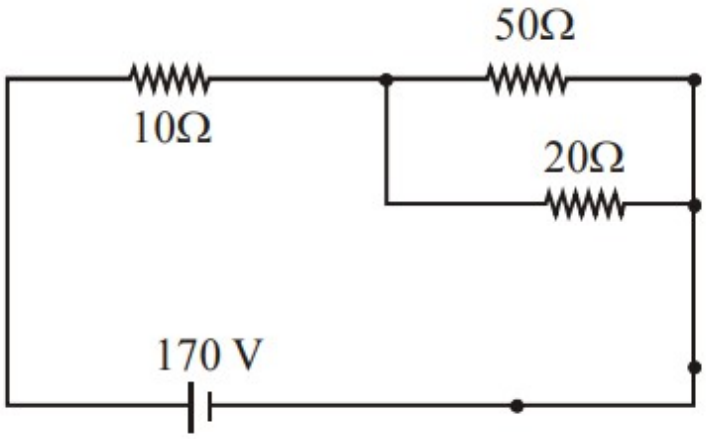


The value of x, to the nearest integer, is _____.

Q87. JEE Main 2021 (Online) 18th March Morning Shift

The voltage across the 10Ω resistor in the given circuit is x volt.

Open solution



The value of 'x' to the nearest integer is _____.

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Q88. JEE Main 2021 (Online) 17th March Morning Shift

The equivalent resistance of series combination of two resistors is 's'. When they are connected in parallel, the equivalent resistance is 'p'. If s = np, then the minimum value for n is _____.
(Round off to the Nearest Integer)

Open solution

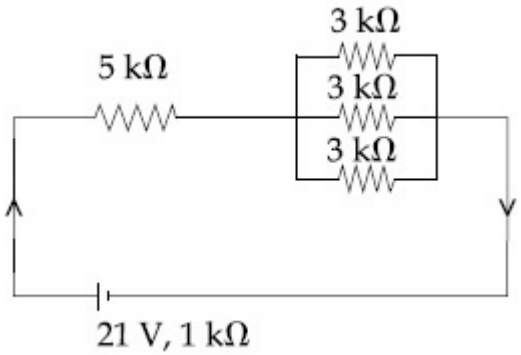
Q89. JEE Main 2021 (Online) 16th March Evening Shift

The energy dissipated by a resistor is 10 mJ in 1 s when an electric current of 2 mA flows through it. The resistance is _____Ω. (Round off to the Nearest Integer)

Open solution

Q90. JEE Main 2021 (Online) 16th March Morning Shift

In the figure given, the electric current flowing through the 5 kΩ resistor is 'x' mA.



The value of x to the nearest integer is _____.

Open solution

Q91. JEE Main 2021 (Online) 26th February Morning Shift

In an electrical circuit, a battery is connected to pass 20C of charge through it in a certain given time. The potential difference between two plates of the battery is maintained at 15V. The workdone by the battery is _____J.

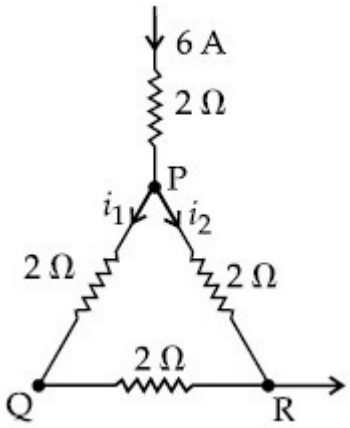
Open solution

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Q92. JEE Main 2021 (Online) 25th February Evening Shift

A current of 6A enters one corner P of an equilateral triangle PQR having 3 wires of resistance 2Ω each and leaves by the corner R. The currents i_1 in ampere is _____.

Open solution



Q93. JEE Main 2021 (Online) 24th February Evening Shift

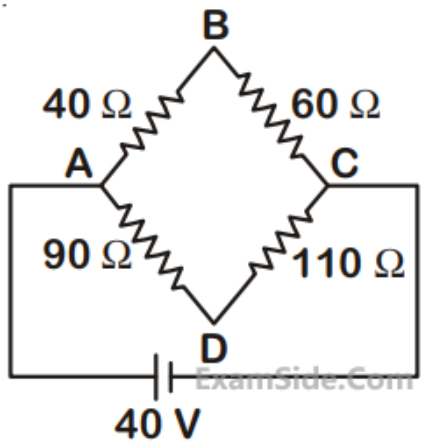
A cylindrical wire of radius 0.5 mm and conductivity 5×10^7 S/m is subjected to an electric field of 10 mV/m. The expected value of current in the wire will be $x^3\pi$ mA. The value of x is _____.

Open solution

Q94. JEE Main 2020 (Online) 4th September Evening Slot

Four resistances 40 Ω, 60 Ω, 90 Ω and 110 Ω make the arms of a quadrilateral ABCD. Across AC is a battery of emf 40 V and internal resistance negligible. The potential difference across BD in V is _____.

Open solution



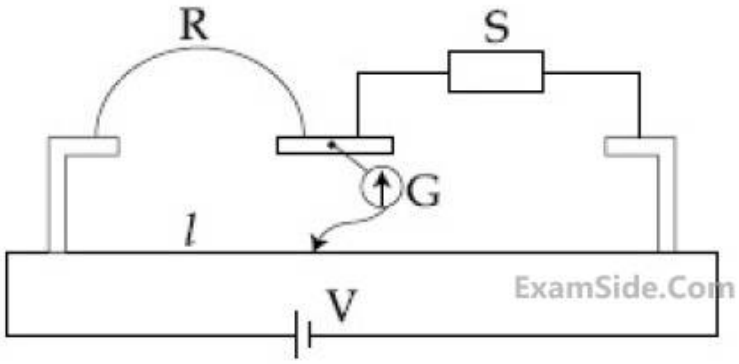
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Q95. JEE Main 2020 (Online) 9th January Evening Slot

In a meter bridge experiment S is a standard resistance. R is a resistance wire. It is found that balancing length is l = 25 cm. If R is replaced by a wire of half length and half diameter that of R of same material, then the balancing distance l' (in cm) will now be_____.

Open solution



Q96. JEE Main 2020 (Online) 8th January Evening Slot

The series combination of two batteries, both of the same emf 10 V, but different internal resistance of 20Ω and 5Ω, is connected to the parallel combination of two resistors 30Ω and R Ω. The voltage difference across the battery of internal resistance 20Ω is zero, the value of R (in Ω) is : _____

Open solution

Q97. JEE Main 2020 (Online) 8th January Morning Slot

Four resistances of 15Ω, 12Ω, 4Ω and 10Ω are connected respectively in cyclic order to form Wheatstone's network. The resistance that is to be connected in parallel with the resistance of 10Ω to balance the network is _____Ω.

Open solution

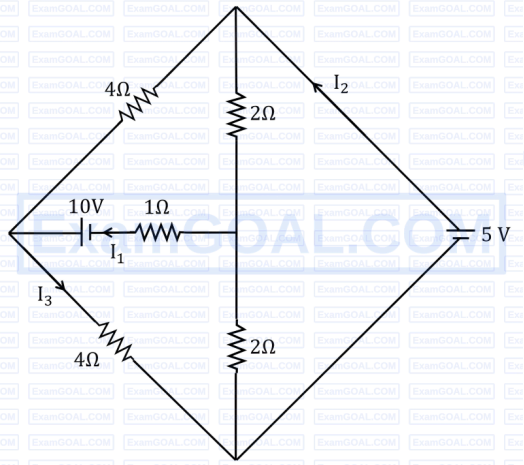
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MCQ (Single Correct Answer) 277 questions

Q98. JEE Main 2026 (Online) 5th April Morning Shift

Open solution

Refer to the figure given below. The values of I_1, I_2 and I_3 are ____.



(A) $I_1 = 2.5\ \text{A}, I_2 = 1.875\ \text{A}, I_3 = 1.875\ \text{A}$

(B) $I_1 = 1.875\ \text{A}, I_2 = 2.5\ \text{A}, I_3 = 1.875\ \text{A}$

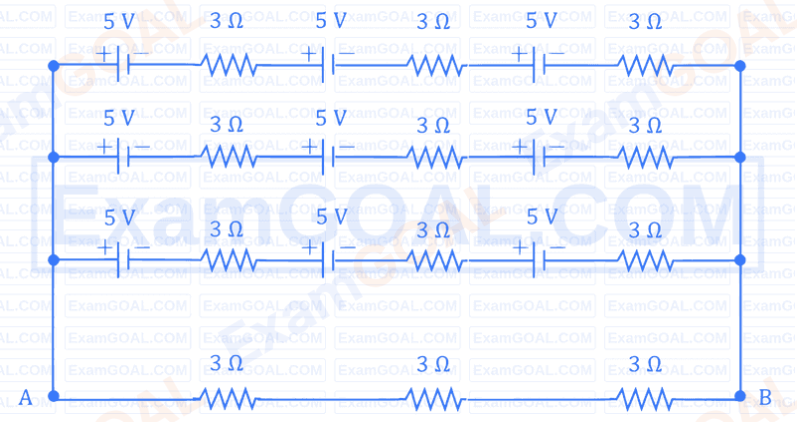
(C) $I_1 = 1.875\ \text{A}, I_2 = 1.875\ \text{A}, I_3 = 2.5\ \text{A}$

(D) $I_1 = 2.5\ \text{A}, I_2 = 2.5\ \text{A}, I_3 = 1.875\ \text{A}$

Q99. JEE Main 2026 (Online) 5th April Morning Shift

Open solution

Refer to the figure given below, current between terminals A and B is ____ A.



(A) 12.5 (B) 1.25 (C) 7.5 (D) 5

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Q100. JEE Main 2026 (Online) 4th April Morning Shift

A voltmeter with internal resistance of $x\Omega$ can be used to measure upto 20 V . In order to increase its measuring range to 30 V , the required modification is to ____ .

Open solution

(A) connect resistor of $\frac{x}{2}\Omega$, in series with voltmeter.

(B) connect resistor of $\frac{x}{2}\Omega$, in parallel to voltmeter.

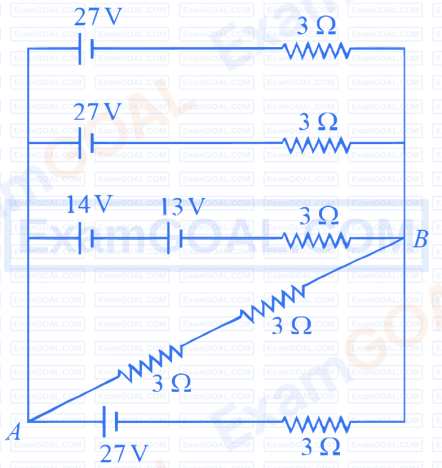
(C) connect a resistor of $x\Omega$ in series with voltmeter.

(D) connect resistor of $2x\Omega$ in parallel to voltmeter.

Q101. JEE Main 2026 (Online) 4th April Morning Shift

The voltage and the current between A and B points shown in the circuit are ____ .

Open solution



(A) $24\ \text{V}, 12\ \text{A}$ (B) $24\ \text{V}, 4\ \text{A}$ (C) $18\ \text{V}, 12\ \text{A}$ (D) $27\ \text{V}, 4\ \text{A}$

Q102. JEE Main 2026 (Online) 2nd April Evening Shift

Two resistors of 200 Ω and 400 Ω are connected in series with a battery of 100 V. A bulb rated at 200 V, 100 W is connected across the 400 Ω resistance. The potential drop across the bulb is _____ V.

Open solution

(A) 25 (B) 50 (C) 66.6 (D) 100

Q103. JEE Main 2026 (Online) 28th January Evening Shift

Which one of the following is **not** a measurable quantity?

Open solution

(A) Resistance (B) Voltage difference

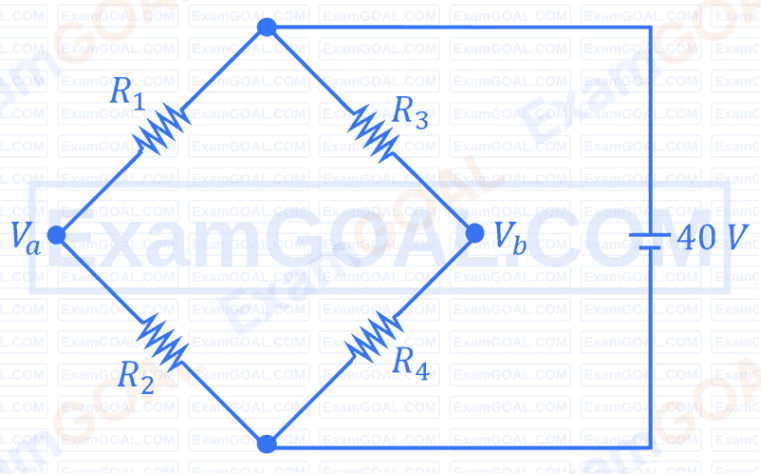
(C) Displacement current (D) Voltage

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Q104. JEE Main 2026 (Online) 28th January Evening Shift

Open solution

A Wheatstone bridge is initially at room temperature and all arms of the bridge have same value of resistances ($R_1 = R_2 = R_3 = R_4$). When R_3 resistance is heated to some temperature, its resistance value has gone up by 10%. The potential difference ($V_a - V_b$) (after R_3 is heated) is _____ V.



(A) 1.05 (B) 0.95 (C) 0 (D) 2

Q105. JEE Main 2026 (Online) 28th January Morning Shift

Open solution

For the two cells having same EMF E and internal resistance r, the current passing through the external resistor 6Ω is same when both the cells are connected either in parallel or in series. The value of internal resistance r is ____ Ω.

(A) 3 (B) 6 (C) 4 (D) 9

Q106. JEE Main 2026 (Online) 28th January Morning Shift

Open solution

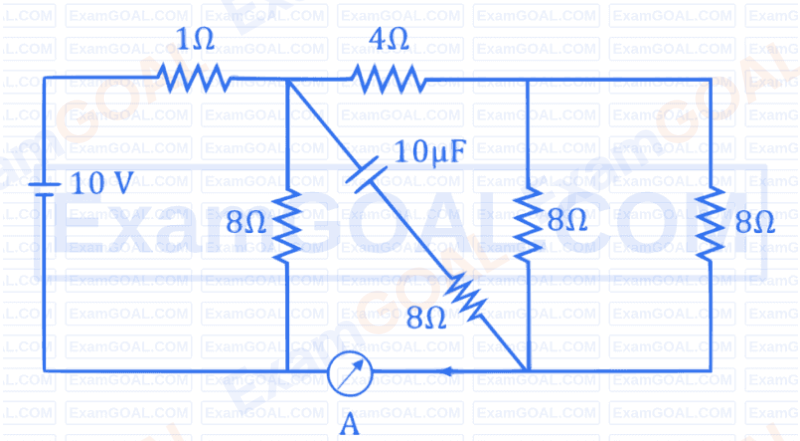
In the potentiometer, when the cell in the secondary circuit is shunted with 4Ω resistance, the balance is obtained at the length 120 cm of wire. Now when the same cell is shunted with 12Ω resistance, the balance is shifted to a length of 180 cm . The internal resistance of cell is ____ Ω

(A) 4 (B) 12 (C) 6 (D) 3

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Q107. JEE Main 2026 (Online) 24th January Evening Shift

The reading of the ammeter (A) in steady state in the following circuit (assuming negligible internal resistance of the ammeter) is ____ A.



Open solution

(A) 1/2 (B) 1 (C) 2 (D) 0

Q108. JEE Main 2026 (Online) 24th January Evening Shift

A moving coil galvanometer of resistance 100Ω shows a full scale deflection for a current of 1 mA . The value of resistance required to convert this galvanometer into an ammeter, showing full scale deflection for a current of 5 mA , is ____ Ω

Open solution

(A) 25 (B) 2.5 (C) 0.5 (D) 10

Q109. JEE Main 2026 (Online) 24th January Evening Shift

A regular hexagon is formed by six wires each of resistance $r\Omega$ and the corners are joined to the centre by wires of same resistance. If the current enters at one corner and leaves at the opposite corner, the equivalent resistance of the hexagon between the two opposite corners will be

Open solution

(A) $\frac{3}{5}r$ (B) $\frac{4}{5}r$ (C) $\frac{5}{8}r$ (D) $\frac{3}{4}r$

Q110. JEE Main 2026 (Online) 24th January Morning Shift

Two resistors of 100Ω each are connected in series with a 9 V battery. A voltmeter of 400Ω resistance is connected to measure the voltage drop across one of the resistors. The voltmeter reading is ____ V.

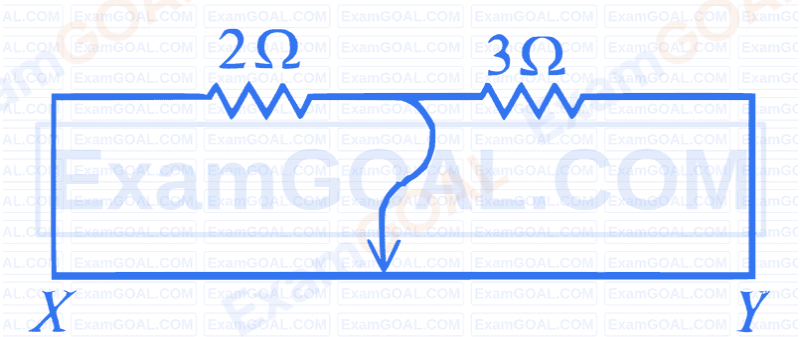
Open solution

(A) 3 (B) 2 (C) 4 (D) 4.5

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Q111. JEE Main 2026 (Online) 24th January Morning Shift

Two resistors 2Ω and 3Ω are connected in the gaps of bridge as shown in figure. The null point is obtained with the contact of jockey at some point on wire XY. When an unknown resistor is connected in parallel with 3Ω resistor, the null point is shifted by 22.5 cm toward Y. The resistance of unknown resistor is ____ Ω.



Open solution

(A) 3 (B) 1 (C) 4 (D) 2

Q112. JEE Main 2026 (Online) 23rd January Evening Shift

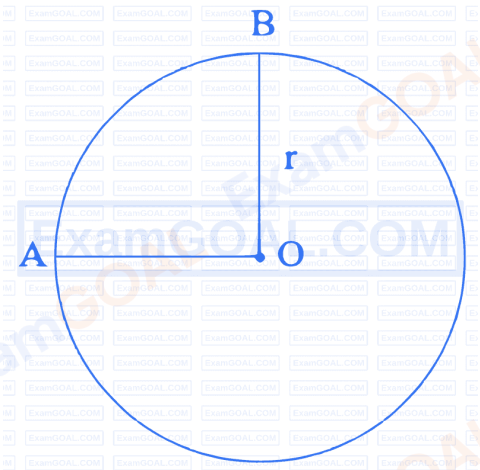
To compare EMF of two cells using potentiometer the balancing lengths obtained are 200 cm and 150 cm . The least count of scale is 1 cm . The percentage error in the ratio of EMFs is ____

Open solution

(A) 1.75 (B) 1.55 (C) 1.17 (D) 1.65

Q113. JEE Main 2026 (Online) 23rd January Morning Shift

A wire of uniform resistance $\lambda\Omega/\mathrm{m}$ is bent into a circle of radius r and another piece of wire with length $2r$ is connected between points A and $B(\mathrm{AOB})$ as shown in figure. The equivalent resistance between points A and B is ____ Ω.



Open solution

(A) $(\pi+1)2r\lambda$ (B) $\frac{6\pi\lambda r}{3\pi+16}$ (C) $\frac{3\pi\lambda r}{8}$ (D) $2\pi\lambda r$

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Q114. JEE Main 2026 (Online) 22nd January Evening Shift

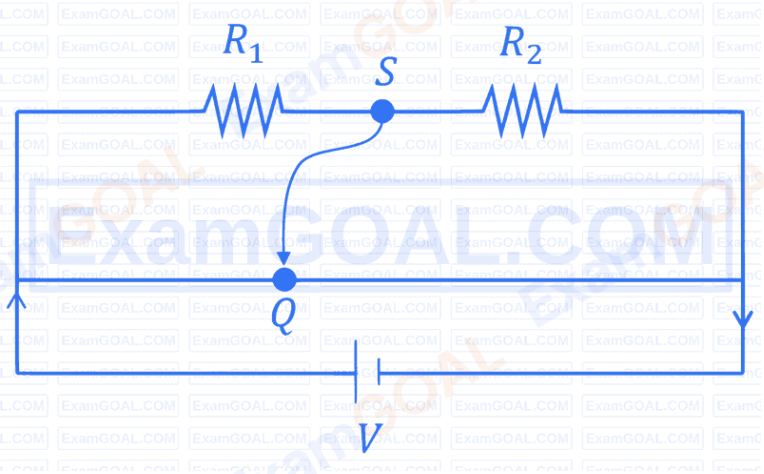
An electric power line having total resistance of 2Ω, delivers 1 kW of power at 250 V . The percentage efficiency of transmission line is ____ .

(A) 96.9 (B) 92.5 (C) 86.5 (D) 100

Open solution

Q115. JEE Main 2026 (Online) 22nd January Morning Shift

A meter bridge with two resistances R_1 and R_2 as shown in figure was balanced (null point) at 40 cm from the point P. The null point changed to 50 cm from the point P, when 16Ω resistance is connected in parallel to R_2. The values of resistances R_1 and R_2 are ____ .



Open solution

(A) $R_2 = 8\Omega, R_1 = \frac{16}{3}\Omega$ (B) $R_2 = 12\Omega, R_1 = \frac{12}{3}\Omega$

(C) $R_2 = 4\Omega, R_1 = \frac{4}{3}\Omega$ (D) $R_2 = 16\Omega, R_1 = \frac{16}{3}\Omega$

Q116. JEE Main 2026 (Online) 21st January Evening Shift

Two known resistances of $R\,\Omega$ and $2R\,\Omega$ and one unknown resistance $X\,\Omega$ are connected in a circuit as shown in the figure. If the equivalent resistance between points A and B in the circuit is $X\,\Omega$, then the value of X is _____ Ω.



Open solution

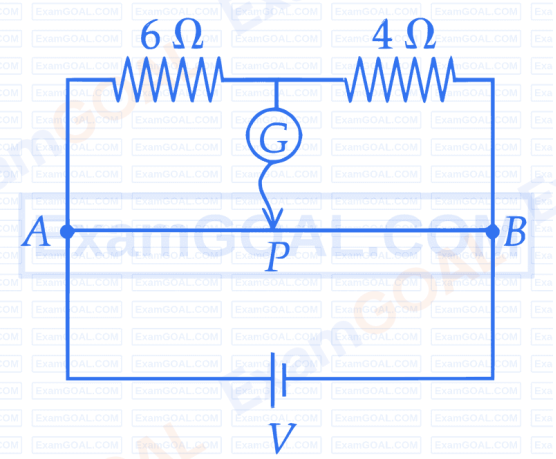
(A) $2(\sqrt{3}-1)R$ (B) R (C) $(\sqrt{3}-1)R$ (D) $(\sqrt{3}+1)R$

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Q117. JEE Main 2026 (Online) 21st January Evening Shift

The total length of potentiometer wire AB is 50 cm in the arrangement as shown in the figure. If P is the point where the galvanometer shows zero reading then the length AP is _____ cm.

Open solution



(A) 25 (B) 30 (C) 15 (D) 20

Q118. JEE Main 2026 (Online) 21st January Evening Shift

A battery with EMF E and internal resistance r is connected across a resistance R. The power consumption in R will be maximum when:

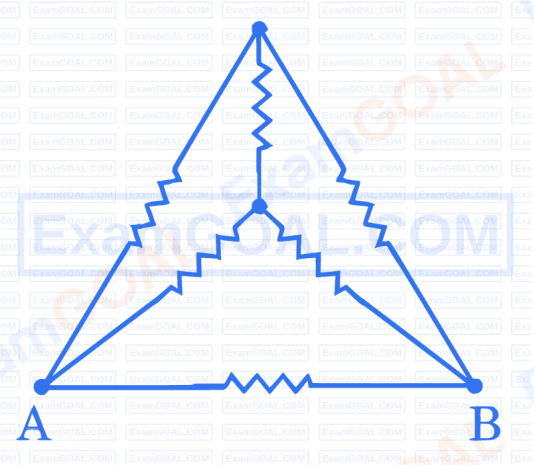
Open solution

(A) $R = \sqrt{2}\,r$ (B) $R = 2r$ (C) $R = \frac{r}{2}$ (D) $R = r$

Q119. JEE Main 2025 (Online) 7th April Morning Shift

A wire of resistance R is bent into a triangular pyramid as shown in figure with each segment having same length. The resistance between points A and B is R/n. The value of n is :

Open solution



(A) 16 (B) 10 (C) 12 (D) 14

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Q120. JEE Main 2025 (Online) 4th April Evening Shift

Open solution

There are ' n ' number of identical electric bulbs, each is designed to draw a power p independently from the mains supply. They are now joined in series across the mains supply. The total power drawn by the combination is :

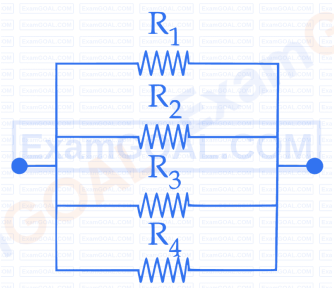
(A) np (B) p (C) $\frac{p}{n^2}$ (D) $\frac{p}{n}$

Q121. JEE Main 2025 (Online) 4th April Evening Shift

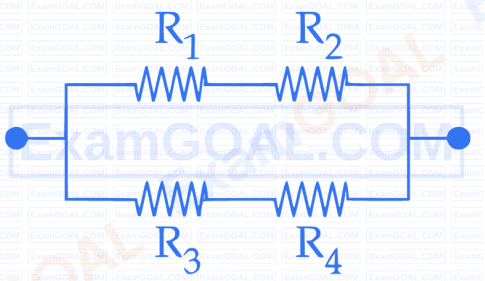
Open solution

From the combination of resistors with resistances values $R_1 = R_2 = R_3 = 5\Omega$ and $R_4 = 10\Omega$, which of the following combination is the best circuit to get an equivalent resistance of 6Ω ?

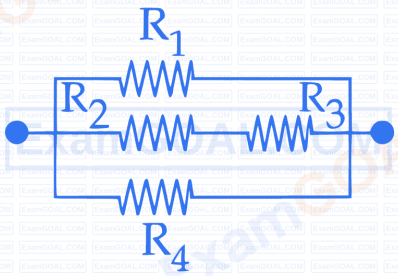
(A)



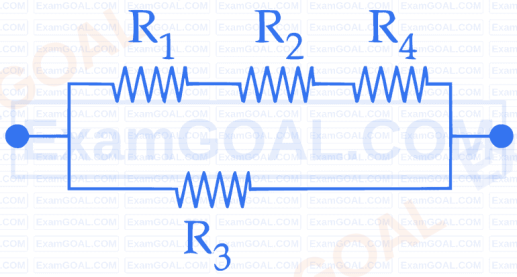
(B)



(C)



(D)



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Q122. JEE Main 2025 (Online) 4th April Morning Shift

Current passing through a wire as function of time is given as $I(t) = 0.02t + 0.01$ A. The charge that will flow through the wire from $t = 1$ s to $t = 2$ s is

(A) 0.02 C (B) 0.07 C (C) 0.06 C (D) 0.04 C

Open solution

Q123. JEE Main 2025 (Online) 3rd April Evening Shift

A motor operating on 100 V draws a current of 1 A . If the efficiency of the motor is 91.6%, then the loss of power in units of $\mathrm{cal/s}$ is

(A) 6.2 (B) 2 (C) 8.4 (D) 4

Open solution

Q124. JEE Main 2025 (Online) 3rd April Morning Shift

A wire of length 25 m and cross-sectional area $5\ \mathrm{mm}^2$ having resistivity of $2 \times 10^{-6} \Omega\ \mathrm{m}$ is bent into a complete circle. The resistance between diametrically opposite points will be

(A) 100Ω (B) 2.5Ω (C) 12.5Ω (D) 50Ω

Open solution

Q125. JEE Main 2025 (Online) 2nd April Morning Shift

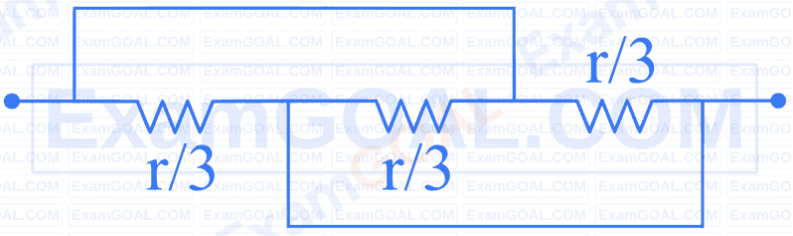
The battery of a mobile phone is rated as $4.2\ \mathrm{V}, 5800\ \mathrm{mAh}$. How much energy is stored in it when fully charged?

(A) 87.7 kJ (B) 43.8 kJ (C) 48.7 kJ (D) 24.4 kJ

Open solution

Q126. JEE Main 2025 (Online) 28th January Morning Shift

Find the equivalent resistance between two ends of the following circuit



(A) r (B) $\frac{r}{6}$ (C) $\frac{r}{3}$ (D) $\frac{r}{9}$

Open solution

Q127. JEE Main 2025 (Online) 28th January Morning Shift

A wire of resistance R is bent into an equilateral triangle and an identical wire is bent into a square. The ratio of resistance between the two end points of an edge of the triangle to that of the square is

(A) $8/9$ (B) $9/8$ (C) $32/27$ (D) $27/32$

Open solution

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Q128. JEE Main 2025 (Online) 23rd January Evening Shift

Open solution

A galvanometer having a coil of resistance 30Ω need 20 mA of current for full-scale deflection. If a maximum current of 3 A is to be measured using this galvanometer, the resistance of the shunt to be added to the galvanometer should be $\frac{30}{X}\Omega$, where X is

(A) 447 (B) 298 (C) 149 (D) 596

Q129. JEE Main 2025 (Online) 23rd January Morning Shift

Open solution

Consider a moving coil galvanometer (MCG):

A. The torsional constant in moving coil galvanometer has dimensions $\left[\mathrm{ML}^2\,\mathrm{T}^{-2}\right]$

B. Increasing the current sensitivity may not necessarily increase the voltage sensitivity.

C. If we increase number of turns (N) to its double $(2\,\mathrm{N})$, then the voltage sensitivity doubles.

D. MCG can be converted into an ammeter by introducing a shunt resistance of large value in parallel with galvanometer.

E. Current sensitivity of MCG depends inversely on number of turns of coil.

Choose the correct answer from the options given below:

(A) A, D Only (B) A, B Only (C) A, B, E Only (D) B, D, E Only

Q130. JEE Main 2025 (Online) 22nd January Morning Shift

Open solution

Given below are two statements :

Statement-I : The equivalent emf of two nonideal batteries connected in parallel is smaller than either of the two emfs.

Statement-II : The equivalent internal resistance of two nonideal batteries connected in parallel is smaller than the internal resistance of either of the two batteries.

In the light of the above statements, choose the correct answer from the options given below.

(A) Statement-I is false but Statement-II is true

(B) Statement-I is true but Statement-II is false

(C) Both Statement-I and Statement-II are false

(D) Both Statement-I and Statement-II are true

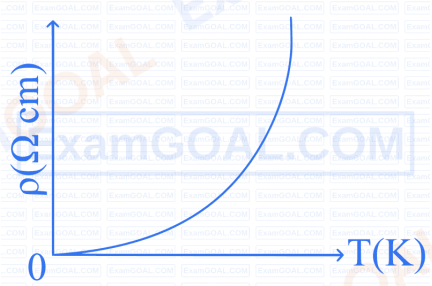
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Q131. JEE Main 2025 (Online) 22nd January Morning Shift

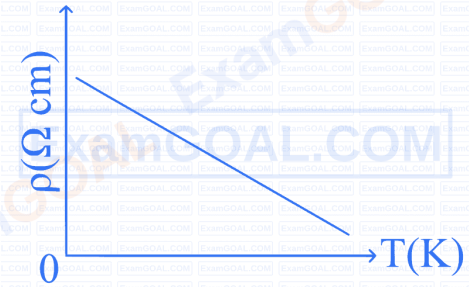
Which of the following resistivity (ρ) v/s temperature (T) curves is most suitable to be used in wire bound standard resistors?

Open solution

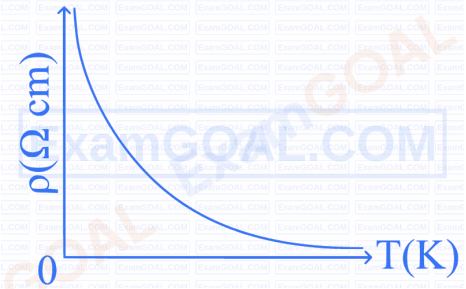
(A)



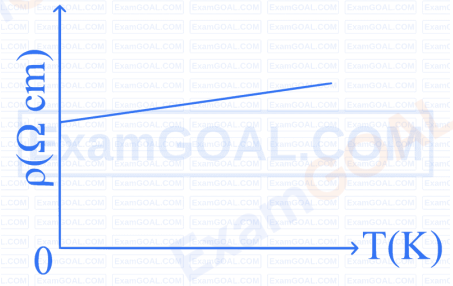
(B)



(C)

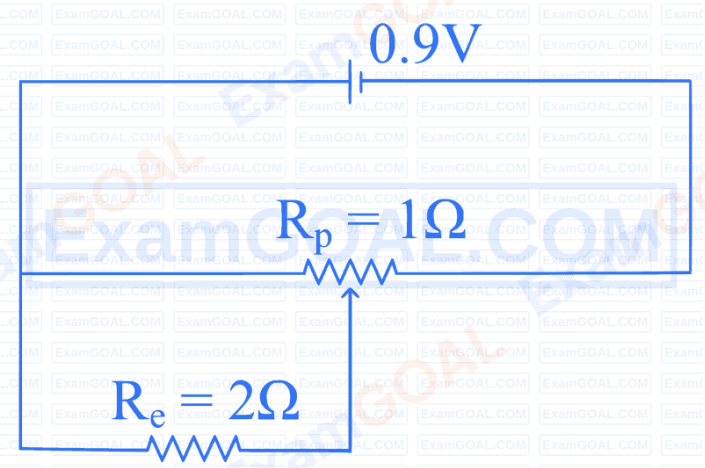


(D)



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Q132. JEE Main 2025 (Online) 22nd January Morning Shift



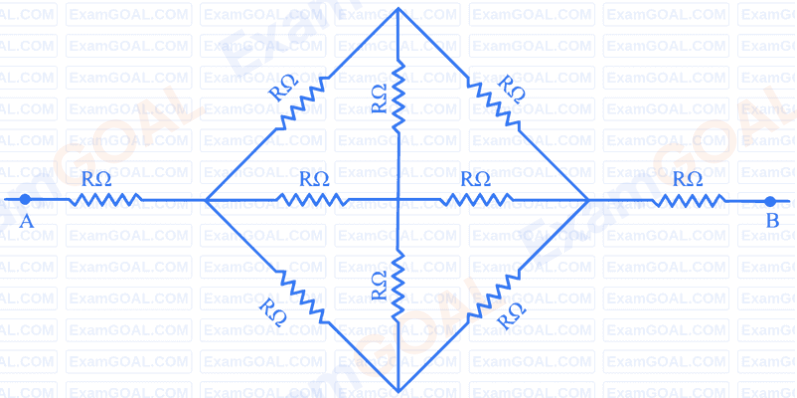
Open solution

Sliding contact of a potentiometer is in the middle of the potentiometer wire having resistance $R_p = 1\Omega$ as shown in the figure. An external resistance of $R_e = 2\Omega$ is connected via the sliding contact. The electric current in the circuit is :

(A) 1.35 A (B) 0.9 A (C) 1.0 A (D) 0.3 A

Q133. JEE Main 2024 (Online) 9th April Evening Shift

The effective resistance between A and B, if resistance of each resistor is R, will be :



Open solution

(A) $\frac{8R}{3}$ (B) $\frac{4R}{3}$ (C) $\frac{2}{3}R$ (D) $\frac{5R}{3}$

Q134. JEE Main 2024 (Online) 9th April Morning Shift

A galvanometer has a coil of resistance 200Ω with a full scale deflection at $20\mu A$. The value of resistance to be added to use it as an ammeter of range $(0 - 20)\text{mA}$ is :

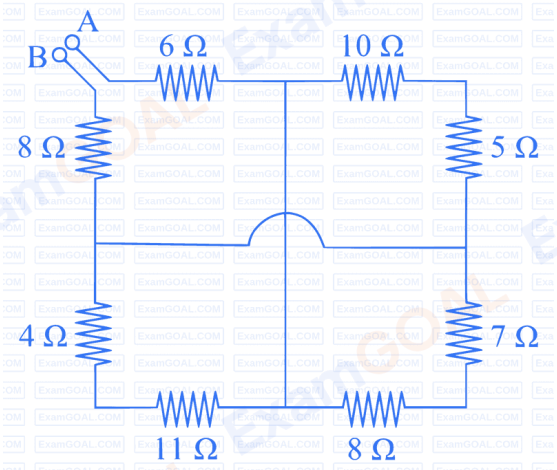
(A) 0.40Ω (B) 0.10Ω (C) 0.20Ω (D) 0.50Ω

Open solution

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Q135. JEE Main 2024 (Online) 9th April Morning Shift

The equivalent resistance between A and B is :



Open solution

(A) 18 Ω (B) 27 Ω (C) 19 Ω (D) 25 Ω

Q136. JEE Main 2024 (Online) 8th April Evening Shift

Water boils in an electric kettle in 20 minutes after being switched on. Using the same main supply, the length of the heating element should be _____ to _____ times of its initial length if the water is to be boiled in 15 minutes.

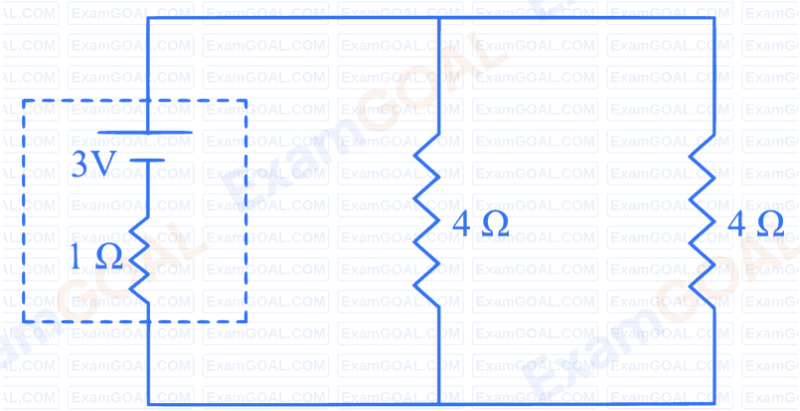
Open solution

(A) increased, $\frac{3}{4}$ (B) increased, $\frac{4}{3}$

(C) decreased, $\frac{3}{4}$ (D) decreased, $\frac{4}{3}$

Q137. JEE Main 2024 (Online) 8th April Morning Shift

In the given circuit, the terminal potential difference of the cell is :



Open solution

(A) 3 V (B) 4 V (C) 2 V (D) 1.5 V

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Q138. JEE Main 2024 (Online) 6th April Evening Shift

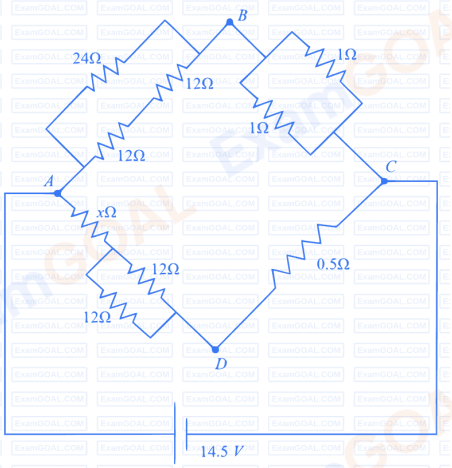
The number of electrons flowing per second in the filament of a $110\ \mathrm{W}$ bulb operating at $220\ \mathrm{V}$ is : (Given $\mathrm{e} = 1.6 \times 10^{-19}\mathrm{C}$)

(A) 1.25×10^{19} (B) 31.25×10^{17} (C) 6.25×10^{18} (D) 6.25×10^{17}

Open solution

Q139. JEE Main 2024 (Online) 6th April Morning Shift

The value of unknown resistance (x) for which the potential difference between B and D will be zero in the arrangement shown, is :

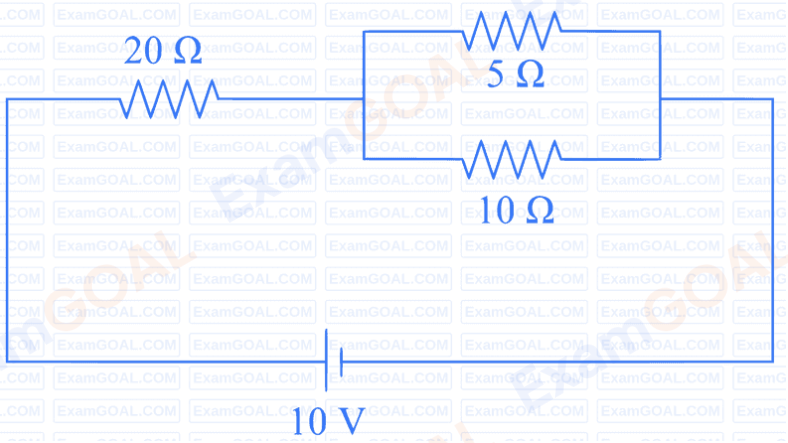


(A) $3\ \Omega$ (B) $42\ \Omega$ (C) $6\ \Omega$ (D) $9\ \Omega$

Open solution

Q140. JEE Main 2024 (Online) 5th April Evening Shift

The ratio of heat dissipated per second through the resistance 5Ω and 10Ω in the circuit given below is:



(A) 4 : 1 (B) 2 : 1 (C) 1 : 2 (D) 1 : 1

Open solution

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Q141. JEE Main 2024 (Online) 5th April Evening Shift

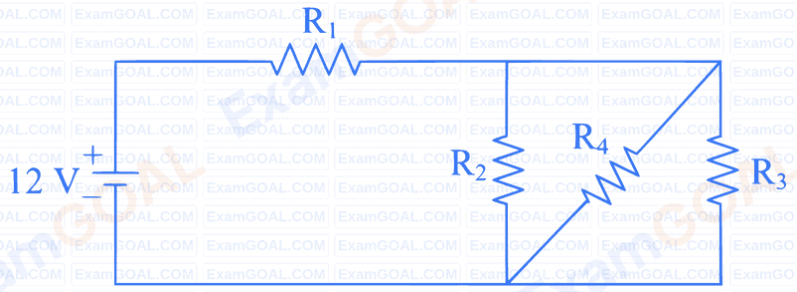
A galvanometer of resistance 100Ω when connected in series with 400Ω measures a voltage of upto $10\ \mathrm{V}$. The value of resistance required to convert the galvanometer into ammeter to read upto $10\ \mathrm{A}$ is $x \times 10^{-2}\Omega$. The value of x is :

Open solution

(A) 2 (B) 20 (C) 800 (D) 200

Q142. JEE Main 2024 (Online) 5th April Morning Shift

In the given figure $\mathrm{R}_1 = 10\Omega, \mathrm{R}_2 = 8\Omega, \mathrm{R}_3 = 4\Omega$ and $\mathrm{R}_4 = 8\Omega$. Battery is ideal with emf $12\ \mathrm{V}$. Equivalent resistance of the circuit and current supplied by battery are respectively :



Open solution

(A) 12Ω and $11.4\ \mathrm{A}$ (B) 10.5Ω and $1.14\ \mathrm{A}$
(C) 10.5Ω and $1\ \mathrm{A}$ (D) 12Ω and $1\ \mathrm{A}$

Q143. JEE Main 2024 (Online) 4th April Evening Shift

An electric bulb rated $50\ \mathrm{W} - 200\ \mathrm{V}$ is connected across a $100\ \mathrm{V}$ supply. The power dissipation of the bulb is:

Open solution

(A) 100 W (B) 50 W (C) 12.5 W (D) 25 W

Q144. JEE Main 2024 (Online) 4th April Morning Shift

To measure the internal resistance of a battery, potentiometer is used. For $R = 10\Omega$, the balance point is observed at $l = 500\ \mathrm{cm}$ and for $\mathrm{R} = 1\Omega$ the balance point is observed at $l = 400\ \mathrm{cm}$. The internal resistance of the battery is approximately :

Open solution

(A) 0.1Ω (B) 0.3Ω (C) 0.2Ω (D) 0.4Ω

Q145. JEE Main 2024 (Online) 1st February Evening Shift

In a metre-bridge when a resistance in the left gap is 2Ω and unknown resistance in the right gap, the balance length is found to be $40\ \mathrm{cm}$. On shunting the unknown resistance with 2Ω, the balance length changes by :

Open solution

(A) $62.5\ \mathrm{cm}$ (B) $22.5\ \mathrm{cm}$ (C) $20\ \mathrm{cm}$ (D) $65\ \mathrm{cm}$

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Q146. JEE Main 2024 (Online) 1st February Evening Shift

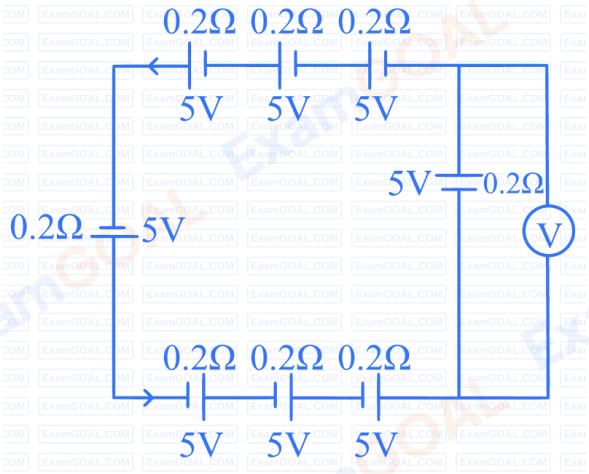
In an ammeter, 5% of the main current passes through the galvanometer. If resistance of the galvanometer is G, the resistance of ammeter will be :

(A) $199\ \mathrm{G}$ (B) $200\ \mathrm{G}$ (C) $\frac{G}{20}$ (D) $\frac{\mathrm{G}}{199}$

Open solution

Q147. JEE Main 2024 (Online) 1st February Morning Shift

The reading in the ideal voltmeter (V) shown in the given circuit diagram is :



(A) $3\ \mathrm{V}$ (B) $10\ \mathrm{V}$ (C) $5\ \mathrm{V}$ (D) $0\ \mathrm{V}$

Open solution

Q148. JEE Main 2024 (Online) 1st February Morning Shift

A galvanometer has a resistance of $50\ \Omega$ and it allows maximum current of $5\ \mathrm{mA}$. It can be converted into voltmeter to measure up to $100\ \mathrm{V}$ by connecting in series a resistor of resistance :

(A) 19500Ω (B) 5975Ω (C) 20050Ω (D) 19950Ω

Open solution

Q149. JEE Main 2024 (Online) 31st January Evening Shift

By what percentage will the illumination of the lamp decrease if the current drops by 20%?

(A) 26% (B) 36% (C) 46% (D) 56%

Open solution

Q150. JEE Main 2024 (Online) 31st January Evening Shift

The resistance per centimeter of a meter bridge wire is r, with $X\Omega$ resistance in left gap. Balancing length from left end is at $40\ \mathrm{cm}$ with 25Ω resistance in right gap. Now the wire is replaced by another wire of $2r$ resistance per centimeter. The new balancing length for same settings will be at

(A) 10 cm (B) 80 cm (C) 40 cm (D) 20 cm

Open solution

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Q151. JEE Main 2024 (Online) 31st January Morning Shift

Two conductors have the same resistances at $0^{\circ}\mathrm{C}$ but their temperature coefficients of resistance are α_1 and α_2. The respective temperature coefficients for their series and parallel combinations are :

Open solution

(A) $\alpha_1 + \alpha_2, \frac{\alpha_1\alpha_2}{\alpha_1+\alpha_2}$

(B) $\frac{\alpha_1+\alpha_2}{2}, \frac{\alpha_1+\alpha_2}{2}$

(C) $\alpha_1 + \alpha_2, \frac{\alpha_1+\alpha_2}{2}$

(D) $\frac{\alpha_1+\alpha_2}{2}, \alpha_1 + \alpha_2$

Q152. JEE Main 2024 (Online) 30th January Evening Shift

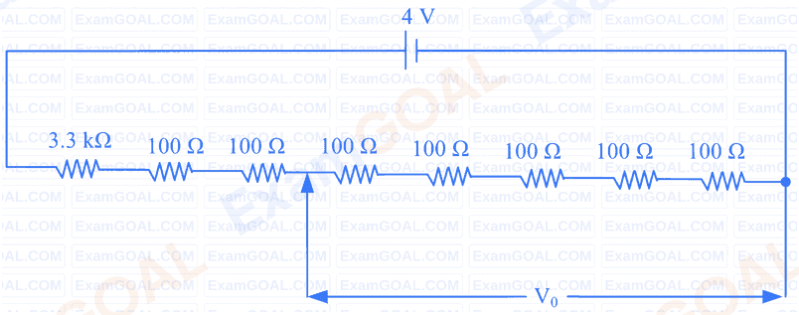
When a potential difference V is applied across a wire of resistance R, it dissipates energy at a rate W. If the wire is cut into two halves and these halves are connected mutually parallel across the same supply, the energy dissipation rate will become:

Open solution

(A) 1/2W (B) 4W (C) 1/4W (D) 2W

Q153. JEE Main 2024 (Online) 30th January Morning Shift

A potential divider circuit is shown in figure. The output voltage V_0 is :



Open solution

(A) 2 mV (B) 4 V (C) 0.5 V (D) 12 mV

Q154. JEE Main 2024 (Online) 30th January Morning Shift

An electric toaster has resistance of 60Ω at room temperature $(27^{\circ}\mathrm{C})$. The toaster is connected to a $220\ \mathrm{V}$ supply. If the current flowing through it reaches $2.75\ \mathrm{A}$, the temperature attained by toaster is around : (if $\alpha = 2 \times 10^{-4}/^{\circ}\mathrm{C}$)

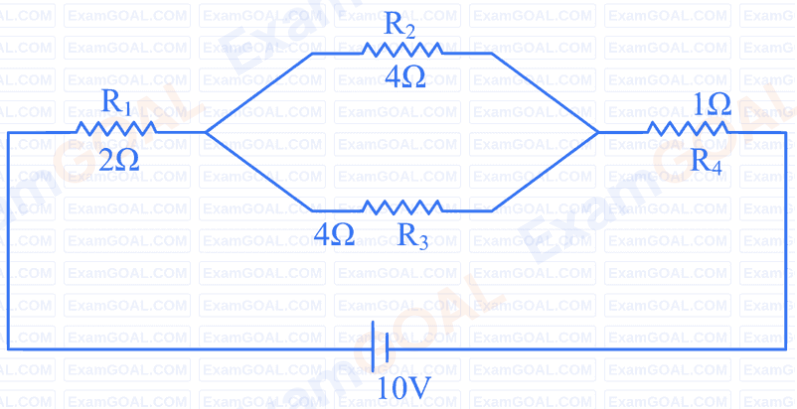
Open solution

(A) 1235 °C (B) 1667 °C (C) 694 °C (D) 1694 °C

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Q155. JEE Main 2024 (Online) 29th January Evening Shift

In the given circuit, the current in resistance R_3 is :



Open solution

(A) 2 A (B) 1.5 A (C) 1 A (D) 2.5 A

Q156. JEE Main 2024 (Online) 29th January Morning Shift

The deflection in moving coil galvanometer falls from 25 divisions to 5 divisions when a shunt of 24Ω is applied. The resistance of galvanometer coil will be :

(A) 48Ω (B) 100Ω (C) 96Ω (D) 12Ω

Open solution

Q157. JEE Main 2024 (Online) 29th January Morning Shift

A galvanometer having coil resistance 10Ω shows a full scale deflection for a current of $3\ \text{mA}$. For it to measure a current of $8\ \text{A}$, the value of the shunt should be:

(A) $3.75 \times 10^{-3}\Omega$ (B) $3 \times 10^{-3}\Omega$ (C) $4.85 \times 10^{-3}\Omega$ (D) $2.75 \times 10^{-3}\Omega$

Open solution

Q158. JEE Main 2024 (Online) 29th January Morning Shift

The electric current through a wire varies with time as $I = I_0 + \beta t$, where $I_0 = 20\ \text{A}$ and $\beta = 3\ \text{A/s}$. The amount of electric charge crossed through a section of the wire in $20\ \text{s}$ is :

(A) 80 C (B) 800 C (C) 1000 C (D) 1600 C

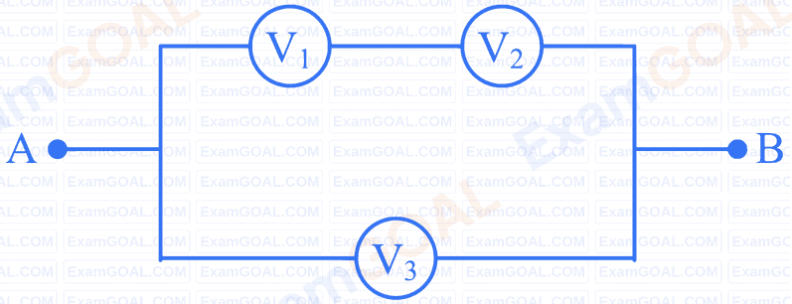
Open solution

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Q159. JEE Main 2024 (Online) 27th January Evening Shift

Three voltmeters, all having different internal resistances are joined as shown in figure. When some potential difference is applied across A and B, their readings are V_1, V_2 and V_3. Choose the correct option.

Open solution



(A) $V_1 = V_2$ (B) $V_1 \neq V_3 - V_2$ (C) $V_1 + V_2 = V_3$ (D) $V_1 + V_2 > V_3$

Q160. JEE Main 2024 (Online) 27th January Evening Shift

A current of $200\mu\text{A}$ deflects the coil of a moving coil galvanometer through 60°. The current to cause deflection through $\frac{\pi}{10}$ radian is :

Open solution

(A) 120 μA (B) 180 μA (C) 30 μA (D) 60 μA

Q161. JEE Main 2024 (Online) 27th January Evening Shift

Wheatstone bridge principle is used to measure the specific resistance (S_1) of given wire, having length L, radius r. If X is the resistance of wire, then specific resistance is ; $S_1 = X\left(\frac{\pi r^2}{L}\right)$. If the length of the wire gets doubled then the value of specific resistance will be :

Open solution

(A) $\frac{S_1}{4}$ (B) $2\,S_1$ (C) $\frac{S_1}{2}$ (D) S_1

Q162. JEE Main 2024 (Online) 27th January Morning Shift

A wire of length $10\ \text{cm}$ and radius $\sqrt{7} \times 10^{-4}\ \text{m}$ is connected across the right gap of a meter bridge. When a resistance of 4.5Ω is connected on the left gap by using a resistance box, the balance length is found to be at $60\ \text{cm}$ from the left end. If the resistivity of the wire is $\text{R} \times 10^{-7}\Omega\text{m}$, then value of R is :

Open solution

(A) 63 (B) 70 (C) 66 (D) 35

Q163. JEE Main 2024 (Online) 27th January Morning Shift

A wire of resistance R and length L is cut into 5 equal parts. If these parts are joined parallely, then resultant resistance will be :

Open solution

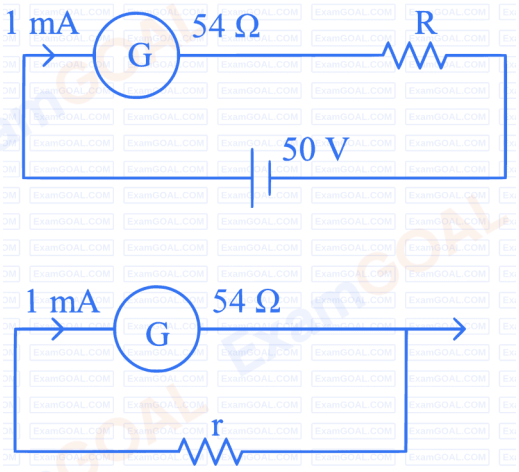
(A) $\frac{1}{25}\text{R}$ (B) $\frac{1}{5}R$ (C) 25 R (D) 5 R

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Q164. JEE Main 2023 (Online) 15th April Morning Shift

Open solution

For designing a voltmeter of range $50\ \mathrm{V}$ and an ammeter of range $10\ \mathrm{mA}$ using a galvanometer which has a coil of resistance 54Ω showing a full scale deflection for $1\ \mathrm{mA}$ as in figure.



(A) for voltmeter $R \approx 50\mathrm{k}\Omega$

(B) for ammeter $\mathrm{r} \approx 0.2\Omega$

(C) for ammeter $\mathrm{r} = 6\Omega$

(D) for voltmeter $R \approx 5\mathrm{k}\Omega$

(E) for voltmeter $R \approx 500\Omega$

Choose the correct answer from the options given below:

(A) (A) and (C)

(B) (A) and (B)

(C) (C) and (E)

(D) (C) and (D)

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Q165. JEE Main 2023 (Online) 15th April Morning Shift

Open solution

Given below are two statements:

Statement I : The equivalent resistance of resistors in a series combination is smaller than least resistance used in the combination.

Statement II : The resistivity of the material is independent of temperature.

In the light of the above statements, choose the correct answer from the options given below :

(A) Statement I is true but Statement II is false

(B) Both Statement I and Statement II are true

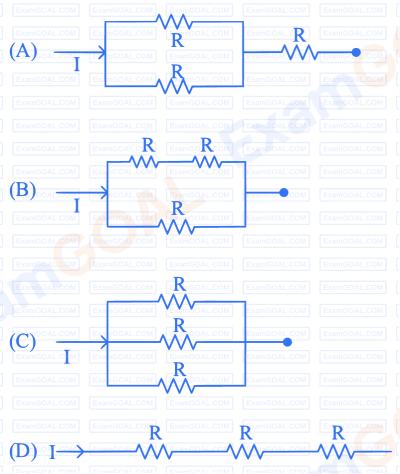
(C) Both Statement I and Statement II are false

(D) Statement I is false but Statement II is true

Q166. JEE Main 2023 (Online) 13th April Morning Shift

Open solution

Different combination of 3 resistors of equal resistance R are shown in the figures. The increasing order for power dissipation is:



(A) $\mathrm{P_B} < \mathrm{P_C} < \mathrm{P_D} < \mathrm{P_A}$

(B) $\mathrm{P_C} < \mathrm{P_B} < \mathrm{P_A} < \mathrm{P_D}$

(C) $\mathrm{P_C} < \mathrm{P_D} < \mathrm{P_A} < \mathrm{P_B}$

(D) $\mathrm{P_A} < \mathrm{P_B} < \mathrm{P_C} < \mathrm{P_D}$

Q167. JEE Main 2023 (Online) 12th April Morning Shift

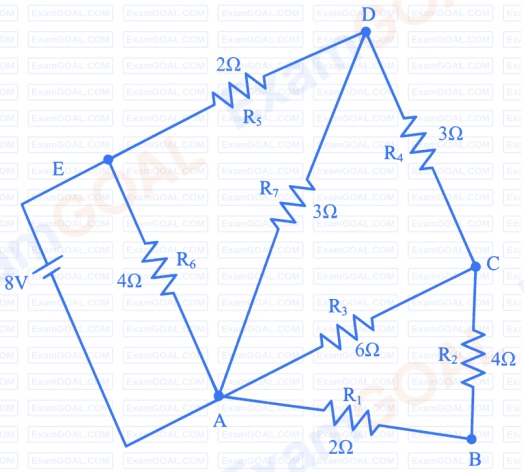
Open solution

A wire of resistance $160\ \Omega$ is melted and drawn in a wire of one-fourth of its length. The new resistance of the wire will be

(A) $640\ \Omega$ (B) $40\ \Omega$ (C) $16\ \Omega$ (D) $10\ \Omega$

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Q168. JEE Main 2023 (Online) 11th April Evening Shift



Open solution

The current flowing through R_2 is :

(A) $\frac{1}{2}$ A (B) $\frac{1}{3}$ A (C) $\frac{1}{4}$ A (D) $\frac{2}{3}$ A

Q169. JEE Main 2023 (Online) 11th April Morning Shift

Two identical heater filaments are connected first in parallel and then in series. At the same applied voltage, the ratio of heat produced in same time for parallel to series will be:

(A) 4 : 1 (B) 1 : 4 (C) 2 : 1 (D) 1 : 2

Open solution

Q170. JEE Main 2023 (Online) 11th April Morning Shift

The current sensitivity of moving coil galvanometer is increased by 25%. This increase is achieved only by changing in the number of turns of coils and area of cross section of the wire while keeping the resistance of galvanometer coil constant. The percentage change in the voltage sensitivity will be:

(A) +25% (B) -50% (C) -25% (D) Zero

Open solution

Q171. JEE Main 2023 (Online) 10th April Evening Shift

In a metallic conductor, under the effect of applied electric field, the free electrons of the conductor

(A) move in the straight line paths in the same direction

(B) move with the uniform velocity throughout from lower potential to higher potential

(C) drift from higher potential to lower potential.

(D) move in the curved paths from lower potential to higher potential

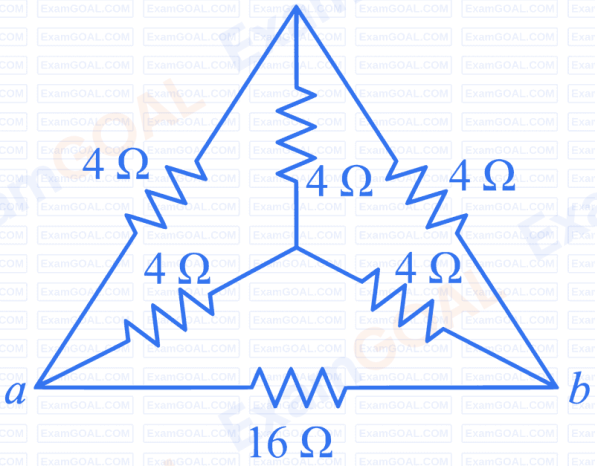
Open solution

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Q172. JEE Main 2023 (Online) 10th April Morning Shift

Open solution

The equivalent resistance of the circuit shown below between points a and b is :



(A) 16Ω (B) 20Ω (C) 3.2Ω (D) 24Ω

Q173. JEE Main 2023 (Online) 8th April Evening Shift

Open solution

The equivalent resistance between A and B as shown in figure is:

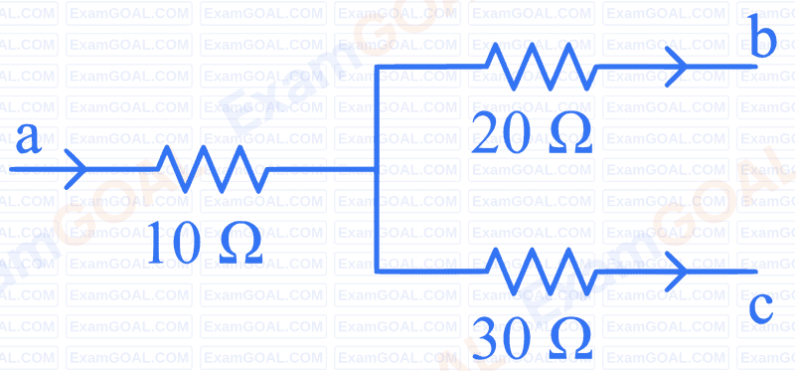


(A) $30\ \text{k}\Omega$ (B) $20\ \text{k}\Omega$ (C) $5\ \text{k}\Omega$ (D) $10\ \text{k}\Omega$

Q174. JEE Main 2023 (Online) 6th April Evening Shift

Open solution

Figure shows a part of an electric circuit. The potentials at points a, b and c are $30\ \text{V}, 12\ \text{V}$ and $2\ \text{V}$ respectively. The current through the $20\ \Omega$ resistor will be,



(A) 0.2 A (B) 0.6 A (C) 0.4 A (D) 1.0 A

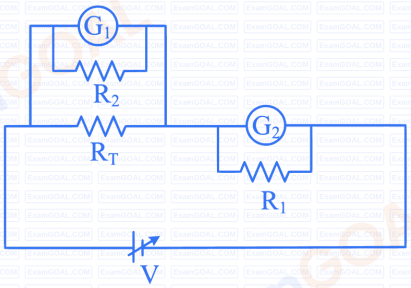
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Q175. JEE Main 2023 (Online) 6th April Evening Shift

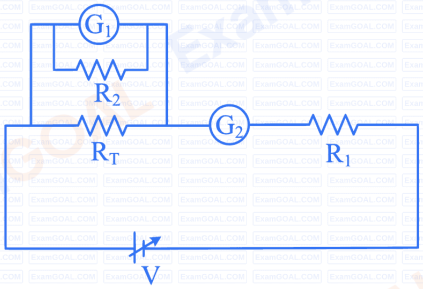
Open solution

A student is provided with a variable voltage source V, a test resistor $R_T = 10\,\Omega$, two identical galvanometers G_1 and G_2 and two additional resistors, $R_1 = 10\,M\Omega$ and $R_2 = 0.001\,\Omega$. For conducting an experiment to verify Ohm's law, the most suitable circuit is:

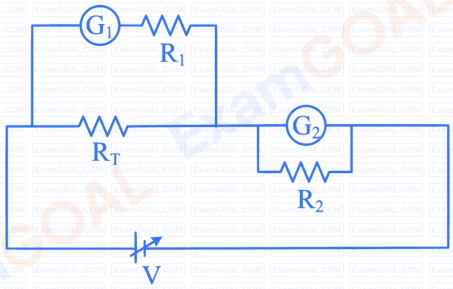
(A)



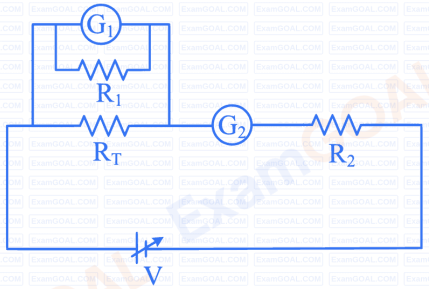
(B)



(C)



(D)



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Q176. JEE Main 2023 (Online) 1st February Evening Shift

Open solution

Given below are two statements : One is labelled as Assertion $\mathbf{A}$ and the other is labelled as Reason $\mathbf{R}$.

Assertion A : For measuring the potential difference across a resistance of 600Ω, the voltmeter with resistance 1000Ω will be preferred over voltmeter with resistance 4000Ω.

Reason R : Voltmeter with higher resistance will draw smaller current than voltmeter with lower resistance.

In the light of the above statements, choose the most appropriate answer from the options given below.

(A) $\mathbf{A}$ is not correct but $\mathbf{R}$ is correct

(B) Both $\mathbf{A}$ and $\mathbf{R}$ are correct and $\mathbf{R}$ is the correct explanation of $\mathbf{A}$

(C) $\mathbf{A}$ is correct but $\mathbf{R}$ is not correct

(D) Both $\mathbf{A}$ and $\mathbf{R}$ are correct but $\mathbf{R}$ is not the correct explanation of $\mathbf{A}$

Q177. JEE Main 2023 (Online) 1st February Evening Shift

Open solution

Equivalent resistance between the adjacent corners of a regular n-sided polygon of uniform wire of resistance R would be :

(A) $\frac{(n-1)R}{n^2}$

(B) $\frac{n^2R}{n-1}$

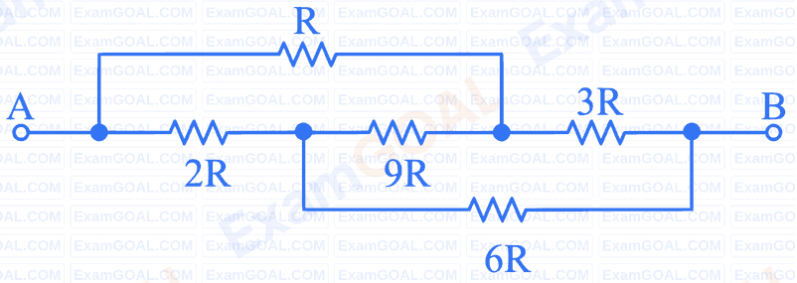
(C) $\frac{(n-1)R}{(2n-1)}$

(D) $\frac{(n-1)R}{n}$

Q178. JEE Main 2023 (Online) 1st February Morning Shift

Open solution

The equivalent resistance between A and B of the network shown in figure;



(A) 21 R

(B) $\frac{8}{3}$ R

(C) $11\frac{2}{3}$ R

(D) 14 R

Q179. JEE Main 2023 (Online) 31st January Evening Shift

Open solution

The number of turns of the coil of a moving coil galvanometer is increased in order to increase current sensitivity by 50%. The percentage change in voltage sensitivity of the galvanometer will be :

(A) 0%

(B) 75%

(C) 100%

(D) 50%

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Q180. JEE Main 2023 (Online) 31st January Evening Shift

The H amount of thermal energy is developed by a resistor in $10\ \mathrm{s}$ when a current of $4\ \mathrm{A}$ is passed through it. If the current is increased to $16\ \mathrm{A}$, the thermal energy developed by the resistor in $10\ \mathrm{s}$ will be :

Open solution

(A) $\frac{\mathrm{H}}{4}$ (B) $16\mathrm{H}$ (C) H (D) $4\mathrm{H}$

Q181. JEE Main 2023 (Online) 31st January Morning Shift

The drift velocity of electrons for a conductor connected in an electrical circuit is $\mathrm{V_d}$. The conductor is now replaced by another conductor with same material and same length but double the area of cross section. The applied voltage remains same. The new drift velocity of electrons will be

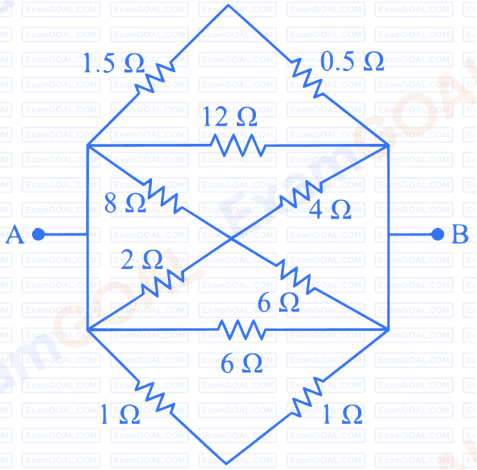
Open solution

(A) $\frac{V_d}{4}$ (B) $\mathrm{V_d}$ (C) $2\ \mathrm{V_d}$ (D) $\frac{V_d}{2}$

Q182. JEE Main 2023 (Online) 30th January Evening Shift

The equivalent resistance between A and B is _____.

Open solution



(A) $\frac{1}{2}\Omega$ (B) $\frac{2}{3}\Omega$ (C) $\frac{3}{2}\Omega$ (D) $\frac{1}{3}\Omega$

Q183. JEE Main 2023 (Online) 30th January Morning Shift

The charge flowing in a conductor changes with time as $\mathrm{Q(t)} = \alpha\mathrm{t} - \beta\mathrm{t}^2 + \gamma\mathrm{t}^3$. Where α, β and γ are constants. Minimum value of current is :

Open solution

(A) $\beta - \frac{\alpha^2}{3\gamma}$ (B) $\alpha - \frac{3\beta^2}{\gamma}$
(C) $\alpha - \frac{\beta^2}{3\gamma}$ (D) $\alpha - \frac{\gamma^2}{3\beta}$

Q184. JEE Main 2023 (Online) 29th January Morning Shift

Ratio of thermal energy released in two resistors R and 3R connected in parallel in an electric circuit is :

Open solution

(A) 1 : 1 (B) 1 : 27 (C) 1 : 3 (D) 3 : 1

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Q185. JEE Main 2023 (Online) 25th January Evening Shift

The resistance of a wire is 5 Ω. Its new resistance in ohm if stretched to 5 times of its original length will be :

(A) 25 (B) 625 (C) 5 (D) 125

Open solution

Q186. JEE Main 2023 (Online) 25th January Morning Shift

A uniform metallic wire carries a current 2 A, when 3.4 V battery is connected across it. The mass of uniform metallic wire is 8.92×10^{-3} kg, density is 8.92×10^{3} kg/m^3 and resistivity is 1.7×10^{-8} Ω-m. The length of wire is :

(A) $l = 100$ m (B) $l = 6.8$ m (C) $l = 5$ m (D) $l = 10$ m

Open solution

Q187. JEE Main 2023 (Online) 24th January Evening Shift

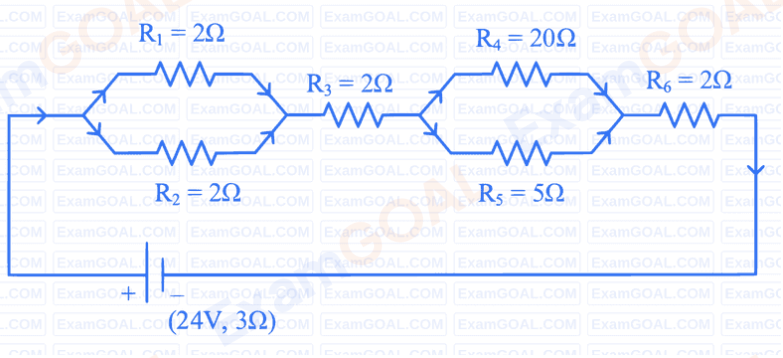
A cell of emf 90 V is connected across series combination of two resistors each of 100Ω resistance. A voltmeter of resistance 400Ω is used to measure the potential difference across each resistor. The reading of the voltmeter will be :

(A) 40 V (B) 90 V (C) 45 V (D) 80 V

Open solution

Q188. JEE Main 2023 (Online) 24th January Morning Shift

As shown in the figure, a network of resistors is connected to a battery of 24V with an internal resistance of 3 Ω. The currents through the resistors R_4 and R_5 are I_4 and I_5 respectively. The values of I_4 and I_5 are :



(A) $I_4 = \frac{8}{5}A$ and $I_5 = \frac{2}{5}A$ (B) $I_4 = \frac{6}{5}A$ and $I_5 = \frac{24}{5}A$

(C) $I_4 = \frac{2}{5}A$ and $I_5 = \frac{8}{5}A$ (D) $I_4 = \frac{24}{5}A$ and $I_5 = \frac{6}{5}A$

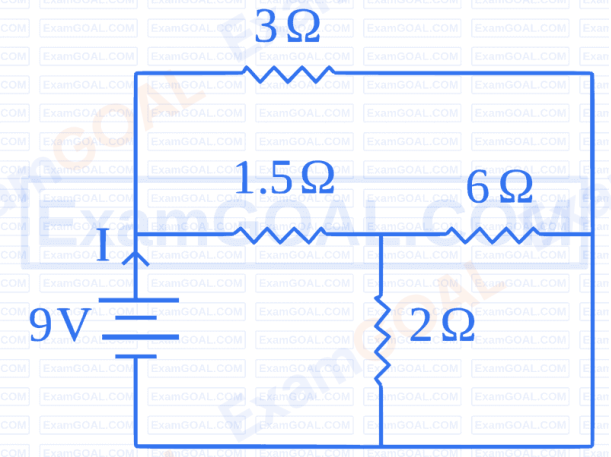
Open solution

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Q189. JEE Main 2022 (Online) 30th July Evening Shift

Open solution

The current I drawn out of the battery connected in the given circuit is:



(A) $5\ \mathrm{A}$ (B) $6\ \mathrm{A}$ (C) $7\ \mathrm{A}$ (D) $8\ \mathrm{A}$

Q190. JEE Main 2022 (Online) 29th July Evening Shift

Open solution

Given below are two statements: one is labelled as Assertion A and the other is labelled as Reason R.

Assertion A: Alloys such as constantan and manganin are used in making standard resistance coils.

Reason R: Constantan and manganin have very small value of temperature coefficient of resistance.

In the light of the above statements, choose the correct answer from the options given below.

(A) Both A and R are true and R is the correct explanation of A.

(B) Both A and R are true but R is NOT the correct explanation of A.

(C) A is true but R is false.

(D) A is false but R is true.

Q191. JEE Main 2022 (Online) 29th July Evening Shift

Open solution

A $1\ \mathrm{m}$ long wire is broken into two unequal parts X and Y. The X part of the wire is streched into another wire W. Length of W is twice the length of X and the resistance of W is twice that of Y. Find the ratio of length of X and Y.

(A) 1 : 4 (B) 1 : 2 (C) 4 : 1 (D) 2 : 1

Q192. JEE Main 2022 (Online) 29th July Morning Shift

Open solution

Two metallic wires of identical dimensions are connected in series. If σ_1 and σ_2 are the conductivities of these wires respectively, the effective conductivity of the combination is :

(A) $\frac{\sigma_1\sigma_2}{\sigma_1+\sigma_2}$ (B) $\frac{2\sigma_1\sigma_2}{\sigma_1+\sigma_2}$

(C) $\frac{\sigma_1+\sigma_2}{2\sigma_1\sigma_2}$ (D) $\frac{\sigma_1+\sigma_2}{\sigma_1\sigma_2}$

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Q193. JEE Main 2022 (Online) 28th July Evening Shift

Open solution

Given below are two statements :

Statement I : A uniform wire of resistance $80\,\Omega$ is cut into four equal parts. These parts are now connected in parallel. The equivalent resistance of the combination will be $5\,\Omega$.

Statement II: Two resistances 2R and 3R are connected in parallel in an electric circuit. The value of thermal energy developed in 3R and 2R will be in the ratio $3:2$.

In the light of the above statements, choose the most appropriate answer from the option given below

(A) Both statement I and statement II are correct

(B) Both statement I and statement II are incorrect

(C) Statement I is correct but statement II is incorrect

(D) Statement I is incorrect but statement II is correct

Q194. JEE Main 2022 (Online) 28th July Morning Shift

Open solution

A wire of resistance R_1 is drawn out so that its length is increased by twice of its original length. The ratio of new resistance to original resistance is :

(A) 9 : 1 **(B)** 1 : 9 **(C)** 4 : 1 **(D)** 3 : 1

Q195. JEE Main 2022 (Online) 27th July Evening Shift

Open solution

(A) The drift velocity of electrons decreases with the increase in the temperature of conductor.

(B) The drift velocity is inversely proportional to the area of cross-section of given conductor.

(C) The drift velocity does not depend on the applied potential difference to the conductor.

(D) The drift velocity of electron is inversely proportional to the length of the conductor.

(E) The drift velocity increases with the increase in the temperature of conductor.

Choose the correct answer from the options given below :

(A) (A) and (B) only **(B)** (A) and (D) only **(C)** (B) and (E) only **(D)** (B) and (C) only

Q196. JEE Main 2022 (Online) 27th July Morning Shift

Open solution

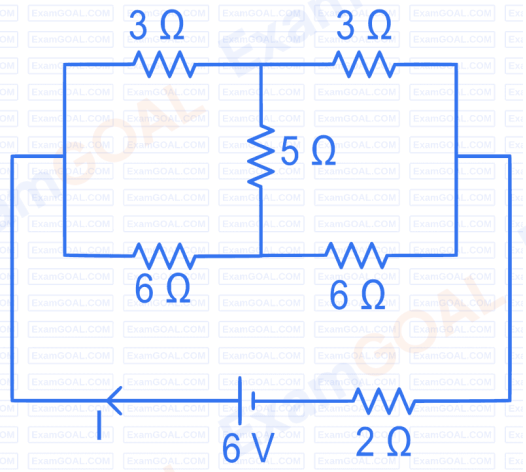
Two sources of equal emfs are connected in series. This combination is connected to an external resistance R. The internal resistances of the two sources are r_1 and r_2 $(r_1 > r_2)$. If the potential difference across the source of internal resistance r_1 is zero, then the value of R will be :

(A) $r_1 - r_2$ **(B)** $\frac{r_1 r_2}{r_1 + r_2}$ **(C)** $\frac{r_1 + r_2}{2}$ **(D)** $r_2 - r_1$

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Q197. JEE Main 2022 (Online) 26th July Evening Shift

A battery of $6\ \mathrm{V}$ is connected to the circuit as shown below. The current I drawn from the battery is :

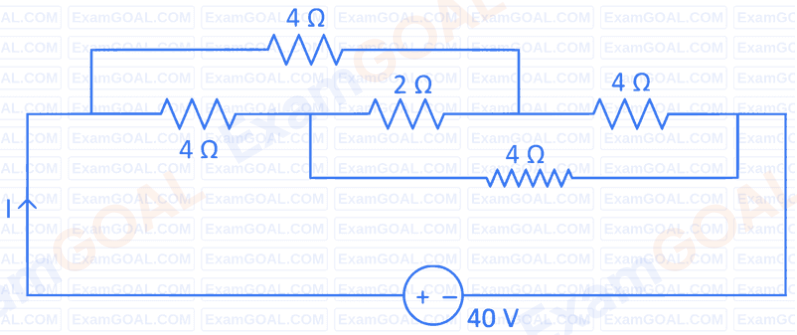


Open solution

(A) 1A (B) 2A (C) $\frac{6}{11}$ A (D) $\frac{4}{3}$ A

Q198. JEE Main 2022 (Online) 26th July Morning Shift

The current I in the given circuit will be :



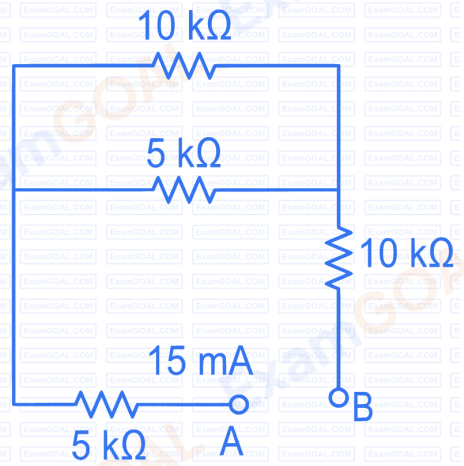
Open solution

(A) 10 A (B) 20 A (C) 4 A (D) 40 A

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Q199. JEE Main 2022 (Online) 25th July Evening Shift

A current of 15 mA flows in the circuit as shown in figure. The value of potential difference between the points A and B will be:



Open solution

(A) 50 V (B) 75 V (C) 150 V (D) 275 V

Q200. JEE Main 2022 (Online) 30th June Morning Shift

An electric cable of copper has just one wire of radius 9 mm. Its resistance is 14 Ω. If this single copper wire of the cable is replaced by seven identical well insulated copper wires each of radius 3 mm connected in parallel, then the new resistance of the combination will be :

Open solution

(A) 9 Ω (B) 18 Ω (C) 28 Ω (D) 126 Ω

Q201. JEE Main 2022 (Online) 29th June Evening Shift

The combination of two identical cells, whether connected in series or parallel combination provides the same current through an external resistance of 2Ω. The value of internal resistance of each cell is

Open solution

(A) 2Ω (B) 4Ω (C) 6Ω (D) 8Ω

Q202. JEE Main 2022 (Online) 28th June Evening Shift

Resistance of the wire is measured as 2 Ω and 3 Ω at 10°C and 30°C respectively. Temperature co-efficient of resistance of the material of the wire is :

Open solution

(A) $0.033\ ^{\circ}C^{-1}$ (B) $-0.033\ ^{\circ}C^{-1}$ (C) $0.011\ ^{\circ}C^{-1}$ (D) $0.055\ ^{\circ}C^{-1}$

Q203. JEE Main 2022 (Online) 27th June Morning Shift

A 72 Ω galvanometer is shunted by a resistance of 8 Ω. The percentage of the total current which passes through the galvanometer is :

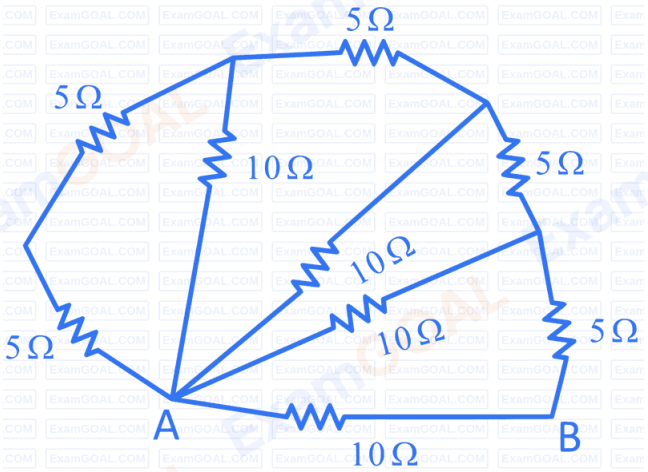
Open solution

(A) 0.1% (B) 10% (C) 25% (D) 0.25%

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Q204. JEE Main 2022 (Online) 26th June Evening Shift

The equivalent resistance between points A and B in the given network is :



Open solution

(A) 65Ω (B) 20Ω (C) 5Ω (D) 2Ω

Q205. JEE Main 2022 (Online) 26th June Morning Shift

An aluminium wire is stretched to make its length, 0.4% larger. The percentage change in resistance is :

(A) 0.4% (B) 0.2% (C) 0.8% (D) 0.6%

Open solution

Q206. JEE Main 2022 (Online) 25th June Evening Shift

Two cells of same emf but different internal resistances r_1 and r_2 are connected in series with a resistance R. The value of resistance R, for which the potential difference across second cell is zero, is :

(A) $r_2 - r_1$ (B) $r_1 - r_2$ (C) r_1 (D) r_2

Open solution

Q207. JEE Main 2022 (Online) 25th June Evening Shift

If n represents the actual number of deflections in a converted galvanometer of resistance G and shunt resistance S. Then the total current I when its figure of merit is K will be:

(A) $\frac{KS}{(S+G)}$ (B) $\frac{(G+S)}{nKS}$ (C) $\frac{nKS}{(G+S)}$ (D) $\frac{nK(G+S)}{S}$

Open solution

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Q208. JEE Main 2022 (Online) 25th June Morning Shift

A teacher in his physics laboratory allotted an experiment to determine the resistance (G) of a galvanometer. Students took the observations for $\frac{1}{3}$ deflection in the galvanometer. Which of the below is true for measuring value of G?

Open solution

(A) $\frac{1}{3}$ deflection method cannot be used for determining the resistance of the galvanometer.

(B) $\frac{1}{3}$ deflection method can be used and in this case the G equals to twice the value of shunt resistances.

(C) $\frac{1}{3}$ deflection method can be used and in this case, the G equals to three times the value of shunt resistances.

(D) $\frac{1}{3}$ deflection method can be used and in this case the G value equals to the shunt resistances.

Q209. JEE Main 2022 (Online) 24th June Evening Shift

What will be the most suitable combination of three resistors A = 2Ω, B = 4Ω, C = 6Ω so that $\left(\frac{22}{3}\right)\Omega$ is equivalent resistance of combination?

Open solution

(A) Parallel combination of A and C connected in series with B.

(B) Parallel combination of A and B connected in series with C.

(C) Series combination of A and C connected in parallel with B.

(D) Series combination of B and C connected in parallel with A.

Q210. JEE Main 2022 (Online) 24th June Morning Shift

Two identical cells each of emf 1.5 V are connected in parallel across a parallel combination of two resistors each of resistance 20 Ω. A voltmeter connected in the circuit measures 1.2 V. The internal resistance of each cell is :

Open solution

(A) 2.5 Ω (B) 4 Ω (C) 5 Ω (D) 10 Ω

Q211. JEE Main 2021 (Online) 1st September Evening Shift

Due to cold weather a 1 m water pipe of cross-sectional area 1 cm^2 is filled with ice at $-10^{\circ}C$. Resistive heating is used to melt the ice. Current of 0.5A is passed through 4 kΩ resistance. Assuming that all the heat produced is used for melting, what is the minimum time required? (Given latent heat of fusion for water/ice = 3.33×10^5 J kg^{-1}, specific heat of ice = 2×10^3 J kg^{-1} and density of ice = 10^3 kg/m^3)

Open solution

(A) 0.353 s (B) 35.3 s (C) 3.53 s (D) 70.6 s

Q212. JEE Main 2021 (Online) 1st September Evening Shift

Two resistors $R_1 = (4 \pm 0.8)\ \Omega$ and $R_2 = (4 \pm 0.4)\ \Omega$ are connected in parallel. The equivalent resistance of their parallel combination will be :

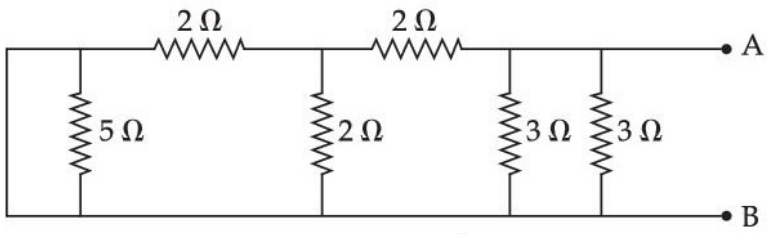
Open solution

(A) $(4 \pm 0.4)\ \Omega$ (B) $(2 \pm 0.4)\ \Omega$ (C) $(2 \pm 0.3)\ \Omega$ (D) $(4 \pm 0.3)\ \Omega$

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Q213. JEE Main 2021 (Online) 31st August Evening Shift

The equivalent resistance of the given circuit between the terminals A and B is :



Open solution

(A) 0Ω (B) 3Ω (C) $\frac{9}{2}\Omega$ (D) 1Ω

Q214. JEE Main 2021 (Online) 31st August Morning Shift

Consider a galvanometer shunted with 5Ω resistance and 2% of current passes through it. What is the resistance of the given galvanometer ?

Open solution

(A) 300 Ω (B) 344 Ω (C) 245 Ω (D) 226 Ω

Q215. JEE Main 2021 (Online) 27th August Evening Shift

For full scale deflection of total 50 divisions, 50 mV voltage is required in galvanometer. The resistance of galvanometer if its current sensitivity is 2 div/mA will be :

Open solution

(A) 1Ω (B) 5Ω (C) 4Ω (D) 2Ω

Q216. JEE Main 2021 (Online) 27th August Morning Shift

Five identical cells each of internal resistance 1Ω and emf 5V are connected in series and in parallel with an external resistance 'R'. For what value of 'R', current in series and parallel combination will remain the same?

Open solution

(A) 1 Ω (B) 25 Ω (C) 5 Ω (D) 10 Ω

Q217. JEE Main 2021 (Online) 26th August Evening Shift

If you are provided a set of resistances 2Ω, 4Ω, 6Ω and 8Ω. Connect these resistances so as to obtain an equivalent resistance of $\frac{46}{3}\Omega$.

Open solution

(A) 4Ω and 6Ω are in parallel with 2Ω and 8Ω in series

(B) 6Ω and 8Ω are in parallel with 2Ω and 4Ω in series

(C) 2Ω and 6Ω are in parallel with 4Ω and 8Ω in series

(D) 2Ω and 4Ω are in parallel with 6Ω and 8Ω in series

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Q218. JEE Main 2021 (Online) 26th August Evening Shift

An electric bulb of 500 watt at 100 volt is used in a circuit having a 200 V supply. Calculate the resistance R to be connected in series with the bulb so that the power delivered by the bulb is 500 W.

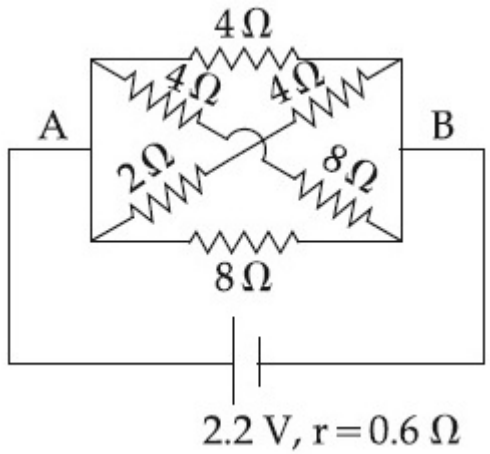
Open solution

(A) 20 Ω (B) 30 Ω (C) 5 Ω (D) 10 Ω

Q219. JEE Main 2021 (Online) 26th August Morning Shift

In the given figure, the emf of the cell is 2.2 V and if internal resistance is 0.6Ω. Calculate the power dissipated in the whole circuit :

Open solution



(A) 1.32 W (B) 0.65 W (C) 2.2 W (D) 4.4 W

Q220. JEE Main 2021 (Online) 26th August Morning Shift

What equal length of an iron wire and a copper-nickel alloy wire, each of 2 mm diameter connected parallel to give an equivalent resistance of 3Ω ?

Open solution

(Given resistivities of iron and copper-nickel alloy wire are 12 $\mu\Omega$ cm and 51 $\mu\Omega$ cm respectively)

(A) 82 m (B) 97 m (C) 110 m (D) 90 m

Q221. JEE Main 2021 (Online) 27th July Evening Shift

The resistance of a conductor at 15°C is 16Ω and at 100°C is 20Ω. What will be the temperature coefficient of resistance of the conductor?

Open solution

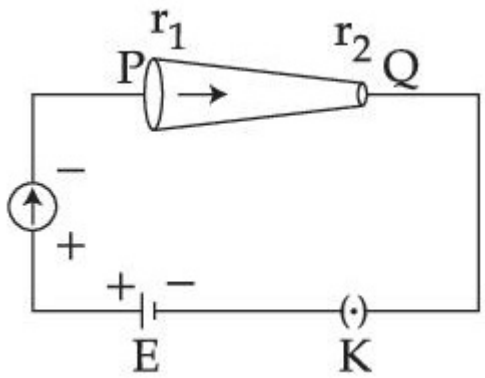
(A) $0.010^{\circ}C^{-1}$ (B) $0.033^{\circ}C^{-1}$ (C) $0.003^{\circ}C^{-1}$ (D) $0.042^{\circ}C^{-1}$

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Q222. JEE Main 2021 (Online) 27th July Morning Shift

Open solution

In the given figure, a battery of emf E is connected across a conductor PQ of length 'l' and different area of cross-sections having radii r_1 and r_2 ($r_2 < r_1$).



Choose the correct option as one moves from P to Q :

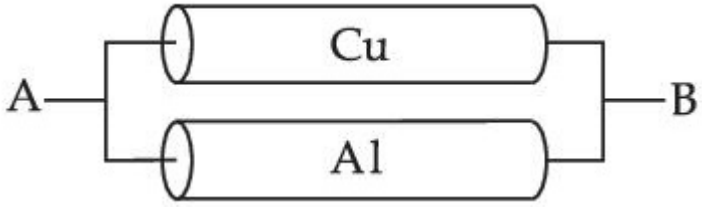
(A) Drift velocity of electron increases.
(B) Electric field decreases.
(C) Electron current decreases.
(D) All of these

Q223. JEE Main 2021 (Online) 22th July Evening Shift

Open solution

A Copper (Cu) rod of length 25 cm and cross-sectional area 3 mm^2 is joined with a similar Aluminium (Al) rod as shown in figure. Find the resistance of the combination between the ends A and B.

(Take Resistivity of Copper = 1.7×10^{-8} Ωm and Resistivity of Aluminium = 2.6×10^{-8} Ωm)



(A) 0.0858 mΩ
(B) 1.420 mΩ
(C) 0.858 mΩ
(D) 2.170 mΩ

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Q224. JEE Main 2021 (Online) 20th July Morning Shift

Open solution

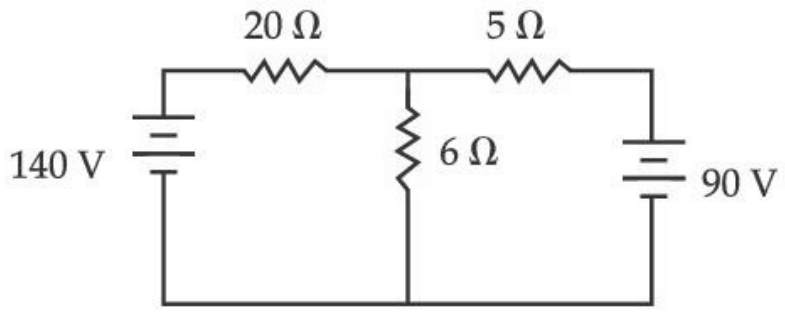
A current of 5 A is passing through a non-linear magnesium wire of cross-section 0.04 m^2. At every point the direction of current density is at an angle of 60° with the unit vector of area of cross-section. The magnitude of electric field at every point of the conductor is :

(Resistivity of magnesium $\rho = 44 \times 10^{-8}\ \Omega m$)

(A) 11×10^{-5} V/m (B) 11×10^{-3} V/m (C) 11×10^{-7} V/m (D) 11×10^{-2} V/m

Q225. JEE Main 2021 (Online) 20th July Morning Shift

Open solution



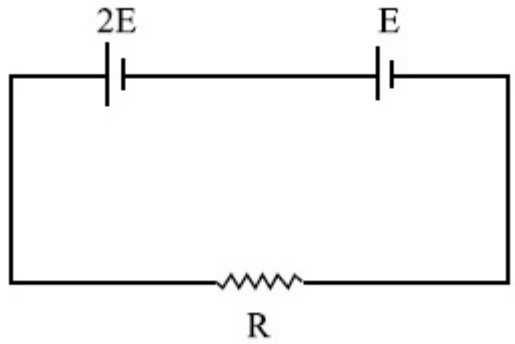
The value of current in the 6 Ω resistance is :

(A) 4A (B) 8A (C) 10A (D) 6A

Q226. JEE Main 2021 (Online) 17th March Evening Shift

Open solution

Two cells of emf 2E and E with internal resistance r_1 and r_2 respectively are connected in series to an external resistor R (see figure). The value of R, at which the potential difference across the terminals of the first cell becomes zero is



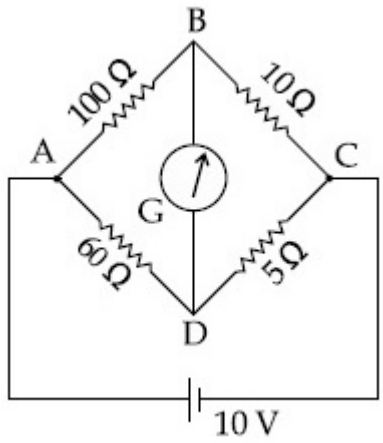
(A) $r_1 - r_2$ (B) $\frac{r_1}{2} - r_2$ (C) $\frac{r_1}{2} + r_2$ (D) $r_1 + r_2$

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Q227. JEE Main 2021 (Online) 17th March Evening Shift

The four arms of a Wheatstone bridge have resistances as shown in the figure. A galvanometer of 15Ω resistance is connected across BD. Calculate the current through the galvanometer when a potential difference of 10V is maintained across AC.

Open solution



(A) 4.87 μA (B) 4.87 mA (C) 2.44 mA (D) 2.44 μA

Q228. JEE Main 2021 (Online) 17th March Morning Shift

A current of 10A exists in a wire of cross-sectional area of 5 mm^2 with a drift velocity of 2×10^{-3} ms^{-1}. The number of free electrons in each cubic meter of the wire is _____.

Open solution

(A) 625×10^{25} (B) 1×10^{23} (C) 2×10^{25} (D) 2×10^{6}

Q229. JEE Main 2021 (Online) 16th March Evening Shift

A resistor develops 500 J of thermal energy in 20 s when a current of 1.5A is passed through it. If the current is increased from 1.5A to 3A, what will be the energy developed in 20 s.

Open solution

(A) 1000 J (B) 2000 J (C) 1500 J (D) 500 J

Q230. JEE Main 2021 (Online) 16th March Morning Shift

A conducting wire of length 'l', area of cross-section A and electric resistivity ρ is connected between the terminals of a battery. A potential difference V is developed between its ends, causing an electric current.

Open solution

If the length of the wire of the same material is doubled and the area of cross-section is halved, the resultant current would be :

(A) $4\frac{VA}{\rho l}$ (B) $\frac{3}{4}\frac{VA}{\rho l}$ (C) $\frac{1}{4}\frac{VA}{\rho l}$ (D) $\frac{1}{4}\frac{\rho l}{VA}$

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Q231. JEE Main 2021 (Online) 26th February Evening Shift

A wire of 1Ω has a length of 1 m. It is stretched till its length increases by 25%. The percentage change in resistance to the nearest integer is :

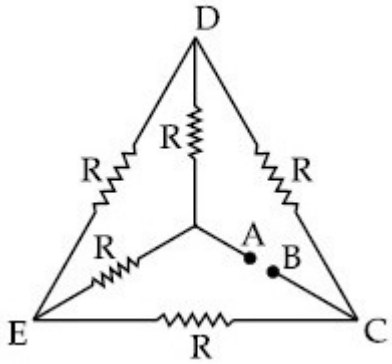
Open solution

(A) 76% (B) 12.5% (C) 25% (D) 56%

Q232. JEE Main 2021 (Online) 26th February Morning Shift

Five equal resistances are connected in a network as shown in figure. The net resistance between the points A and B is :

Open solution



(A) $\frac{3R}{2}$ (B) $\frac{R}{2}$ (C) 2R (D) R

Q233. JEE Main 2021 (Online) 24th February Morning Shift

A current through a wire depends on time as

$i = \alpha_0 t + \beta t^2$

Open solution

where α_0 = 20 A/s and β = 8 As^{-2}. Find the charge crossed through a section of the wire in 15 s.

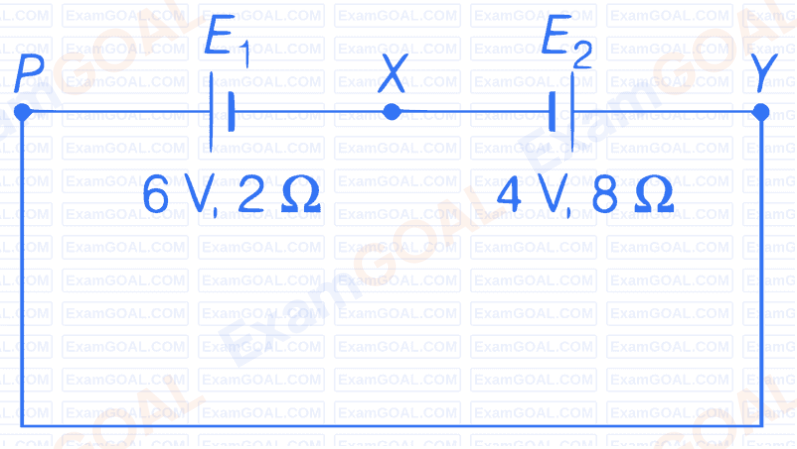
(A) 2250 C (B) 2100 C (C) 260 C (D) 11250 C

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Q234. JEE Main 2021 (Online) 24th February Morning Shift

A cell E_1 of emf 6V and internal resistance 2Ω is connected with another cell E_2 of emf 4V and internal resistance 8Ω (as shown in the figure). The potential difference across points X and Y is :

Open solution



(A) 10.0 V (B) 2.0 V (C) 5.6 V (D) 3.6 V

Q235. JEE Main 2020 (Online) 6th September Evening Slot

A circuit to verify Ohm's law uses ammeter and voltmeter in series or parallel connected correctly to the resistor. In the circuit :

Open solution

(A) Ammeter is always connected in series and voltmeter in parallel

(B) Both ammeter and voltmeter must be connected in series

(C) Both ammeter and voltmeter must be connected in parallel

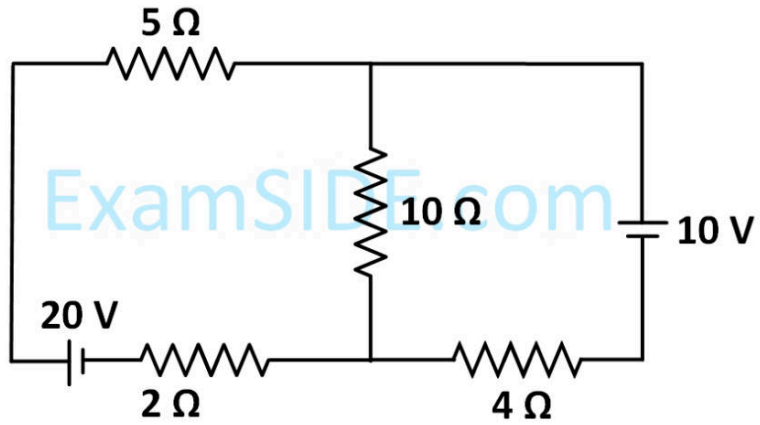
(D) Ammeter is always used in parallel and voltmeter is series

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Q236. JEE Main 2020 (Online) 6th September Evening Slot

Open solution

In the figure shown, the current in the 10 V battery is close to :



(A) 0.36 A from negative to positive terminal

(B) 0.42 A from positive to negative terminal

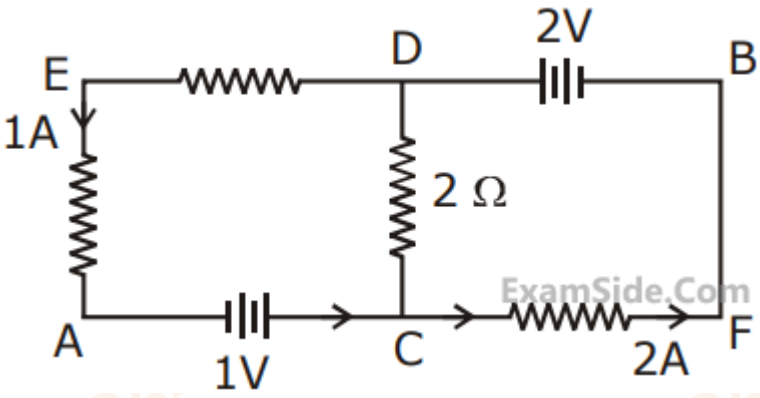
(C) 0.71 A from positive to negative terminal

(D) 0.21 A from positive to negative terminal

Q237. JEE Main 2020 (Online) 5th September Evening Slot

Open solution

In the circuit, given in the figure currents in different branches and value of one resistor are shown. Then potential at point B with respect to the point A is :



(A) +2 V (B) -2 V (C) +1 V (D) -1 V

Q238. JEE Main 2020 (Online) 5th September Evening Slot

Open solution

A galvanometer is used in laboratory for detecting the null point in electrical experiments. If, on passing a current of 6 mA it produces a deflection of 2°, its figure of merit is close to :

(A) 6×10^{-3} A/div (B) 666° A/div

(C) 3×10^{-3} A/div (D) 333° A/div

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Q239. JEE Main 2020 (Online) 5th September Morning Slot

An electrical power line, having a total resistance of 2 Ω, delivers 1 kW at 220 V. The efficiency of the transmission line is approximately :

(A) 85% (B) 96% (C) 72% (D) 91%

Open solution

Q240. JEE Main 2020 (Online) 5th September Morning Slot

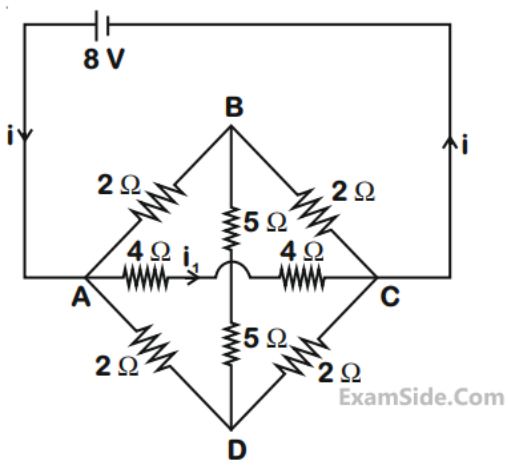
A galvanometer of resistance G is converted into a voltmeter of range 0 – 1 V by connecting a resistance R_1 in series with it. The additional resistance that should be connected in series with R_1 to increase the range of the voltmeter to 0 – 2 V will be :

(A) G (B) R_1 (C) $R_1 + G$ (D) $R_1 - G$

Open solution

Q241. JEE Main 2020 (Online) 4th September Evening Slot

The value of current i_1 flowing from A to C in the circuit diagram is :



(A) 1 A (B) 4 A (C) 5 A (D) 2 A

Open solution

Q242. JEE Main 2020 (Online) 4th September Morning Slot

A battery of 3.0 V is connected to a resistor dissipating 0.5 W of power. If the terminal voltage of the battery is 2.5 V, the power dissipated within the internal resistance is :

(A) 0.50 W (B) 0.072 W (C) 0.10 W (D) 0.125 W

Open solution

Q243. JEE Main 2020 (Online) 3rd September Evening Slot

Two resistors 400Ω and 800Ω are connected in series across a 6 V battery. The potential difference measured by a voltmeter of 10 kΩ across 400 Ω resistor is close to :

(A) 2.05 V (B) 1.95 V (C) 2 V (D) 1.8 V

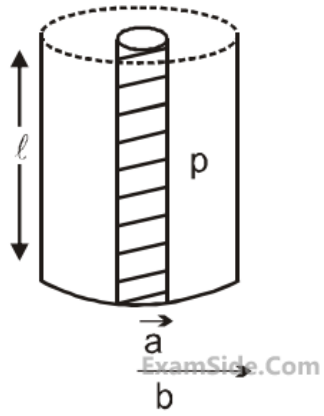
Open solution

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Q244. JEE Main 2020 (Online) 3rd September Morning Slot

Open solution

Model a torch battery of length l to be made up of a thin cylindrical bar of radius 'a' and a concentric thin cylindrical shell of radius 'b' filled in between with an electrolyte of resistivity ρ (see figure). If the battery is connected to a resistance of value R, the maximum Joule heating in R will take place for :



(A) $R = \frac{2\rho}{\pi l}\ln\left(\frac{b}{a}\right)$

(B) $R = \frac{\rho}{\pi l}\ln\left(\frac{b}{a}\right)$

(C) $R = \frac{\rho}{2\pi l}\ln\left(\frac{b}{a}\right)$

(D) $R = \frac{\rho}{2\pi l}\left(\frac{b}{a}\right)$

Q245. JEE Main 2020 (Online) 2nd September Morning Slot

Open solution

Consider four conducting materials copper, tungsten, mercury and aluminium with resistivity ρ_C, ρ_T, ρ_M and ρ_A respectively. Then :

(A) $\rho_C > \rho_A > \rho_T$

(B) $\rho_M > \rho_A > \rho_C$

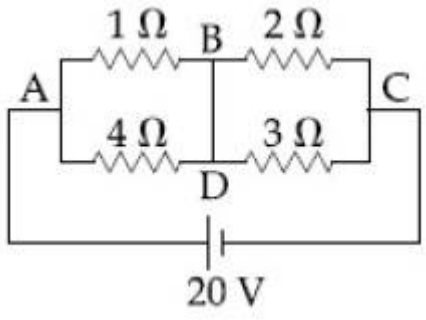
(C) $\rho_A > \rho_T > \rho_C$

(D) $\rho_A > \rho_M > \rho_C$

Q246. JEE Main 2020 (Online) 9th January Morning Slot

Open solution

In the given circuit diagram, a wire is joining points B and D. The current in this wire is :



(A) 4 A

(B) 2 A

(C) zero

(D) 0.4 A

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Q247. JEE Main 2020 (Online) 8th January Evening Slot

A galvanometer having a coil resistance 100 Ω gives a full scale deflection when a current of 1 mA is passed through it. What is the value of the resistance which can convert this galvanometer into a voltmeter giving full scale deflection for a potential difference of 10 V?

Open solution

(A) 8.9 kΩ (B) 10 kΩ (C) 9.9 kΩ (D) 7.9 kΩ

Q248. JEE Main 2020 (Online) 7th January Evening Slot

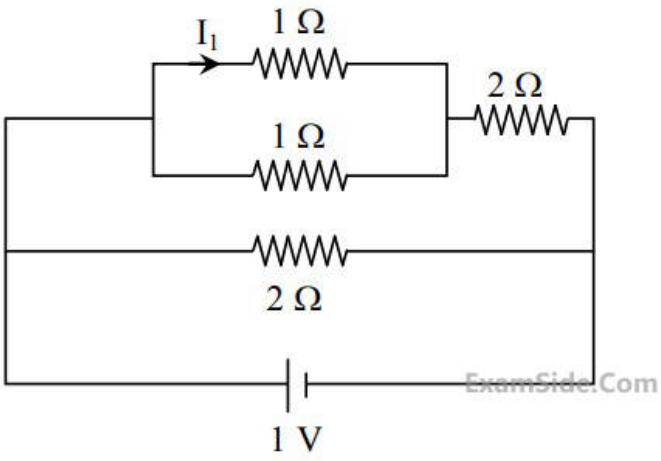
In a building there are 15 bulbs of 45 W, 15 bulbs of 100 W, 15 small fans of 10 W and 2 heaters of 1 kW. The voltage of electric main is 220 V. The minimum fuse capacity (rated value) of the building will be :

Open solution

(A) 15 A (B) 20 A (C) 25 A (D) 10 A

Q249. JEE Main 2020 (Online) 7th January Morning Slot

The current I_1 (in A) flowing through 1 Ω resistor in the following circuit is :



Open solution

(A) 0.4 (B) 0.25 (C) 0.2 (D) 0.5

Q250. JEE Main 2019 (Online) 12th April Evening Slot

A moving coil galvanometer, having a resistance G, produces full scale deflection when a current Ig flows through it. This galvanometer can be converted into (i) an ammeter of range 0 to $I_0 (I_0 > I_g)$ by connecting a shunt resistance R_A to it and (ii) into a voltmeter of range 0 to V ($V = GI_0$) by connecting a series resistance R_V to it. Then,

Open solution

(A) $R_A R_V = G^2$ and $\frac{R_A}{R_V} = \frac{I_g}{(I_0 - I_g)}$

(B) $R_A R_V = G^2 \left(\frac{I_g}{I_0 - I_g}\right)$ and $\frac{R_A}{R_V} = \left(\frac{I_0 - I_g}{I_g}\right)^2$

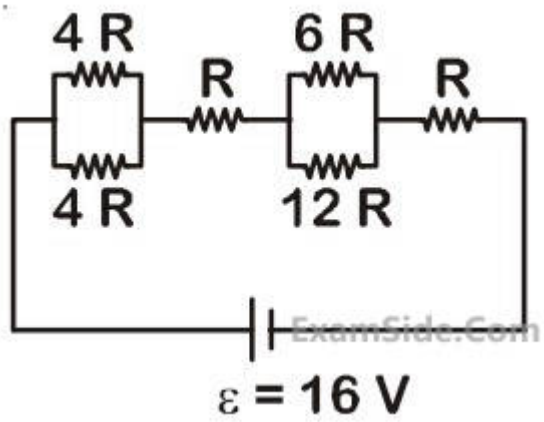
(C) $R_A R_V = G^2 \left(\frac{I_0 - I_g}{I_g}\right)$ and $\frac{R_A}{R_V} = \left(\frac{I_g}{I_0 - I_g}\right)^2$

(D) $R_A R_V = G^2$ and $\frac{R_A}{R_V} = \left(\frac{I_g}{I_0 - I_g}\right)^2$

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Q251. JEE Main 2019 (Online) 12th April Morning Slot

The resistive network shown below is connected to a D.C. source of 16 V. The power consumed by the network is 4 Watt. The value of R is:



Open solution

(A) 16 Ω (B) 1 Ω (C) 8 Ω (D) 6 Ω

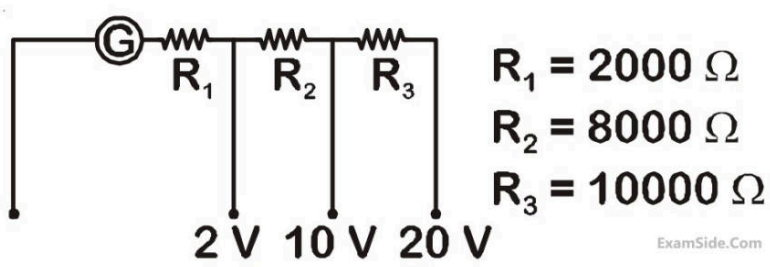
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Q252. JEE Main 2019 (Online) 12th April Morning Slot

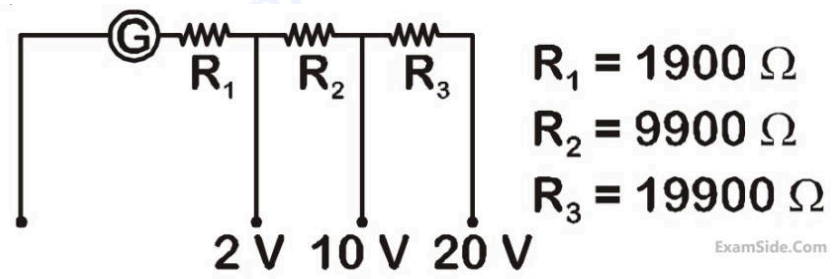
Open solution

A galvanometer of resistance 100 Ω has 50 divisions on its scale and has sensitivitv of 20 μA/division. It is to be converted to a voltmeter with three ranges of 0-2V, 0-10 V and 0-20 V. The appropriate circuit to do so is

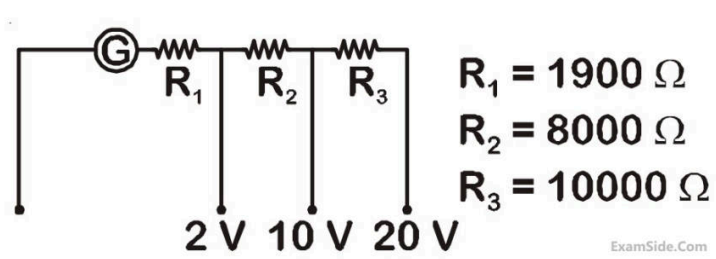
(A)



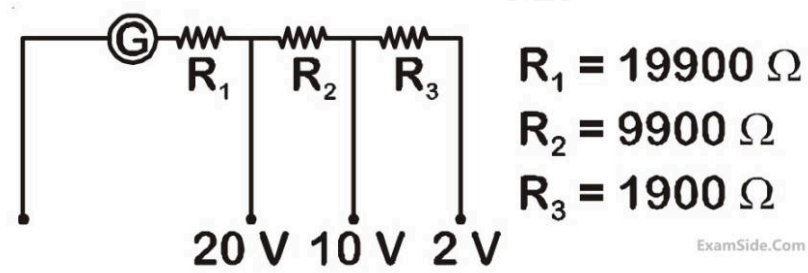
(B)



(C)



(D)

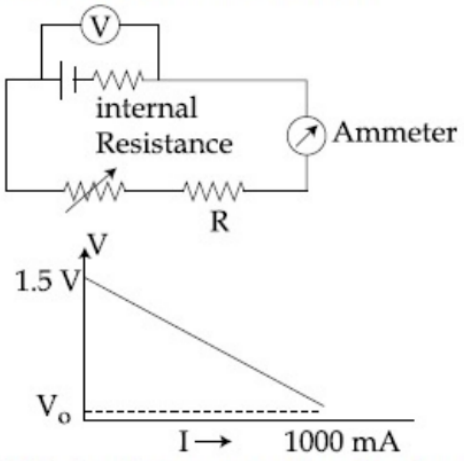


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Q253. JEE Main 2019 (Online) 12th April Morning Slot

Open solution

To verify Ohm's law, a student connects the voltmeter across the battery as, shown in the figure. The measured voltage is plotted as a function of the current, and the following graph is obtained :



If V_0 is almost zero, identify the correct statement :

(A) The value of the resistance R is 1.5 Ω

(B) The emf of the battery is l.5 V and its internal resistance is 1.5 Ω

(C) The emf of the battery is l.5 V and the value of R is 1.5 Ω

(D) The potential difference across the battery is 1.5 V when it sends a current of 1000 mA

Q254. JEE Main 2019 (Online) 10th April Evening Slot

Open solution

Space between two concentric conducting spheres of radii a and b (b > a) is filled with a medium of resistivity ρ. The resistance between the two spheres will be :

(A) $\frac{\rho}{2\pi}\left(\frac{1}{a}+\frac{1}{b}\right)$ (B) $\frac{\rho}{4\pi}\left(\frac{1}{a}+\frac{1}{b}\right)$

(C) $\frac{\rho}{2\pi}\left(\frac{1}{a}-\frac{1}{b}\right)$ (D) $\frac{\rho}{4\pi}\left(\frac{1}{a}-\frac{1}{b}\right)$

Q255. JEE Main 2019 (Online) 10th April Morning Slot

Open solution

A current of 5 A passes through a copper conductor (resistivity = 1.7×10^{-8} Ωm) of radius of cross-section 5 mm. Find the mobility of the charges if their drift velocity is 1.1×10^{-3} m/s.

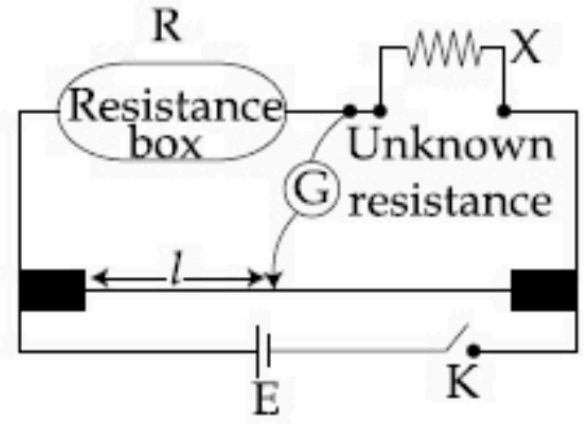
(A) 1.3 m^2/Vs (B) 1.0 m^2/Vs (C) 1.8 m^2/Vs (D) 1.5 m^2/Vs

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Q256. JEE Main 2019 (Online) 10th April Morning Slot

Open solution

In a meter bridge experiment, the circuit diagram and the corresponding observation table are shown in figure



S. No.	R(Ω)	l(cm)
1.	1000	60
2.	100	13
3.	10	1.5
4.	1	1.0

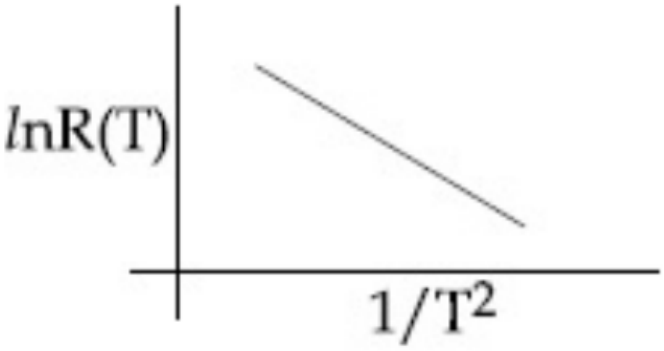
Which of the readings is inconsistent?

(A) 4 (B) 3 (C) 2 (D) 1

Q257. JEE Main 2019 (Online) 10th April Morning Slot

Open solution

In an experiment, the resistance of a material is plotted as a function of temperature (in some range). As shown in the figure, it is a straight line. One may conclude that :



(A) $R(T) = R_0 e^{-T^2/T_0^2}$ (B) $R(T) = \frac{R_0}{T^2}$ (C) $R(T) = R_0 e^{T^2/T_0^2}$ (D) $R(T) = R_0 e^{-T_0^2/T^2}$

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Q258. JEE Main 2019 (Online) 10th April Morning Slot

A moving coil galvanometer allows a full scale current of 10^{-4} A. A series resistance of 2 MΩ is required to convert the above galvanometer into a voltmeter of range 0-5 V. Therefore the value of shunt resistance required to convert the above galvanometer into an ammeter of range 0-10 mA is :

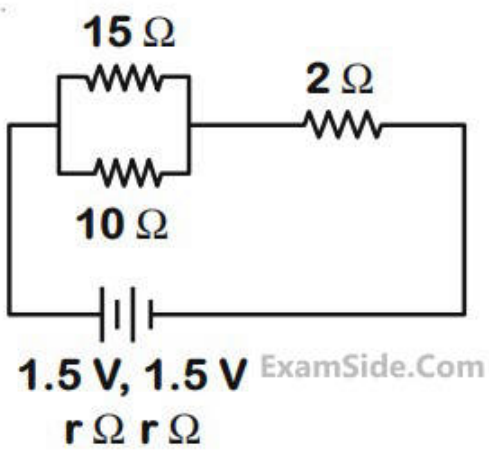
Open solution

(A) 200 Ω
(B) 500 Ω
(C) 100 Ω
(D) None of the options are correct

Q259. JEE Main 2019 (Online) 10th April Morning Slot

In the given circuit, an ideal voltmeter connected across the 10Ω resistance reads 2V. The internal resistance r, of each cell is:

Open solution



(A) 1.5 Ω
(B) 1 Ω
(C) 0.5 Ω
(D) 0 Ω

Q260. JEE Main 2019 (Online) 9th April Evening Slot

A metal wire of resistance 3 Ω is elongated to make a uniform wire of double its previous length. This new wire is now bent and the ends joined to make a circle. If two points on this circle make an angle 60° at the centre, the equivalent resistance between these two points will be :-

Open solution

(A) 5 /2 Ω
(B) 12/5 Ω
(C) 7/2 Ω
(D) 5 / 3 Ω

Q261. JEE Main 2019 (Online) 9th April Evening Slot

The resistance of a galvanometer is 50 ohm and the maximum current which can be passed through it is 0.002 A. What resistance must be connected to it in order to convert it into an ammeter of range 0 – 0.5 A ?

Open solution

(A) 0.02 ohm
(B) 0.2 ohm
(C) 0.002 ohm
(D) 0.5 ohm

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Q262. JEE Main 2019 (Online) 9th April Evening Slot

In a conductor, if the number of conduction electrons per unit volume is 8.5×10^{28} m^{-3} and mean free time is 25fs (femto second), it's approximate resistivity is :-
($m_e = 9.1 \times 10^{-31}$ kg)

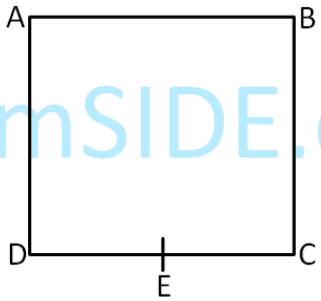
Open solution

(A) $10^{-8}\ \Omega m$ (B) $10^{-7}\ \Omega m$ (C) $10^{-5}\ \Omega m$ (D) $10^{-6}\ \Omega m$

Q263. JEE Main 2019 (Online) 9th April Morning Slot

A wire of resistance R is bent to form a square ABCD as shown in the figure. The effective resistance between E and C is : (E is mid-point of arm CD)

Open solution



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(A) $\frac{7}{64}R$ (B) $\frac{1}{16}R$ (C) R (D) $\frac{3}{4}R$

Q264. JEE Main 2019 (Online) 9th April Morning Slot

A moving coil galvanometer has resistance 50Ω and it indicates full deflection at 4mA current. A voltmeter is made using this galvanometer and a 5 kΩ resistance. The maximum voltage, that can be measured using this voltmeter, will be close to :

Open solution

(A) 15 V (B) 10 V (C) 40 V (D) 20 V

Q265. JEE Main 2019 (Online) 8th April Evening Slot

A cell of internal resistance r drives current through an external resistance R. The power delivered by the cell to the external resistance will be maximum when :-

Open solution

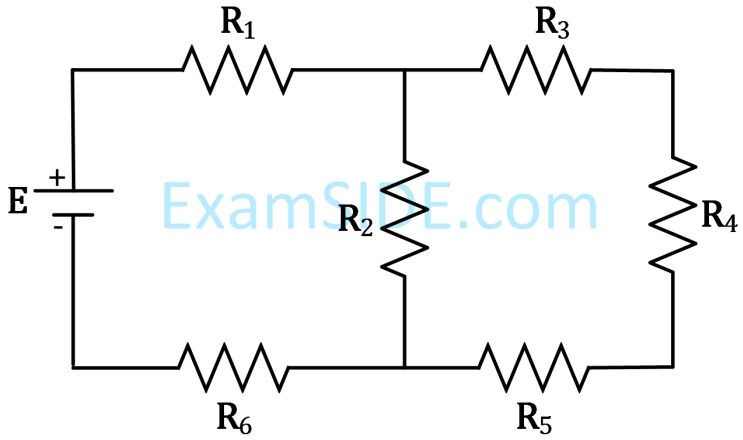
(A) R = 1000 r (B) R = r (C) R = 2r (D) R = 0.001 r

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Q266. JEE Main 2019 (Online) 8th April Evening Slot

Open solution

In the figure shown, what is the current (in Ampere) drawn from the battery ?
You are given: $R_1 = 15\Omega$, $R_2 = 10\ \Omega$, $R_3 = 20\ \Omega$, $R_4 = 5\Omega$, $R_5 = 25\Omega$, $R_6 = 30\ \Omega$, E = 15 V

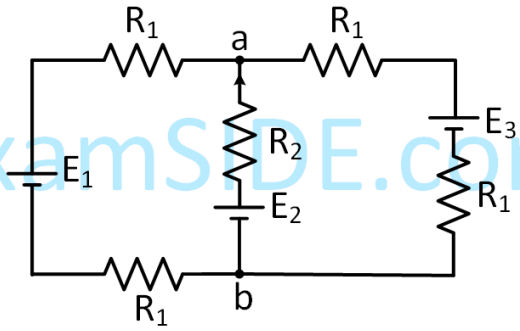


(A) 13/24 (B) 7/18 (C) 9/32 (D) 20/3

Q267. JEE Main 2019 (Online) 8th April Morning Slot

Open solution

For the circuit shown, with $R_1 = 1.0\ \Omega$, $R_2 = 2.0\ \Omega$, $E_1 = 2$ V and $E_2 = E_3 = 4$ V, the potential difference between the points 'a' and 'b' is approximately (in V):



(A) 3.3 (B) 2.7 (C) 2.3 (D) 3.7

Q268. JEE Main 2019 (Online) 12th January Evening Slot

Open solution

A galvanometer, whose resistance is 50 ohm, has 25 divisions in it. When a current of 4×10^{-4} A passes through it, its needle (pointer) deflects by one division. To use this galvanometer as a voltmeter of range 2.5 V, it should be connected to a resistance of :

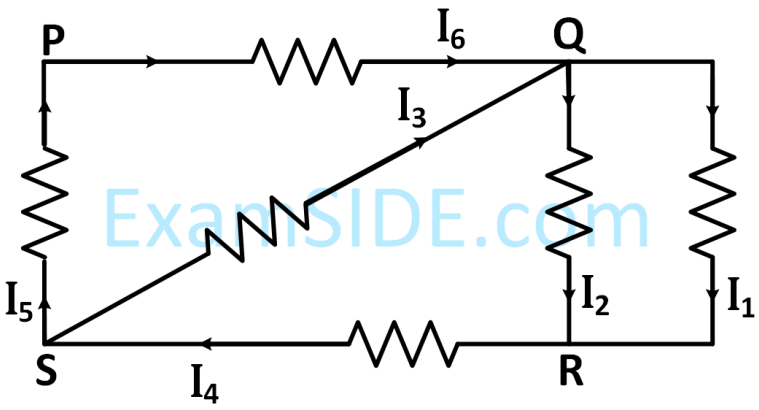
(A) 200 ohm (B) 250 ohm (C) 6200 ohm (D) 6250 ohm

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Q269. JEE Main 2019 (Online) 12th January Evening Slot

Open solution

In the given circuit diagram, the currents, $I_1 = -0.3$ A, $I_4 = 0.8$ A and $I_5 = 0.4$ A, are flowing as shown. The currents I_2, I_3 and I_6, respectively, are :



(A) 1.1 A, – 0.4 A, 0.4 A

(B) – 0.4 A, 0.4 A, 1.1 A

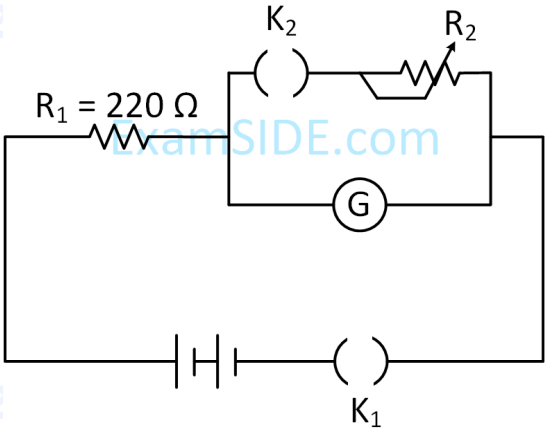
(C) 0.4 A, 1.1 A, 0.4 A

(D) 1.1 A, 0.4 A, 0.4 A

Q270. JEE Main 2019 (Online) 12th January Morning Slot

Open solution

The galvanometer deflection, when key K_1 is closed but K_2 is open, equals θ_0 (see figure). On closing K_2 also and adjusting R_2 to 5Ω, the deflection in galvanometer becomes $\frac{\theta_0}{5}$. . The resistance of the galvanometer is, then, given by [Neglect the internal resistance of battery] :



(A) 5 Ω

(B) 25 Ω

(C) 12 Ω

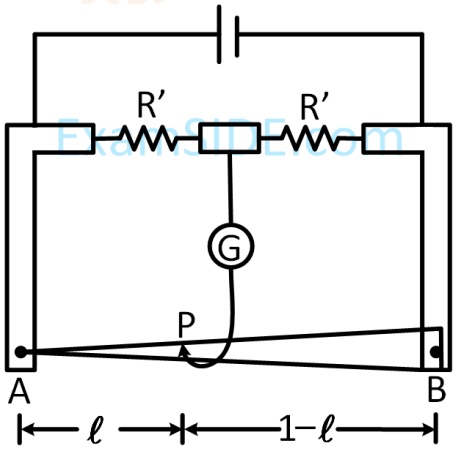
(D) 22 Ω

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Q271. JEE Main 2019 (Online) 12th January Morning Slot

Open solution

In a meter bridge, the wire of length 1 m has a non-uniform cross-section such that, the variation $\frac{dR}{d\ell}$ of its resistance R with length ℓ is $\frac{dR}{d\ell} \propto \frac{1}{\sqrt{\ell}}$. Two equal resistances are connected as shown in the figure. The galvanometer has zero deflection when the jockey is at point P. What is the length AP ?



(A) 0.3 m (B) 0.25 m (C) 0.35 m (D) 0.2 m

Q272. JEE Main 2019 (Online) 12th January Morning Slot

Open solution

Two electric bulbs, rated at (25 W, 220 V) and (100 W, 220 V), are connected in series across a 220 V voltage source. If the 25 W and 100 W bulbs draw powers P_1 and P_2 respectively, then :

(A) P_1 = 4W, P_2 = 16 W (B) P_1 = 16W, P_2 = 4 W

(C) P_1 = 9W, P_2 = 16 W (D) P_1 = 16W, P_2 = 9 W

Q273. JEE Main 2019 (Online) 11th January Evening Slot

Open solution

A galvanometer having a resistance of 20 Ω and 30 divisions on both sides has figure of merit 0.005 ampere/division. The resistance that should be connected in series such that it can be used as a voltmeter upto 15 volt, is:

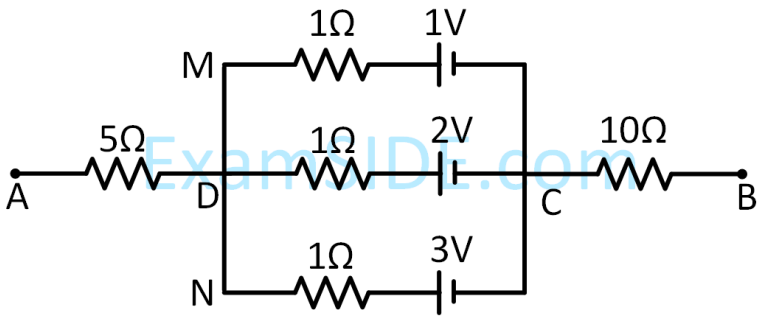
(A) 120 Ω (B) 125 Ω (C) 80 Ω (D) 100 Ω

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Q274. JEE Main 2019 (Online) 11th January Evening Slot

In the circuit shown, the potential difference between A and B is :

Open solution

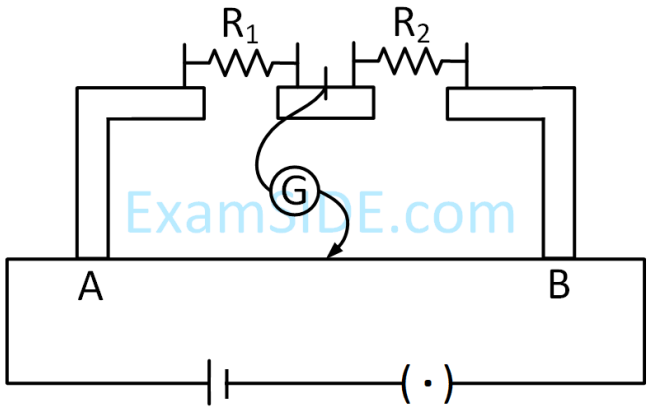


(A) 6 V (B) 3 V (C) 2 V (D) 1 V

Q275. JEE Main 2019 (Online) 11th January Evening Slot

In the experimental set up of metre bridge shown in the figure, the null point is obtained at a distance of 40 cm from A. If a 10 Ω resistor is connected in series with R_1, the null point shifts by 10 cm. The resistance that should be connected in parallel with $(R_1 + 10)$ Ω such that the null point shifts back to its initial position is :

Open solution



(A) 40 Ω (B) 30 Ω (C) 20 Ω (D) 60 Ω

Q276. JEE Main 2019 (Online) 11th January Morning Slot

Two equal resistances when connected in series to a battery, consume electric power of 60 W. If these resistances are now connected in parallel combination to the same battery, the electric power consumed will be :

Open solution

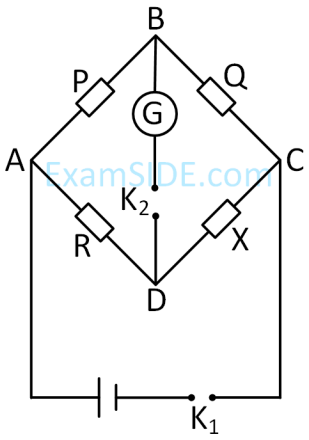
(A) 240 W (B) 60 W (C) 30 W (D) 120 W

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Q277. JEE Main 2019 (Online) 11th January Morning Slot

In a Wheatstone bridge(see fig.), Resistances P and Q are approximately equal. When R = 400 Ω, the bridge is balanced. On interchanging P and Q, the value of R, for balance, is 405 Ω. The value of X is close to :

Open solution



(A) 402.5 ohm (B) 401.5 ohm (C) 403.5 ohm (D) 404.5 ohm

Q278. JEE Main 2019 (Online) 10th January Evening Slot

A current of 2 mA was passed through an unknown resistor which dissipated a power of 4.4 W. Dissipated power when an ideal power supply of 11 V is connected across it is -

(A) 11×10^{-5} W (B) 11×10^{-3} W (C) 11×10^{5} W (D) 11×10^{-4} W

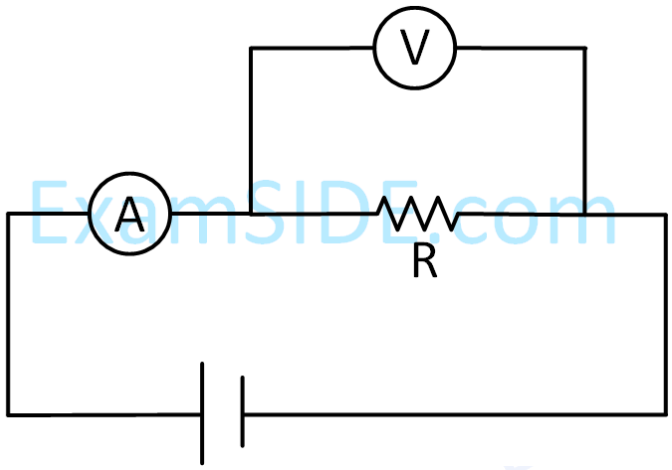
Open solution

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Q279. JEE Main 2019 (Online) 10th January Evening Slot

The actual value of resistance R, shown in the figure is 30Ω. This is measuered in an experiment as shown using the standard formula R = $\frac{V}{I}$, where V and I are the readings of the voltmeter and ammeter, respectively. If the measured value of R is 5% less, then the internal resistance of the voltmeter is -

Open solution



(A) 570 Ω (B) 600 Ω (C) 350 Ω (D) 35 Ω

Q280. JEE Main 2019 (Online) 10th January Morning Slot

A uniform metallic wire has a resistance of 18 Ω and is bent into an equilateral triangle. Then, the resistance between any two vertices of the triangle is -

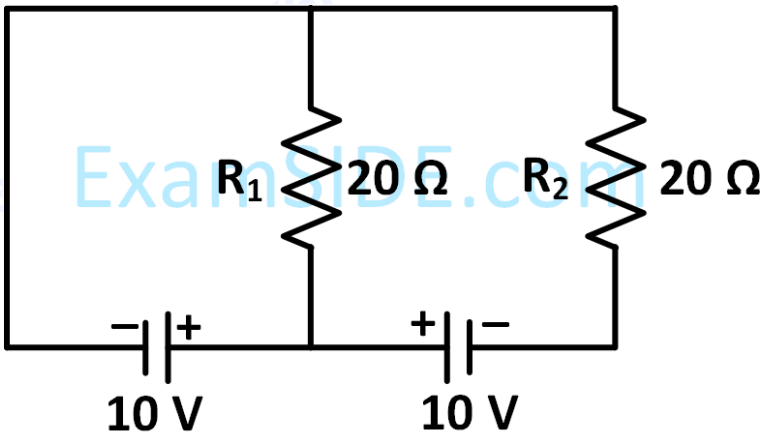
Open solution

(A) 12 Ω (B) 2 Ω (C) 4 Ω (D) 8 Ω

Q281. JEE Main 2019 (Online) 10th January Morning Slot

In the given circuit the cells have zero internal resistance. The currents (in Amperes) passing through resistance R_1 and R_2 respectively, are -

Open solution



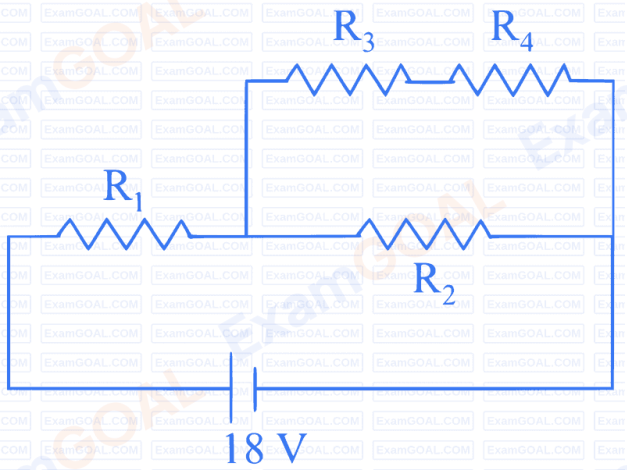
(A) 0.5, 0 (B) 0, 1 (C) 1, 2 (D) 2, 2

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Q282. JEE Main 2019 (Online) 9th January Evening Slot

In the given circuit the internal resistance of the 18 V cell is negligible. If R_1 = 400 Ω, R_3 = 100 Ω and R_4 = 500 Ω and the reading of an ideal voltmeter across R_4 is 5V, then the value of R_2 will be :

Open solution

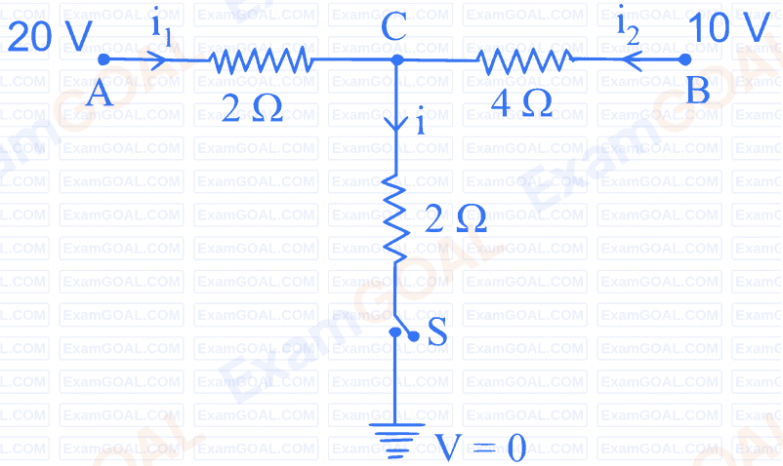


(A) 300 Ω (B) 450 Ω (C) 550 Ω (D) 230 Ω

Q283. JEE Main 2019 (Online) 9th January Morning Slot

When the switch S, in the circuit shown, is closed, then the value of current i will be :

Open solution



(A) 3A (B) 5A (C) 4A (D) 2A

Q284. JEE Main 2019 (Online) 9th January Morning Slot

A copper wire is stretched to make it 0.5% longer. The percentage change in its electrical resistance if its volume remains unchanged is :

Open solution

(A) 2.0 % (B) 2.5 % (C) 1.0 % (D) 0.5 %

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Q285. JEE Main 2019 (Online) 9th January Morning Slot

Drift speed of electrons, when 1.5 A of current flows in a copper wire of cross section 5 mm^2, is v. If the electron density in copper is $9 \times 10^{28}/m^3$ the value of v. in mm/s is close to (Take charge of electron to be = 1.6×10^{-19}C)

Open solution

(A) 0.02 (B) 3 (C) 2 (D) 0.2

Q286. JEE Main 2018 (Online) 16th April Morning Slot

A heating element has a resistance of 100 Ω at room temperature. When it is connected to a supply of 220 V, a steady current of 2 A passes in it and temperature is 500°C more than room temperature. What is the temperature coefficient of resistance of the heating element ?

Open solution

(A) $0.5 \times 10^{-4}\,{}^{o}C^{-1}$ (B) $5 \times 10^{-4}\,{}^{o}C^{-1}$

(C) $1 \times 10^{-4}\,{}^{o}C^{-1}$ (D) $2 \times 10^{-4}\,{}^{o}C^{-1}$

Q287. JEE Main 2018 (Online) 16th April Morning Slot

A galvanometer with its coil resistance 25 Ω requires a current of 1 mA for its full deflection. In order to construct an ammeter to read upto a current of 2 A, the approximate value of the shunt resistance should be :

Open solution

(A) $2.5 \times 10^{-3}\,\Omega$ (B) $1.25 \times 10^{-2}\Omega$ (C) $1.25 \times 10^{-3}\Omega$ (D) $2.5 \times 10^{-2}\Omega$

Q288. JEE Main 2018 (Online) 16th April Morning Slot

In a circuit for finding the resistance of a galvanometer by half deflection method, a 6 V battery and a high resistance of 11 kΩ are used. The figure of merit of the galvanometer is 60 μA/division. In the absence of shunt resistance, the galvanometer produces a deflection of θ = 9 divisions when current flows in the circuit. The value of the shunt resistance that can cause the deflection of $\theta/2$, is closest to :

Open solution

(A) 500 Ω (B) 220 Ω (C) 55 Ω (D) 110 Ω

Q289. JEE Main 2018 (Offline)

Two batteries with e.m.f 12 V and 13 V are connected in parallel across a load resistor of 10 Ω. The internal resistances of the two batteries are 1 Ω and 2 Ω respectively. The voltage across the load lies between :

Open solution

(A) 11.7 V and 11.8 V (B) 11.6 V and 11.7 V

(C) 11.5 V and 11.6 V (D) 11.4 V and 11.5 V

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Q290. JEE Main 2018 (Offline)

On interchanging the resistances, the balance point of a meter bridge shifts to the left by 10 cm. The resistance of their series combination is 1 kΩ. How much was the resistance on the left slot before interchanging the resistances?

Open solution

(A) 910 Ω (B) 990 Ω (C) 505 Ω (D) 550 Ω

Q291. JEE Main 2018 (Online) 15th April Evening Slot

A copper rod of cross-sectional area A carries a uniform current I through it. At temperature T, if the volume charge density of the rod is ρ, how long will the charges take to travel a distance d ?

Open solution

(A) $\frac{2\rho\, d\, A}{\mathrm{I}}$ (B) $\frac{2\rho\, d\, A}{\mathrm{I}\, T}$

(C) $\frac{\rho\, d\, A}{\mathrm{I}}$ (D) $\frac{\rho\, d\, A}{\mathrm{I}\, T}$

Q292. JEE Main 2018 (Online) 15th April Evening Slot

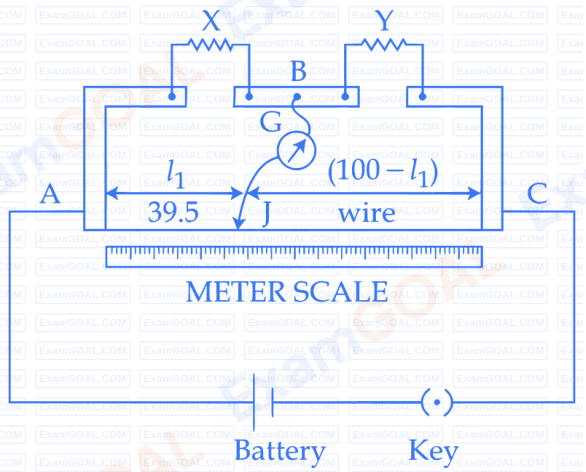
A constant voltages is applied between two ends of a metallic wire. If the length is halved and the radius of the wire is doubled, the rate of heat developed in the wire will be :

Open solution

(A) Doubled (B) Halved

(C) Unchanged (D) Increased 8 times

Q293. JEE Main 2018 (Online) 15th April Morning Slot



Open solution

In a meter bridge, as shown in the figure, it is given that resistance $Y = 12.5\ \Omega$ and that the balance is obtained at a distance $39.5\ cm$ from end A (by Jockey J). After interchanging the resistances X and Y, a new balance point is found at a distance l_2 from end A. What are the value of X and l_2 ?

(A) $8.16\ \Omega$ and $60.5\ cm$ (B) $19.15\ \Omega$ and $39.5\ cm$

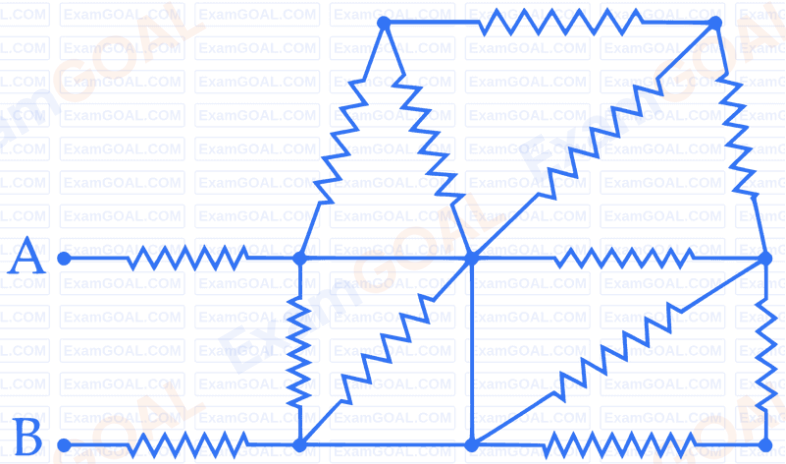
(C) $8.16\ \Omega$ and $39.5\ cm$ (D) $19.15\ \Omega$ and $60.5\ cm$

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Q294. JEE Main 2018 (Online) 15th April Morning Slot

In the given circuit all resistances are of value $R\,ohm$ each. The equivalent resistance between A and B is :

Open solution



(A) $2R$ (B) $3R$ (C) $\frac{5R}{3}$ (D) $\frac{5R}{2}$

Q295. JEE Main 2017 (Online) 9th April Morning Slot

A uniform wire of length l and radius r has a resistance of 100 Ω. It is recast into a wire of radius $\frac{r}{2}$. The resistance of new wire will be :

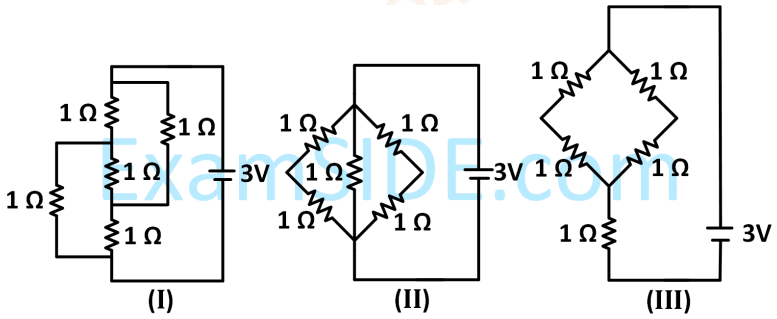
Open solution

(A) 1600 Ω (B) 400 Ω (C) 200 Ω (D) 100 Ω

Q296. JEE Main 2017 (Online) 9th April Morning Slot

The figure shows three circuits I, II and III which are connected to a 3V battery. If the powers dissipated by the configurations I, II and III are P_1 , P_2 and P_3 respectively, then :

Open solution



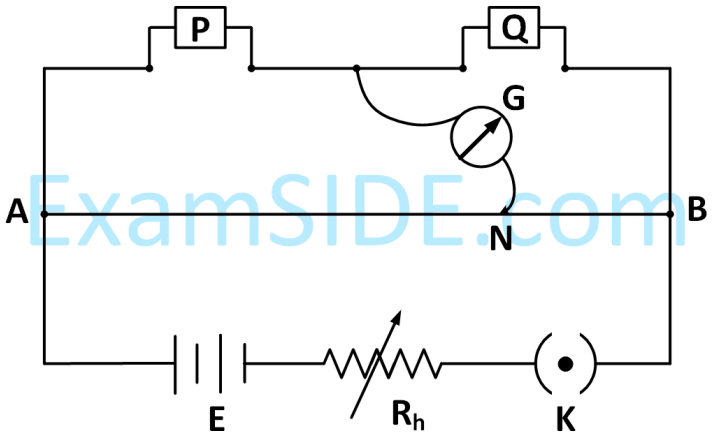
(A) $P_1 > P_2 > P_3$ (B) $P_1 > P_3 > P_2$ (C) $P_2 > P_1 > P_3$ (D) $P_3 > P_2 > P_1$

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Q297. JEE Main 2017 (Online) 9th April Morning Slot

Open solution

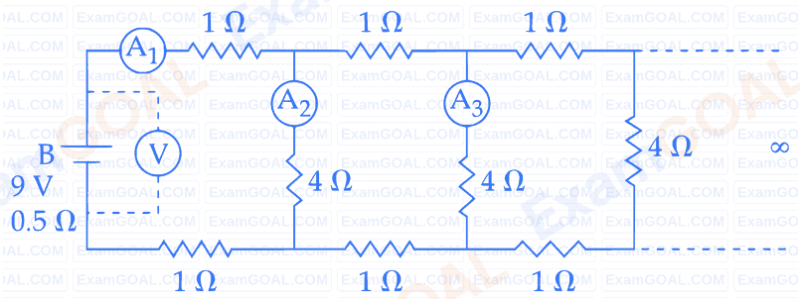
In a meter bridge experiment resistances are connected as shown in the figure. Initially resistance P = 4 Ω and the neutral point N is at 60 cm from A. Now an unknown resistance R is connected in series to P and the new position of the neutral point is at 80 cm from A. The value of unknown resistance R is :



(A) $\frac{33}{5}$ Ω (B) 6 Ω (C) 7 Ω (D) $\frac{20}{3}$ Ω

Q298. JEE Main 2017 (Online) 8th April Morning Slot

Open solution



A 9 V battery with internal resistance of 0.5 Ω is connected across an infinite network as shown in the figure. All ammeters A_1, $A_{2,3}$ and voltmeter V are ideal.

Choose correct statement.

(A) Reading of A_1 is 2 A

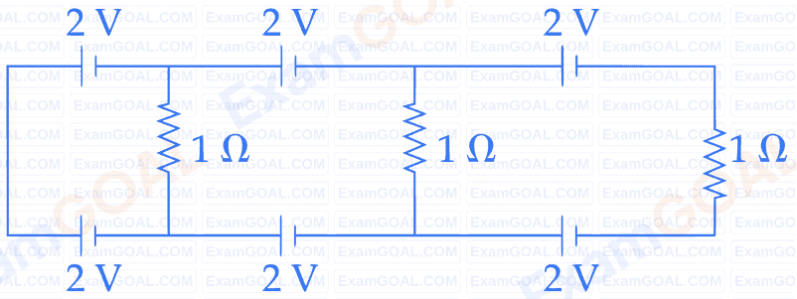
(B) Reading of A_1 is 18 A

(C) Reading of V is 9 V

(D) Reading of V is 7 V

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Q299. JEE Main 2017 (Offline)



Open solution

In the given circuit, the current in each resistance is:

(A) 0 A (B) 1 A (C) 0.25 A (D) 0.5 A

Q300. JEE Main 2017 (Offline)

Which of the following statements is **false**?

(A) Kirchhoff's second law represents energy conservation.

(B) Wheatstone bridge is the most sensitive when all the four resistances are of the same order of magnitude.

(C) In a balanced wheatstone bridge if the cell and the galvanometer are exchanged, the null point is disturbed.

(D) A rheostat can be used as a potential divider.

Open solution

Q301. JEE Main 2017 (Offline)

When a current of 5 mA is passed through a galvanometer having a coil of resistance 15Ω, it shows full scale deflection. The value of the resistance to be put in series with the galvanometer to convert it into a voltmeter of range 0 – 10V is:

(A) $4.005 \times 10^3\ \Omega$ (B) $1.985 \times 10^3\ \Omega$ (C) $2.535 \times 10^3\ \Omega$ (D) $2.045 \times 10^3\ \Omega$

Open solution

Q302. JEE Main 2016 (Online) 10th April Morning Slot

The resistance of an electrical toaster has a temperature dependence given by $R(T) = R_0 [1 + \alpha(T - T_0)]$ in its range of operation. At T_0 = 300 K, R = 100 Ω and at T = 500 K, R = 120 Ω. The toaster is connected to a voltage source at 200 V and its temperature is raised at a constant rate from 300 to 500 K in 30 s. The total work done in raising the temperature is :

(A) $400 \ln \frac{1.5}{1.3} J$ (B) $200 \ln \frac{2}{3} J$

(C) $60000 \ln \frac{6}{5} J$ (D) 300 J

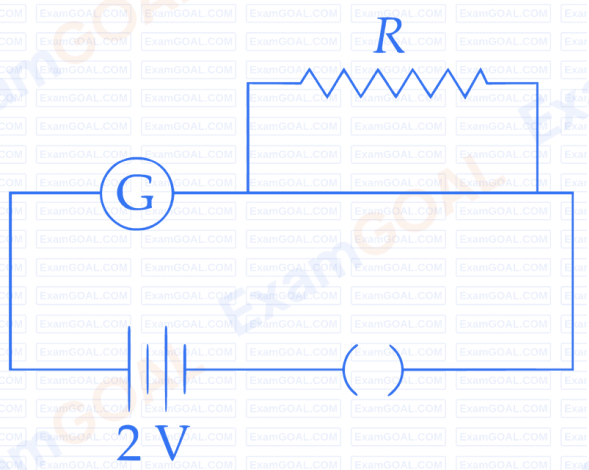
Open solution

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Q303. JEE Main 2016 (Online) 10th April Morning Slot

A galvanometer has a 50 division scale. Battery has no internal resistance. It is found that there is deflection of 40 divisions when R = 2400 Ω. Deflection becomes 20 divisions when resistance taken from resistance box is 4900 Ω. Then we can conclude :

Open solution



(A) Resistance of galvanometer is 200 Ω

(B) Full scale deflection current is 2 mA.

(C) Current sensitivity of galvanometer is 20 μA/division.

(D) Resistance required on R.B. for a deflection of 10 divisions is 9800 Ω.

Q304. JEE Main 2016 (Online) 9th April Morning Slot

A 50 Ω resistance is connected to a battery of 5 V. A galvanometer of resistance 100 Ω is to be used as an ammeter to measure current through the resistance, for this a resistance r_s is connected to the galvanometer. Which of the following connections should be employed if the measured current is within 1% of the current without the ammeter in the circuit ?

Open solution

(A) r_s = 0.5 Ω in parallel with the galvanometer

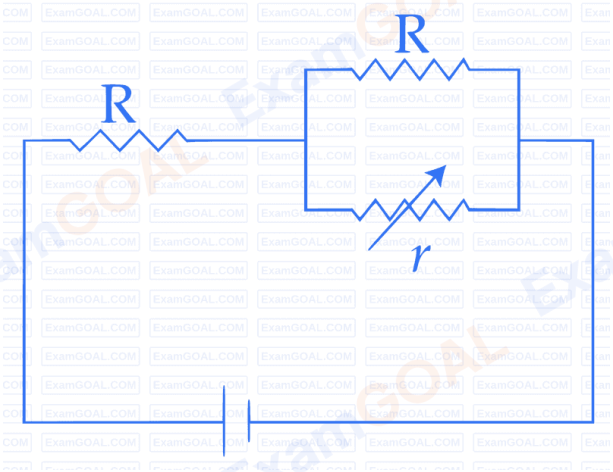
(B) r_s = 0.5 Ω in series with the galvanometer

(C) r_s = 1 Ω in series with galvanometer

(D) r_s =1 Ω in parallel with galvanometer

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Q305. JEE Main 2016 (Online) 9th April Morning Slot



Open solution

In the circuit shown, the resistance r is a variable resistance. If for r = *fR*, the heat generation in r is maximum then the value of *f* is :

(A) $\frac{1}{4}$ (B) $\frac{1}{2}$ (C) $\frac{3}{4}$ (D) 1

Q306. JEE Main 2016 (Offline)

A galvanometer having a coil resistance of $100\,\Omega$ gives a full scale deflection, when a currect of 1 mA is passed through it. The value of the resistance, which can convert this galvanometer into ammeter giving a full scale deflection for a current of 10 A, is :

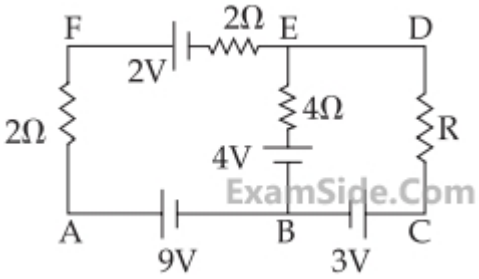
Open solution

(A) $0.1\,\Omega$ (B) $3\,\Omega$ (C) $0.01\,\Omega$ (D) $2\,\Omega$

Q307. JEE Main 2015 (Online) 11th April Morning Slot

In the electric network shown, when no current flows through the 4 Ω resistor in the arm EB, the potential difference between the points A and D will be :

Open solution



(A) 3 V (B) 4 V (C) 5 V (D) 6 V

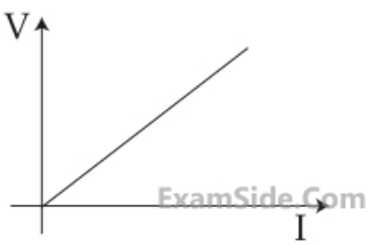
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Q308. JEE Main 2015 (Online) 10th April Morning Slot

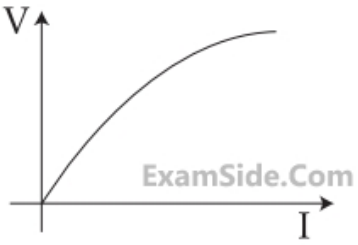
Open solution

Suppose the drift velocity v_d in a material varied with the applied electric field E as $v_d \propto \sqrt{E}$. Then V - I graph for a wire made of such a material is best given by :

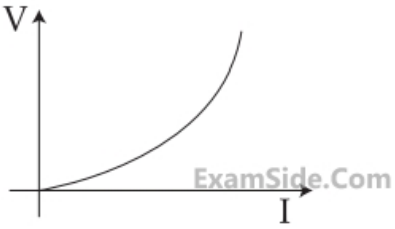
(A)



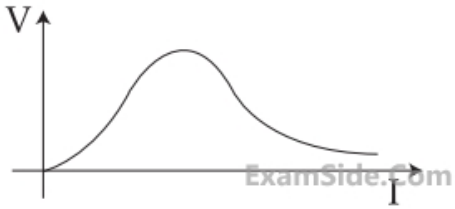
(B)



(C)



(D)

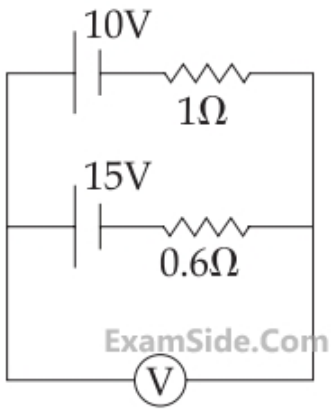


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Q309. JEE Main 2015 (Online) 10th April Morning Slot

A 10V battery with internal resistance 1Ω and a 15V battery with internal resistance 0.6 Ω are connected in parallel to a voltmeter (see figure). The reading in the voltmeter will be close to :

Open solution



(A) 11.9 V (B) 12.5 V (C) 13.1 V (D) 24.5 V

Q310. JEE Main 2015 (Offline)

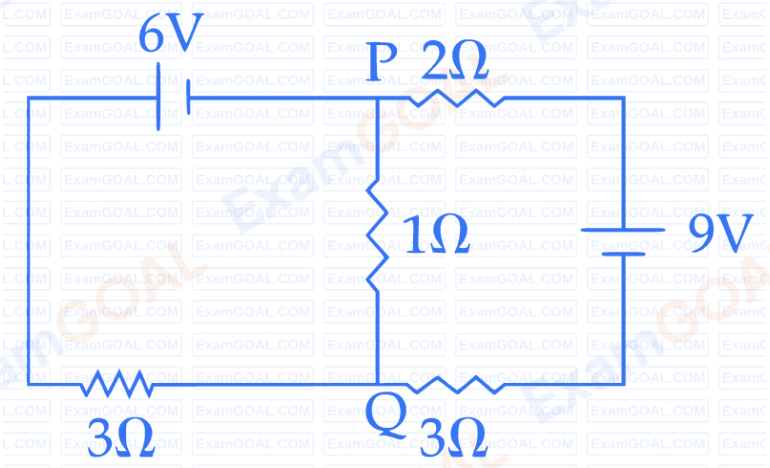
When $5V$ potential difference is applied across a wire of length $0.1\ m$, the drift speed of electrons is $2.5 \times 10^{-4}\ ms^{-1}$. If the electron density in the wire is $8 \times 10^{28}\ m^{-3}$, the resistivity of the material is close to :

Open solution

(A) $1.6 \times 10^{-6}\Omega m$ (B) $1.6 \times 10^{-5}\Omega m$ (C) $1.6 \times 10^{-8}\Omega m$ (D) $1.6 \times 10^{-7}\Omega m$

Q311. JEE Main 2015 (Offline)

Open solution



In the circuit shown, the current in the 1Ω resistor is :

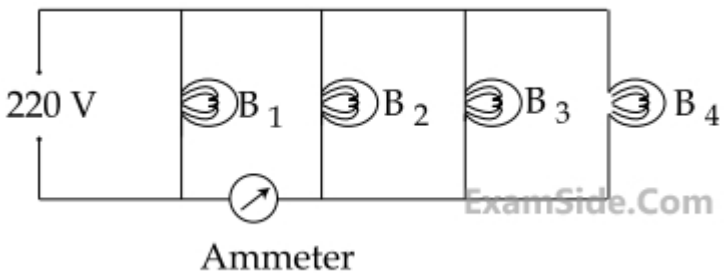
(A) $0.13\ A$, from Q to P (B) $0.13\ A$, from P to Q

(C) $1.3A$ from P to Q (D) $0A$

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Q312. JEE Main 2014 (Online) 19th April Morning Slot

Open solution



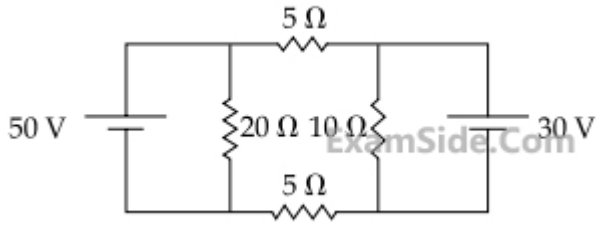
Four bulbs B_1, B_2, B_3 and B_4 of 100 W each are connected to 220 V main as shown in the figure. The reading in an ideal ammeter will be :

(A) 0.45 A **(B)** 0.90 A **(C)** 1.35 A **(D)** 1.80 A

Q313. JEE Main 2014 (Online) 11th April Morning Slot

Open solution

In the circuit shown, current (in A) through the 50 V and 30 V batteries are, respectively :



(A) 2.5 and 3 **(B)** 3.5 and 2 **(C)** 4.5 and 1 **(D)** 3 and 2.5

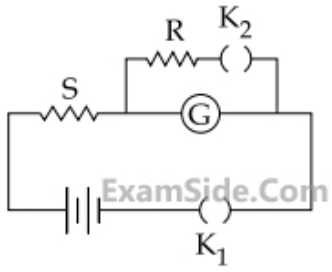
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Q314. JEE Main 2014 (Online) 11th April Morning Slot

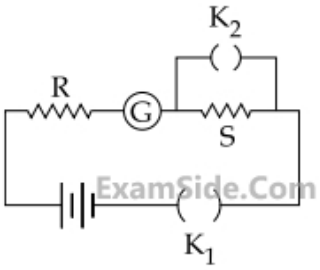
Open solution

In the circuit diagrams (A, B, C and D) shown below, R is a high resistance and S is a resistance of the order of galvanometer resistance G. The correct circuit, corresponding to the half deflection method for finding the resistance and figure of merit of galvanometer, is the circuit labelled as :

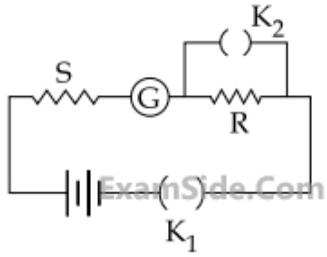
(A)



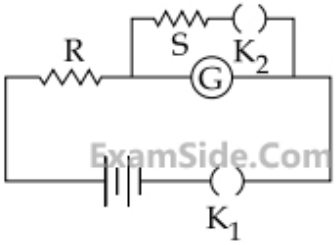
(B)



(C)



(D)



Q315. JEE Main 2014 (Online) 9th April Morning Slot

Open solution

A d.c. main supply of e.m.f. 220 V is connected across a storage battery of e.m.f. 200 V through a resistance of 1 Ω. The battery terminals are connected to an external resistance 'R'. The minimum value of 'R', so that a current passes through the battery to charge it is :

(A) 7 Ω (B) 9 Ω (C) 11 Ω (D) Zero

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Q316. JEE Main 2014 (Offline)

In a large building, there are 15 bulbs of $40\ W$, 5 bulbs of $100\ W$, 5 fans of $80\ W$ and 1 heater of $1\ kW$. The voltage of electric mains is $220\ V$. The minimum capacity of the main fuse of the building will be:

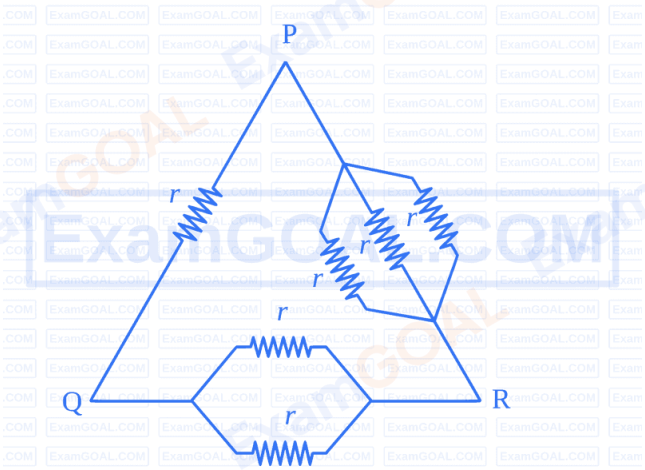
Open solution

(A) $8\ A$ (B) $10\ A$ (C) $12\ A$ (D) $14\ A$

Q317. JEE Main 2013 (Online) 25th April Morning Slot

Six equal resistances are connected between points P, Q and R as shown in the figure. Then net resistance will be maximum between :

Open solution

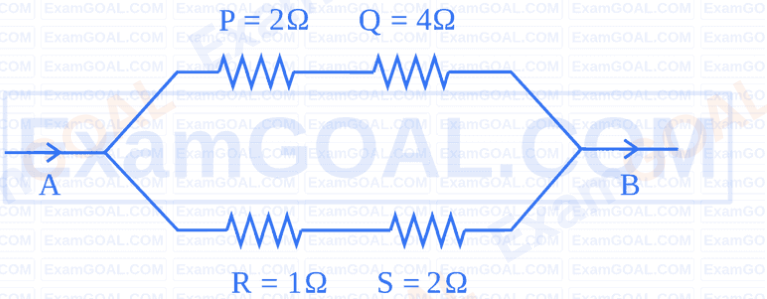


(A) P and R (B) P and Q (C) Q and R (D) Any two points

Q318. JEE Main 2013 (Online) 23th April Morning Slot

Which of the four resistances P, Q, R and S generate the greatest amount of heat when a current flows from A to B?

Open solution



(A) Q (B) S (C) P (D) R

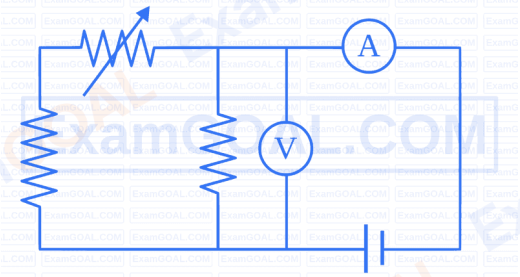
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Q319. JEE Main 2013 (Online) 23th April Morning Slot

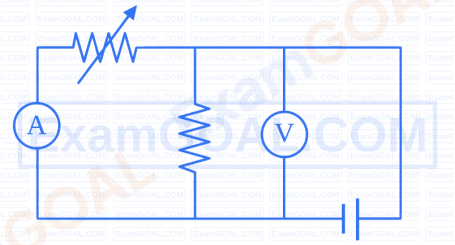
Open solution

Correct set up to verify Ohm's law is :

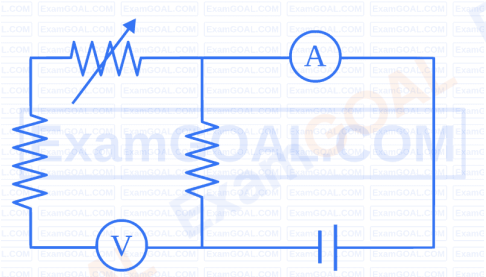
(A)



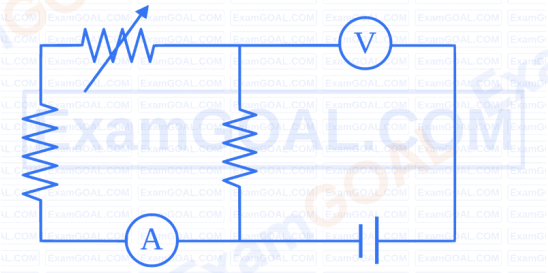
(B)



(C)



(D)

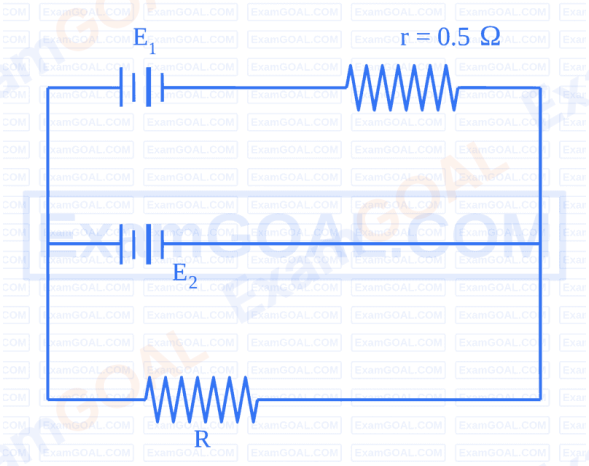


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Q320. JEE Main 2013 (Online) 22th April Morning Slot

A dc source of emf $E_1 = 100$ V and internal resistance $r = 0.5\Omega$, a storage battery of emf $E_2 = 90$ V and an external resistance R are connected as shown in figure. For what value of R no current will pass through the battery?

Open solution

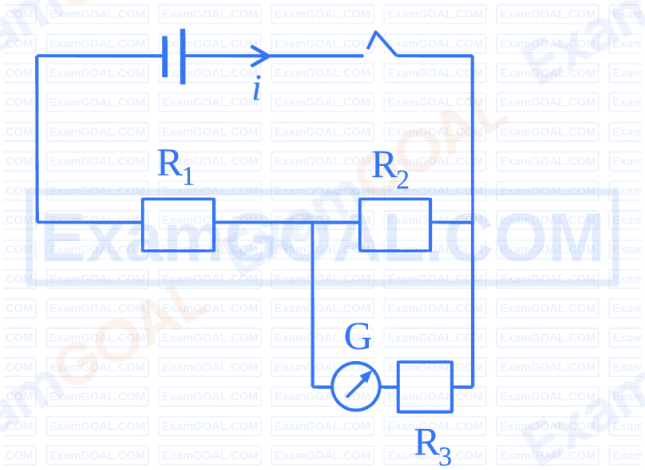


(A) 5.5Ω (B) 3.5Ω (C) 4.5Ω (D) 2.5Ω

Q321. JEE Main 2013 (Online) 22th April Morning Slot

To find the resistance of a galvanometer by the half deflection method, the following circuit is used with resistances $R_1 = 9970\ \Omega$, $R_2 = 30\ \Omega$ and $R_3 = 0$. The deflection in the galvanometer is d. With $R_3 = 107\ \Omega$, the deflection changed to $\frac{d}{2}$. The galvanometer resistance is approximately:

Open solution



(A) 107Ω (B) 137Ω (C) 107/2Ω (D) 77Ω

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Q322. JEE Main 2013 (Online) 9th April Morning Slot

In a metre bridge experiment null point is obtained at 40 cm from one end of the wire when resistance X is balanced against another resistance Y. If X < Y, then the new position of the null point from the same end, if one decides to balance a resistance of 3X against Y, will be close to :

Open solution

(A) 80 cm (B) 75 cm (C) 67 cm (D) 50 cm

Q323. JEE Main 2013 (Online) 9th April Morning Slot

A letter 'A' is constructed of a uniform wire with resistance 1.0Ω per cm. The sides of the letter are 20 cm and the cross piece in the middle is 10 cm long. The apex angle is 60°. The resistance between the ends of the legs is close to:

Open solution

(A) 50.0Ω (B) 10Ω (C) 36.7Ω (D) 26.7Ω

Q324. JEE Main 2013 (Online) 9th April Morning Slot

A shunt of resistance 1 Ω is connected across a galvanometer of 120 Ω resistance. A current of 5.5 ampere gives full scale deflection in the galvanometer. The current that will give full scale deflection in the absence of the shunt is nearly :

Open solution

(A) 5.5 ampere (B) 0.5 ampere (C) 0.004 ampere (D) 0.045 ampere

Q325. JEE Main 2013 (Offline)

The supply voltage to room is $120V$. The resistance of the lead wires is 6Ω. A $60\ W$ bulb is already switched on. What is the decrease of voltage across the bulb, when a $240\ W$ heater is switched on in parallel to the bulb?

Open solution

(A) zero (B) 2.9 Volt (C) 13.3 Volt (D) 10.04 Volt

Q326. JEE Main 2013 (Offline)

This questions has Statement - I and Statement - II. Of the four choices given after the Statements, choose the one that best describes into two Statements.

Statement - I : Higher the range, greater is the resistance of ammeter.

Statement - II : To increase the range of ammeter, additional shunt needs to be used across it.

Open solution

(A) Statement - I is true, Statement - II is true, Statement - II is the correct explanation of statement - I.

(B) Statement - I is true, Statement - II is true, Statement - II is not the correct explanation of statement - I.

(C) Statement - I is true, Statement - II is false

(D) Statement - I is false, Statement - II is true

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Q327. AIEEE 2012 (Online) 26th May

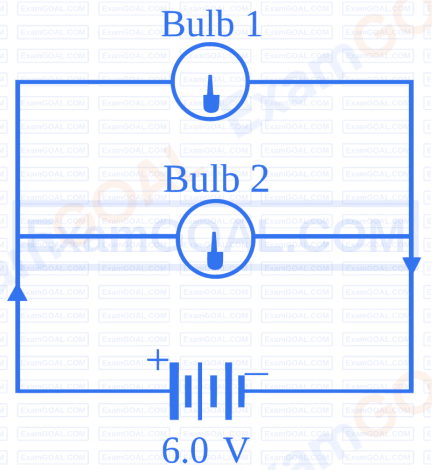
The resistance of a wire is R. It is bent at the middle by 180° and both the ends are twisted together to make a shorter wire. The resistance of the new wire is

(A) $2R$ (B) $R/2$ (C) $R/4$ (D) $R/8$

Open solution

Q328. AIEEE 2012 (Online) 19th May

A 6.0 volt battery is connected to two light bulbs as shown in figure. Light bulb 1 has resistance 3 ohm while light bulb 2 has resistance 6 ohm. Battery has negligible internal resistance. Which bulb will glow brighter?



(A) Bulb 1 will glow more first and then its brightness will become less than bulb 2

(B) Bulb 1

(C) Bulb 2

(D) Both glow equally

Open solution

Q329. AIEEE 2012 (Online) 12th May

Three resistors of $4\ \Omega$, $6\ \Omega$ and $12\ \Omega$ are connected in parallel and the combination is connected in series with a 1.5 V battery of $1\ \Omega$ internal resistance. The rate of Joule heating in the $4\ \Omega$ resistor is

(A) 0.55 W (B) 0.33 W (C) 0.25 W (D) 0.86 W

Open solution

Q330. AIEEE 2012 (Online) 12th May

In a sensitive meter bridge apparatus the bridge wire should possess

(A) high resistivity and low temperature coefficient.

(B) low resistivity and high temperature coefficient.

(C) low resistivity and low temperature coefficient.

(D) high resistivity and high temperature coefficient.

Open solution

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Q331. AIEEE 2012 (Online) 7th May

This question has Statement 1 and Statement 2. Of the four choices given after the Statements, choose the one that best describes the two Statements.
Statement 1: The possibility of an electric bulb fusing is higher at the time of switching ON.
Statement 2: Resistance of an electric bulb when it is not lit up is much smaller than when it is lit up.

(A) Statement 1 is true, Statement 2 is false

(B) Statement 1 is false, Statement 2 is true, Statement 2 is not a correct explanation of Statement 1.

(C) Statement 1 is true, Statement 2 is true, Statement 2 is a correct explanation of Statement 1.

(D) Statement 1 is false, Statement 2 is true.

Open solution

Q332. AIEEE 2012

Two electric bulbs marked $25W - 220\ V$ and $100W - 220V$ are connected in series to a $440\ V$ supply. Which of the bulbs will fuse?

(A) Both (B) $100\ W$ (C) $25\ W$ (D) Neither

Open solution

Q333. AIEEE 2011

If a wire is stretched to make it 0.1% longer, its resistance will:

(A) increase by 0.2%

(B) decrease by 0.2%

(C) decrease by 0.05%

(D) increase by 0.05%

Open solution

Q334. AIEEE 2010

Two conductors have the same resistance at $0^{\circ}C$ but their temperature coefficients of resistance are α_1 and α_2. The respective temperature coefficients of their series and parallel combinations are nearly

(A) $\frac{\alpha_1+\alpha_2}{2}$, $\alpha_1+\alpha_2$

(B) $\alpha_1+\alpha_2$, $\frac{\alpha_1+\alpha_2}{2}$

(C) $\alpha_1+\alpha_2$, $\frac{\alpha_1\alpha_2}{\alpha_1+\alpha_2}$

(D) $\frac{\alpha_1+\alpha_2}{2}$, $\frac{\alpha_1+\alpha_2}{2}$

Open solution

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Q335. AIEEE 2009

Open solution

Statement - 1: The temperature dependence of resistance is usually given as $R = R_0 (1 + \alpha \Delta t)$. The resistance of wire changes from 100Ω to 150Ω when its temperature is increased from $27^\circ C$ to $227^\circ C$. This implies that $\alpha = 2.5 \times 10^{-3}/C$.

Statement - 2: $R = R_0 (1 + \alpha \Delta t)$ is valid only when the change in the temperature ΔT is small and $\Delta T = (R - R_0) << R_0$.

(A) Statement - 1 is true, Statement - 2 is true; Statement - 2 is the correct explanation of Statement - 1

(B) Statement - 1 is true, Statement - 2 is true; Statement - 2 is not the correct explanation of Statement - 1

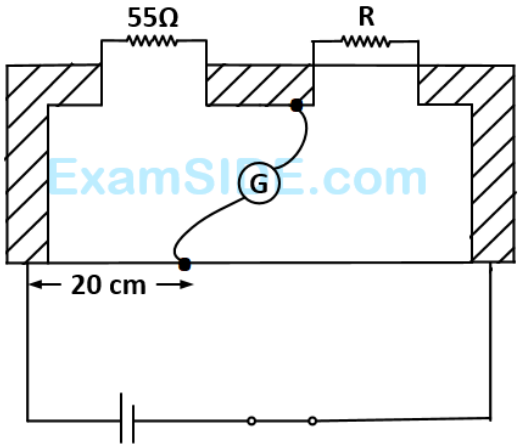
(C) Statement - 1 is false, Statement - 2 is true

(D) Statement - 1 is true, Statement - 2 is false

Q336. AIEEE 2008

Open solution

Shown in the figure below is a meter-bridge set up with null deflection in the galvanometer.



The value of the unknown resister R is

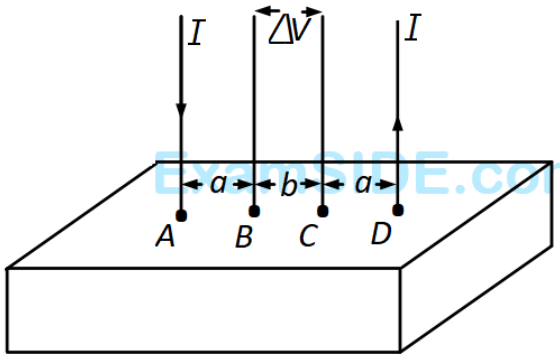
(A) 13.75Ω (B) 220Ω (C) 110Ω (D) 55Ω

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Q337. AIEEE 2008

Open solution

Consider a block of conducting material of resistivity $'\rho'$ shown in the figure. Current $'I'$ enters at $'A'$ and leaves from $'D'$. We apply superposition principle to find voltage $'\Delta V'$ developed between $'B'$ and $'C'$. The calculation is done in the following steps:
(i) Take current $'I'$ entering from $'A'$ and assume it to spread over a hemispherical surface in the block.
(ii) Calculate field $E(r)$ at distance $'r'$ from A by using Ohm's law $E = \rho j$, where j is the current per unit area at $'r'$.
(iii) From the $'r'$ dependence of $E(r)$, obtain the potential $V(r)$ at r.
(iv) Repeat (i), (ii) and (iii) for current $'I'$ leaving $'D'$ and superpose results for $'A'$ and $'D'$.



ΔV measured between B and C is

(A) $\frac{\rho I}{\pi a} - \frac{\rho I}{\pi(a+b)}$

(B) $\frac{\rho I}{a} - \frac{\rho I}{(a+b)}$

(C) $\frac{\rho I}{2\pi a} - \frac{\rho I}{2\pi(a+b)}$

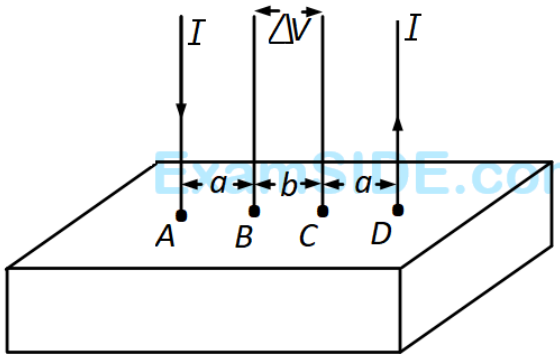
(D) $\frac{\rho I}{2\pi(a-b)}$

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Q338. AIEEE 2008

Open solution

Consider a block of conducting material of resistivity $'\rho'$ shown in the figure. Current $'I'$ enters at $'A'$ and leaves from $'D'$. We apply superposition principle to find voltage $'\Delta V'$ developed between $'B'$ and $'C'$. The calculation is done in the following steps:
(i) Take current $'I'$ entering from $'A'$ and assume it to spread over a hemispherical surface in the block.
(ii) Calculate field $E(r)$ at distance $'r'$ from A by using Ohm's law $E = \rho j$, where j is the current per unit area at $'r'$.
(iii) From the $'r'$ dependence of $E(r)$, obtain the potential $V(r)$ at r.
(iv) Repeat (i), (ii) and (iii) for current $'I'$ leaving $'D'$ and superpose results for $'A'$ and $'D'$.



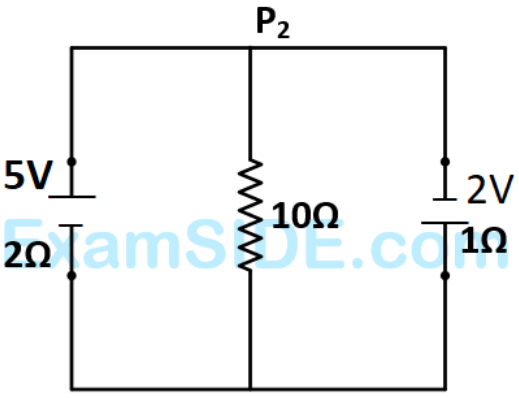
For current entering at A, the electric field at a distance $'r'$ from A is

(A) $\frac{\rho I}{8\pi r^2}$ (B) $\frac{\rho I}{r^2}$ (C) $\frac{\rho I}{2\pi r^2}$ (D) $\frac{\rho I}{4\pi r^2}$

Q339. AIEEE 2008

Open solution

A $5V$ battery with internal resistance 2Ω and a $2V$ battery with internal resistance 1Ω are connected to a 10Ω resistor as shown in the figure.



The current in the 10Ω resistor is

(A) $0.27\,A$, P_2 to P_1 (B) $0.03\,A$, P_1 to P_2 (C) $0.03\,A$, P_2 to P_1 (D) $0.27\,A$, P_1 to P_2

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Q340. AIEEE 2007

The resistance of a wire is 5 ohm at $50^{\circ}C$ and 6 ohm at $100^{\circ}C$. The resistance of the wire at $0^{\circ}C$ will be

(A) 3 ohm (B) 2 ohm (C) 1 ohm (D) 4 ohm

Open solution

Q341. AIEEE 2006

A thermocouple is made from two metals, Antimony and Bismuth. If one junction of the couple is kept hot and the other is kept cold, then, an electric current will

(A) flow from Antimony to Bismuth at the hot junction

(B) flow from Bismuth to Antimony at the cold junction

(C) not flow through the thermocouple

(D) flow from Antimony to Bismuth at the cold junction

Open solution

Q342. AIEEE 2006

The resistance of bulb filament is 100Ω at a temperature of $100^{\circ}C$. If its temperature coefficient of resistance be 0.005 per $^{\circ}C$, its resistance will become $200\,\Omega$ at a temperature of

(A) $300^{\circ}C$ (B) $400^{\circ}C$ (C) $500^{\circ}C$ (D) $200^{\circ}C$

Open solution

Q343. AIEEE 2006

In a Wheatstone's bridge, three resistance P, Q and R connected in the three arms and the fourth arm is formed by two resistances S_1 and S_2 connected in parallel. The condition for the bridge to be balanced will be

(A) $\frac{P}{Q} = \frac{2R}{S_1+S_2}$ (B) $\frac{P}{Q} = \frac{R(S_1+S_2)}{S_1S_2}$

(C) $\frac{P}{Q} = \frac{R(S_1+S_2)}{2S_1S_2}$ (D) $\frac{P}{Q} = \frac{R}{S_1+S_2}$

Open solution

Q344. AIEEE 2006

An electric bulb is rated 220 volt - 100 watt. The power consumed by it when operated on 110 volt will be

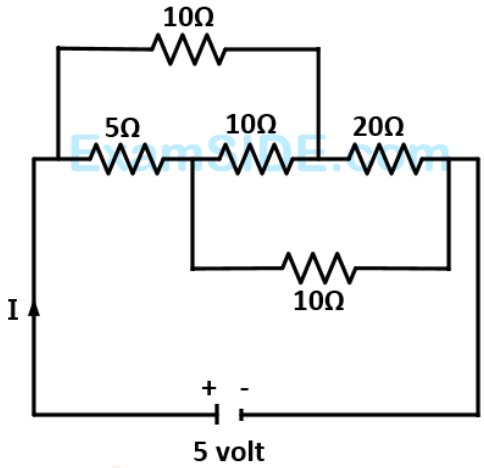
(A) 75 watt (B) 40 watt (C) 25 Watt (D) 50 Watt

Open solution

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Q345. AIEEE 2006

The current I drawn from the 5 volt source will be



Open solution

(A) $0.33\ A$ (B) $0.5\ A$ (C) $0.67\ A$ (D) $0.17\ A$

Q346. AIEEE 2006

A material 'B' has twice the specific resistance of 'A'. A circular wire made of 'B' has twice the diameter of a wire made of 'A'. Then for the two wires to have the same resistance, the ratio ℓ_B/ℓ_A of their respective lengths must be

Open solution

(A) 2 (B) 1 (C) $\frac{1}{2}$ (D) $\frac{1}{4}$

Q347. AIEEE 2006

The Kirchhoff's first law $(\sum i = 0)$ and second law $(\sum iR = \sum E)$, where the symbols have their usual meanings, are respectively based on

Open solution

(A) conservation of charge, conservation of energy

(B) conservation of charge, conservation of momentum

(C) conservation of energy, conservation of charge

(D) conservation of momentum, conservation of charge

Q348. AIEEE 2005

A heater coil is cut into two equal parts and only one part is now used in the heater. The heat generated will now be

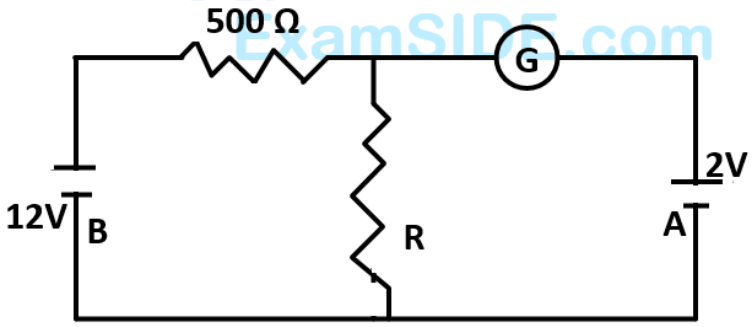
Open solution

(A) four times (B) doubled (C) halved (D) one fourth

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Q349. AIEEE 2005

In the circuit, the galvanometer G shows zero deflection. If the batteries A and B have negligible internal resistance, the value of the resistor R will be -



Open solution

(A) 100Ω (B) 200Ω (C) 1000Ω (D) 500Ω

Q350. AIEEE 2005

A moving coil galvanometer has 150 equal divisions. Its current sensitivity is 10- divisions per milliampere and voltage sensitivity is 2 divisions per millivolt. In order that each division reads 1 volt, the resistance in $ohms$ needed to be connected in series with the coil will be -

Open solution

(A) 10^5 (B) 10^3 (C) 9995 (D) 99995

Q351. AIEEE 2005

Two sources of equal emf are connected to an external resistance R. The internal resistances of the two sources are R_1 and R_2 $(R_2 > R_1)$. If the potential difference across the source having internal resistance R_2 is zero, then

Open solution

(A) $R = R_2 - R_1$ (B) $R = R_2 \times (R_1 + R_2)/(R_2 - R_1)$

(C) $R = R_1R_2/(R_2 - R_1)$ (D) $R = R_1R_2/(R_1 - R_2)$

Q352. AIEEE 2005

Two voltmeters, one of copper and another of silver, are joined in parallel. When a total charge q flows through the voltmeters, equal amount of metals are deposited. If the electrochemical equivalents of copper and silver are Z_1 and Z_2 respectively the charge which flows through the silver voltmeter is

Open solution

(A) $\frac{q}{1+\frac{Z_2}{Z_1}}$ (B) $\frac{q}{1+\frac{Z_1}{Z_2}}$

(C) $q\frac{Z_2}{Z_1}$ (D) $q\frac{Z_1}{Z_2}$

Q353. AIEEE 2005

The resistance of hot tungsten filament is about 10 times the cold resistance. What will be resistance of $100\ W$ and $200\ V$ lamp when not in use ?

Open solution

(A) 20Ω (B) 40Ω (C) 200Ω (D) 400Ω

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Q354. AIEEE 2005

An energy source will supply a constant current into the load if its internal resistance is

(A) very large as compared to the load resistance

(B) equal to the resistance of the load

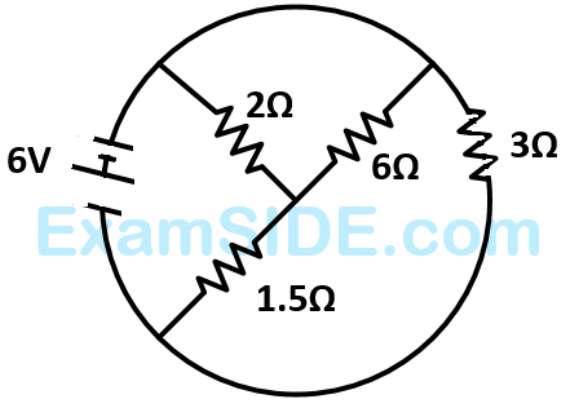
(C) non-zero but less than the resistance of the load

(D) zero

Open solution

Q355. AIEEE 2004

The total current supplied to the circuit by the battery is



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(A) $4A$ (B) $2A$ (C) $1A$ (D) $6A$

Open solution

Q356. AIEEE 2004

The resistance of the series combination of two resistances is S. When they are joined in parallel the total resistance is P. If $S = nP$ then the minimum possible value of n is

(A) 2 (B) 3 (C) 4 (D) 1

Open solution

Q357. AIEEE 2004

An electric current is passed through a circuit containing two wires of the same material, connected in parallel. If the lengths and radii are in the ratio of $\frac{4}{3}$ and $\frac{2}{3}$, then the ratio of the current passing through the wires will be

(A) $8/9$ (B) $1/3$ (C) 3 (D) 2

Open solution

Q358. AIEEE 2004

In a meter bridge experiment null point is obtained at $20\ cm$, from one end of the wire when resistance X is balanced against another resistance Y. If $X < Y$, then where will be the new position of the null point from the same end, if one decides to balance a resistance of $4\ X$ against Y

(A) $40\ cm$ (B) $80\ cm$ (C) $50\ cm$ (D) $70\ cm$

Open solution

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Q359. AIEEE 2004

Thermistors are usually made of

(A) metal oxides with high temperature coefficient of resistivity

(B) metals with high temperature coefficient of resistivity

(C) metals with low temperature coefficient of resistivity

(D) semiconducting materials having low temperature

Open solution

Q360. AIEEE 2004

The thermo emf of a thermocouple varies with temperature θ of the hot junction as $E = a\theta + b\theta^2$ in volts where the ratio a/b is $700^\circ C$. If the cold junction is kept at $0^\circ C$, then the neutral temperature is

(A) $1400^\circ C$

(B) $350^\circ C$

(C) $700^\circ C$

(D) No neutral temperature is possible for this termocouple.

Open solution

Q361. AIEEE 2004

The electrochemical equivalent of a metal is $3.3 \times 10^{-7}\ kg$ per Coulomb. The mass of the metal liberated at the cathode when a $3A$ current is passed for 2 seconds will be

(A) $6.6 \times 10^{-7}\ kg$ (B) $9.9 \times 10^{-7}\ kg$ (C) $19.8 \times 10^{-7}\ kg$ (D) $1.1 \times 10^{-7}\ kg$

Open solution

Q362. AIEEE 2004

The Kirchhoff's first law $(\sum i = 0)$ and second law $(\sum iR = \sum E)$, where the symbols have their usual meanings, are respectively based on

(A) conservation of charge, conservation of momentum

(B) conservation of energy, conservation of charge

(C) conservation of momentum, conservation of charge

(D) conservation of charge, conservation of energy

Open solution

Q363. AIEEE 2004

A material $'B'$ has twice the specific resistance of $'A'$. A circular wire made of $'B'$ has twice the diameter of a wire made of $'A'$. Then for the two wires to have the same resistance, the ratio l_B/l_A of their respective lengths must be

(A) 1 (B) $\frac{1}{2}$ (C) $\frac{1}{4}$ (D) 2

Open solution

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Q364. AIEEE 2003

The length of a given cylindrical wire is increased by 100%. Due to the consequent decrease in diameter the change in the resistance of the wire will be

(A) 200% (B) 100% (C) 50% (D) 300%

Open solution

Q365. AIEEE 2003

The thermo $e.m.f.$ of a thermo -couple is $25\ \mu V/^{\circ}C$ at room temperature. A galvanometer of $40\ ohm$ resistance, capable of detecting current as low as $10^{-5}\ A$, is connected with the thermo couple. The smallest temperature difference that can be detected by this system is

(A) 16^0C (B) 12^0C (C) 8^0C (D) 20^0C

Open solution

Q366. AIEEE 2003

The negative Zn pole of a Daniell cell, sending a constant current through a circuit, decreases in mass by $0.13g$ in 30 minutes. If the electrochemical equivalents of Zn and Cu are 32.5 and 31.5 respectively, the increase in the mass of the positive Cu pole in this time is

(A) $0.180\ g$ (B) $0.141\ g$ (C) $0.126\ g$ (D) $0.242\ g$

Open solution

Q367. AIEEE 2003

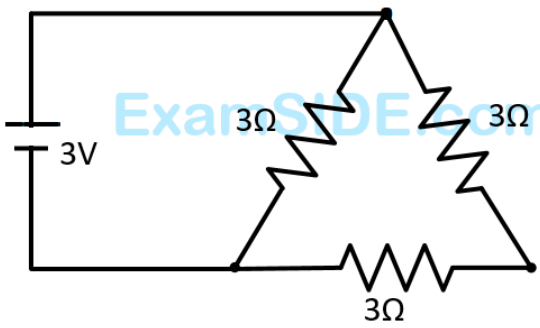
An ammeter reads upto 1 ampere. Its internal resistance is $0.81\ ohm$. To increase the range to $10\ A$ the value of the required shunt is

(A) $0.03\,\Omega$ (B) $0.3\,\Omega$ (C) $0.9\,\Omega$ (D) $0.09\,\Omega$

Open solution

Q368. AIEEE 2003

A 3 volt battery with negligible internal resistance is connected in a circuit as shown in the figure. The current I, in the circuit will be



(A) $1\ A$ (B) $1.5\ A$ (C) $2\ A$ (D) $1/3\ A$

Open solution

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Q369. AIEEE 2003

A 220 volt, 1000 watt bulb is connected across a 110 $volt$ mains supply. The power consumed will be

(A) 750 watt (B) 500 watt (C) 250 watt (D) 1000 watt

Open solution

Q370. AIEEE 2002

If an ammeter is to be used in place of a voltmeter, then we must connect with the ammeter a

(A) low resistance in parallel (B) high resistance in parallel

(C) high resistance in series (D) low resistance in series

Open solution

Q371. AIEEE 2002

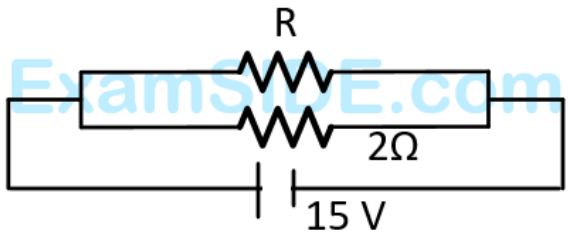
A wire when connected to 220 V mains supply has power dissipation P_1. Now the wire is cut into two equal pieces which are connected in parallel to the same supply. Power dissipation in this case is P_2. Then $P_2 : P_1$ is

(A) 1 (B) 4 (C) 2 (D) 3

Open solution

Q372. AIEEE 2002

If in the circuit, power dissipation is $150W$, then R is



(A) $2\,\Omega$ (B) $6\,\Omega$ (C) $5\,\Omega$ (D) $4\,\Omega$

Open solution

Q373. AIEEE 2002

The mass of product liberated on anode in an electrochemical cell depends on (where t is the time period for which the current is passed).

(A) $(It)^{1/2}$ (B) It (C) I/t (D) I^2t

Open solution

Q374. AIEEE 2002

If θ_i is the inversion temperature, θ_n is the neutral temperature, θ_c is the temperature of the cold junction, then

(A) $\theta_i + \theta_c = \theta_n$ (B) $\theta_i - \theta_c = 2\theta_n$

(C) $\frac{\theta_i+\theta_c}{2} = \theta_n$ (D) $\theta_c - \theta_i = 2\theta_n$

Open solution

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Out of syllabus 34 questions

These questions are on topics no longer in the current syllabus. They are kept for practice, after everything else.

Q375. JEE Main 2023 (Online) 1st February Morning Shift · **Numerical**

In an experiment to find emf of a cell using potentiometer, the length of null point for a cell of emf $1.5\ \mathrm{V}$ is found to be $60\ \mathrm{cm}$. If this cell is replaced by another cell of emf E, the length-of null point increases by $40\ \mathrm{cm}$. The value of E is $\frac{x}{10}V$. The value of x is _____.

Open solution

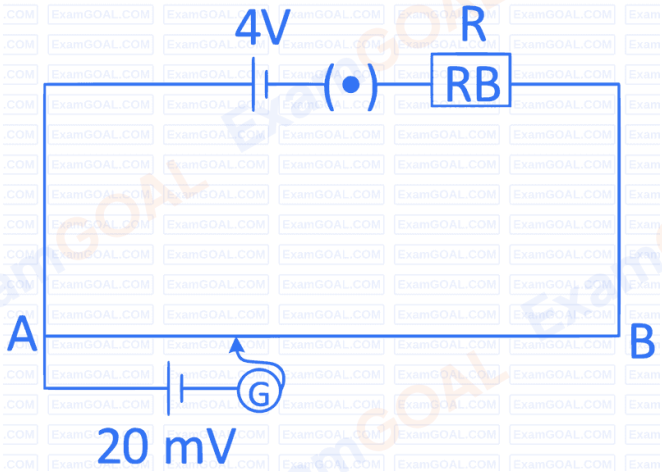
Q376. JEE Main 2023 (Online) 29th January Evening Shift · **Numerical**

A null point is found at 200 cm in potentiometer when cell in secondary circuit is shunted by 5Ω. When a resistance of 15Ω is used for shunting, null point moves to 300 cm. The internal resistance of the cell is _____Ω.

Open solution

Q377. JEE Main 2022 (Online) 28th July Morning Shift · **Numerical**

As shown in the figure, a potentiometer wire of resistance $20\ \Omega$ and length $300\ \mathrm{cm}$ is connected with resistance box (R.B.) and a standard cell of emf $4\ \mathrm{V}$. For a resistance 'R' of resistance box introduced into the circuit, the null point for a cell of $20\ \mathrm{mV}$ is found to be $60\ \mathrm{cm}$. The value of 'R' is _____ Ω.



Open solution

Q378. JEE Main 2022 (Online) 26th July Evening Shift · **Numerical**

A potentiometer wire of length $300\ \mathrm{cm}$ is connected in series with a resistance 780 Ω and a standard cell of emf $4\mathrm{V}$. A constant current flows through potentiometer wire. The length of the null point for cell of emf $20\ \mathrm{mV}$ is found to be $60\ \mathrm{cm}$. The resistance of the potentiometer wire is _____ Ω.

Open solution

Q379. JEE Main 2022 (Online) 25th July Evening Shift · **Numerical**

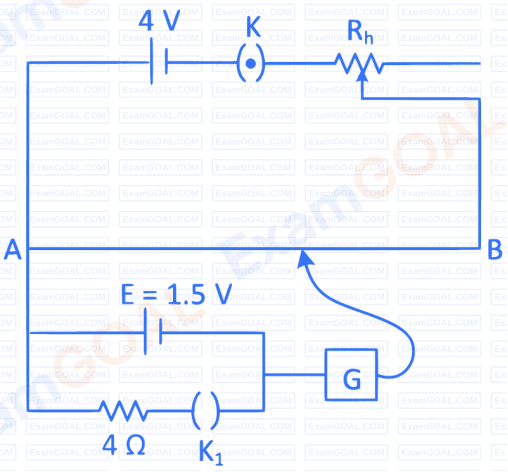
In a potentiometer arrangement, a cell of emf 1.20 V gives a balance point at 36 cm length of wire. This cell is now replaced by another cell of emf 1.80 V. The difference in balancing length of potentiometer wire in above conditions will be _____ cm.

Open solution

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Q380. JEE Main 2022 (Online) 30th June Morning Shift · **Numerical**

The circuit diagram of potentiometer used to measure the internal resistance of a cell (E) is shown in figure. The key 'K' is kept closed so as to send constant current through potentiometer wire. When key 'K_1' is kept open the null point is found to be at 120 cm on the potentiometer wire. When the key 'K_1' is closed the null point is shifted at 80 cm at the potentiometer wire. The internal resistance of the given cell is _____ Ω.



Open solution

Q381. JEE Main 2022 (Online) 27th June Morning Shift · **Numerical**

A cell, shunted by a 8 Ω resistance, is balanced across a potentiometer wire of length 3 m. The balancing length is 2 m when the cell is shunted by 4 Ω resistance. The value of internal resistance of the cell will be _____ Ω.

Open solution

Q382. JEE Main 2022 (Online) 24th June Evening Shift · **Numerical**

A potentiometer wire of length 10 m and resistance 20 Ω is connected in series with a 25 V battery and an external resistance 30 Ω. A cell of emf E in secondary circuit is balanced by 250 cm long potentiometer wire. The value of E (in volt) is $\frac{x}{10}$. The value of x is _____.

Open solution

Q383. JEE Main 2022 (Online) 24th June Morning Shift · **Numerical**

In a potentiometer arrangement, a cell gives a balancing point at 75 cm length of wire. This cell is now replaced by another cell of unknown emf. If the ratio of the emf's of two cells respectively is 3 : 2, the difference in the balancing length of the potentiometer wire in above two cases will be _____ cm.

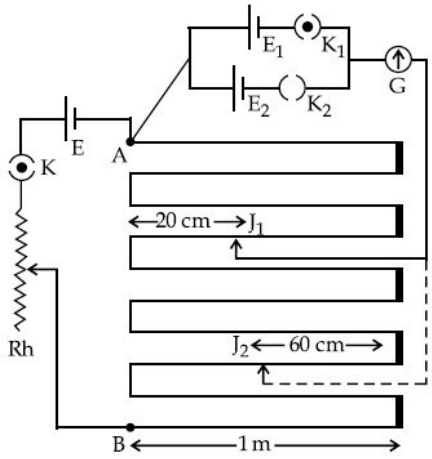
Open solution

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Q384. JEE Main 2021 (Online) 25th February Morning Shift · **Numerical**

Open solution

In the given circuit of potentiometer, the potential difference E across AB (10 m length) is larger than E_1 and E_2 as well. For key K_1 (closed), the jockey is adjusted to touch the wire at point J_1 so that there is no deflection in the galvanometer. Now the first battery (E_1) is replaced by second battery (E_2) for working by making K_1 open and K_2 closed. The galvanometer gives then null deflection at J_2. The value of $\frac{E_1}{E_2}$ is $\frac{a}{b}$, where a = _____.



Q385. JEE Main 2020 (Online) 7th January Evening Slot · **Numerical**

Open solution

The balancing length for a cell is 560 cm in a potentiometer experiment. When an external resistance of 10 Ω is connected in parallel to the cell, the balancing length changes by 60 cm. If the internal resistance of the cell is $\frac{N}{10}$ Ω, where N is an integer then value of N is .....

Q386. JEE Main 2023 (Online) 29th January Evening Shift · **MCQ (Single Correct Answer)**

Open solution

With the help of potentiometer, we can determine the value of emf of a given cell. The sensitivity of the potentiometer is

(A) directly proportional to the length of the potentiometer wire

(B) directly proportional to the potential gradient of the wire

(C) inversely proportional to the potential gradient of the wire

(D) inversely proportional to the length of the potentiometer wire

Choose the correct option for the above statements :

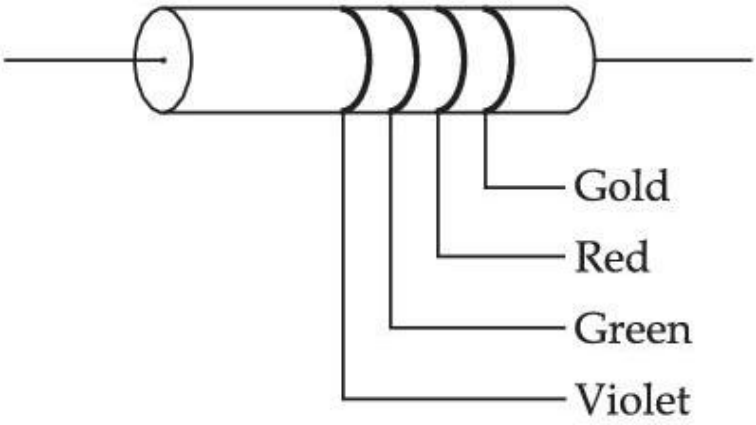
(A) A only **(B)** B and D only **(C)** A and C only **(D)** C only

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Q387. JEE Main 2021 (Online) 27th August Evening Shift · MCQ (Single Correct Answer)

Open solution

The Colour coding on a carbon resistor is shown in the given figure. The resistance value of the given resistor is :



(A) $(5700 \pm 285)\ \Omega$

(B) $(7500 \pm 750)\ \Omega$

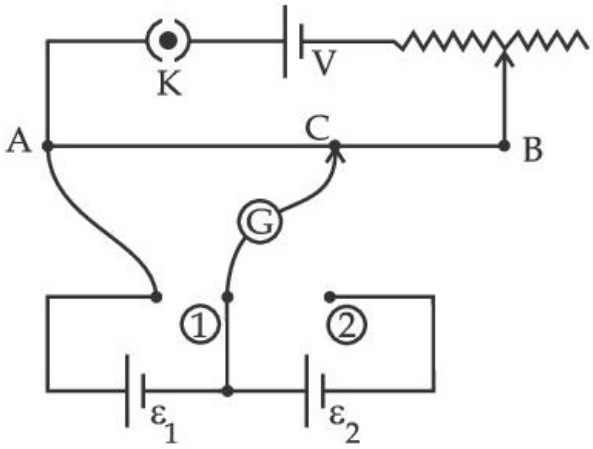
(C) $(5700 \pm 375)\ \Omega$

(D) $(7500 \pm 375)\ \Omega$

Q388. JEE Main 2021 (Online) 25th July Evening Shift · MCQ (Single Correct Answer)

Open solution

In the given potentiometer circuit arrangement, the balancing length AC is measured to be 250 cm. When the galvanometer connection is shifted from point (1) to point (2) in the given diagram, the balancing length becomes 400 cm. The ratio of the emf of two cells, $\frac{\varepsilon_1}{\varepsilon_2}$ is :



(A) $\frac{5}{3}$

(B) $\frac{8}{5}$

(C) $\frac{4}{3}$

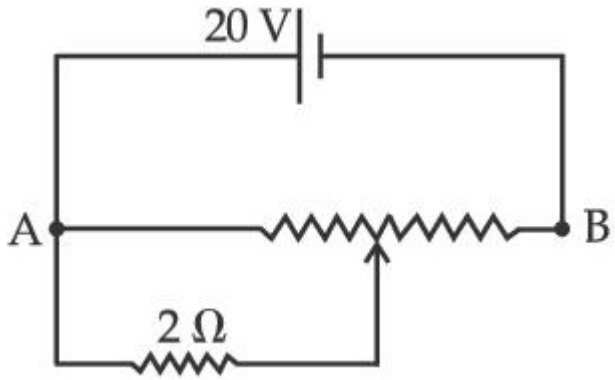
(D) $\frac{3}{2}$

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Q389. JEE Main 2021 (Online) 25th July Evening Shift · MCQ (Single Correct Answer)

The given potentiometer has its wire of resistance 10Ω. When the sliding contact is in the middle of the potentiometer wire, the potential drop across 2Ω resistor is :

Open solution

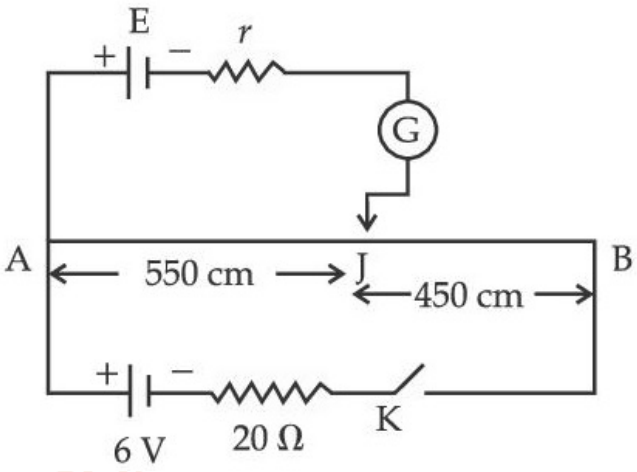


(A) 10 V (B) 5 V (C) $\frac{40}{9}$ V (D) $\frac{40}{11}$ V

Q390. JEE Main 2021 (Online) 25th July Morning Shift · MCQ (Single Correct Answer)

In the given figure, there is a circuit of potentiometer of length AB = 10 m. The resistance per unit length is 0.1 Ω per cm. Across AB, a battery of emf E and internal resistance 'r' is connected. The maximum value of emf measured by this potentiometer is :

Open solution



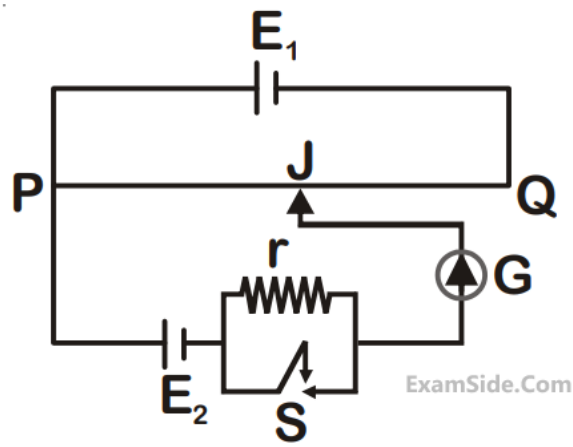
(A) 5 V (B) 2.25 V (C) 6 V (D) 2.75 V

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Q391. JEE Main 2020 (Online) 2nd September Evening Slot · MCQ (Single Correct Answer)

Open solution

A potentiometer wire PQ of 1 m length is connected to a standard cell E_1. Another cell E_2 of emf 1.02 V is connected with a resistance 'r' and switch S (as shown in figure). With switch S open, the null position is obtained at a distance of 49 cm from Q. The potential gradient in the potentiometer wire is :



(A) 0.04 V/cm (B) 0.01 V/cm (C) 0.02 V/cm (D) 0.03 V/cm

Q392. JEE Main 2020 (Online) 8th January Morning Slot · MCQ (Single Correct Answer)

Open solution

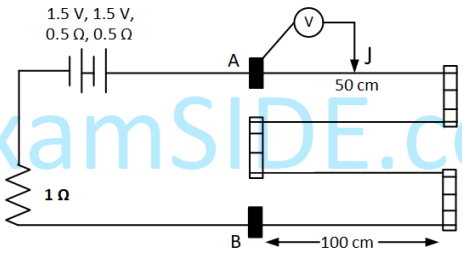
The length of a potentiometer wire is 1200 cm and it carries a current of 60 mA. For a cell of emf 5V and internal resistance of 20Ω, the null point on it is found to be a 1000 cm. The resistance of whole wire is :

(A) 80Ω (B) 60Ω (C) 120Ω (D) 100Ω

Q393. JEE Main 2019 (Online) 8th April Evening Slot · MCQ (Single Correct Answer)

Open solution

In the circuit shown, a four-wire potentiometer is made of a 400 cm long wire, which extends between A and B. The resistance per unit length of the potentiometer wire is r = 0.01 Ω/cm. If an ideal voltmeter is connected as shown with jockey J at 50 cm from end A, the expected reading of the voltmeter will be :-



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(A) 0.25 V (B) 0.20 V (C) 0.50V (D) 0.75 V

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Q394. JEE Main 2019 (Online) 8th April Morning Slot · MCQ (Single Correct Answer)

A 200 Ω resistor has a certain color code. If one replaces the red color by green in the code, the new resistance will be :

(A) 500 Ω (B) 100 Ω (C) 200 Ω (D) 300 Ω

Open solution

Q395. JEE Main 2019 (Online) 12th January Morning Slot · MCQ (Single Correct Answer)

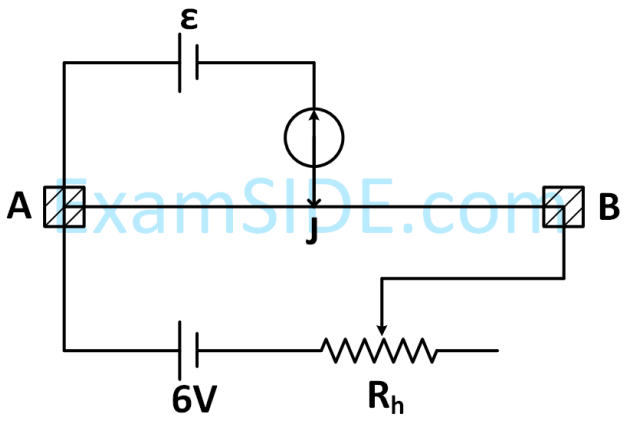
An ideal battery of 4 V and resistance R are connected in series in the primary circuit of a potentiometer of length 1 m and resistance 5 Ω. The value of R, to give a potential difference of 5 mV across 10 cm of potentiometer wire, is :

(A) 480 Ω (B) 495 Ω (C) 490 Ω (D) 395 Ω

Open solution

Q396. JEE Main 2019 (Online) 11th January Morning Slot · MCQ (Single Correct Answer)

The resistance of the meter bridge AB in given figure is 4 Ω. With a cell of emf ε = 0.5 V and rheostat resistance R_h = 2 Ω the null point is obtained at some point J. When the cell is replaced by another one of emf $\varepsilon = \varepsilon_2$ the same null point J is found for R_h = 6 Ω. The emf ε_2 is :



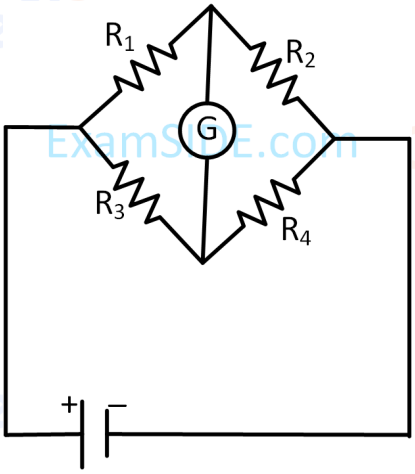
(A) 0.3 V (B) 0.6 V (C) 0.5 V (D) 0.4 V

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Q397. JEE Main 2019 (Online) 10th January Evening Slot · MCQ (Single Correct Answer)

The Wheatstone bridge shown in figure, here, gets balanced when the carbon resistor used as R_1 has the colour code (Orange, Red, Brown). The resistors R_2 and R_4 are 80Ω and 40Ω, respectively. Assuming that the colour code for the carbon resistors gives their accurate values, the colour code for the carbon resistor, used as R_3, would be -

Open solution



(A) Brown, Blue, Brown
(B) Grey, Black, Brown
(C) Red, Green, Brown
(D) Brown, Blue, Black

Q398. JEE Main 2019 (Online) 10th January Morning Slot · MCQ (Single Correct Answer)

A 2 W carbon resistor is color coded with green, black, red and brown respectively. The maximum current which can be passed through this resistor is -

Open solution

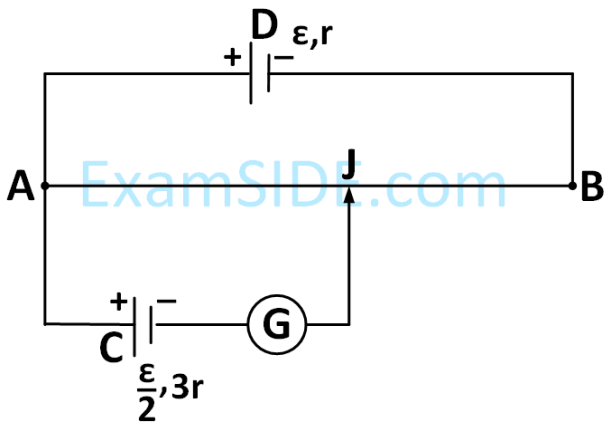
(A) 0.4 mA
(B) 20 mA
(C) 63 mA
(D) 100 mA

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Q399. JEE Main 2019 (Online) 10th January Morning Slot · MCQ (Single Correct Answer)

A potentiometer wire AB having length L and resistance 12 r is joined to a cell D of emf ε and internal resistance r. A cell C having emf $\varepsilon/2$ and internal resistance 3r is connected. The length AJ at which the galvanometer as shown in figure shows no deflection is –

Open solution



(A) $\frac{11}{12}L$ (B) $\frac{13}{24}L$ (C) $\frac{5}{12}L$ (D) $\frac{11}{24}L$

Q400. JEE Main 2019 (Online) 9th January Evening Slot · MCQ (Single Correct Answer)

A carbon resistance has a following colour code. What is the value of the resistance ?

Open solution



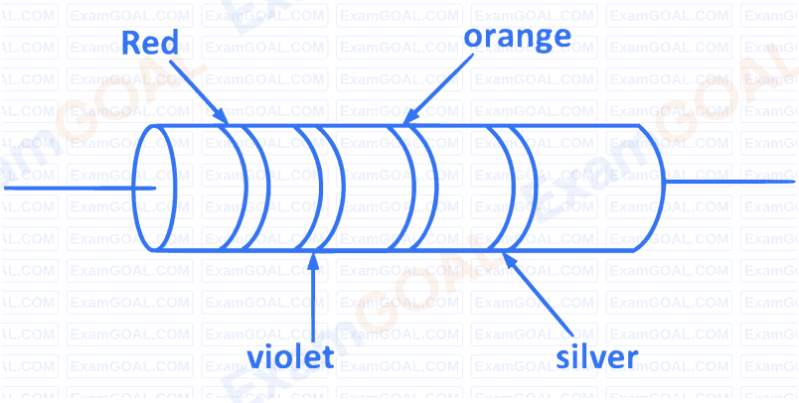
(A) 530 kΩ ± 5% (B) 5.3 MΩ ± 5% (C) 6.4 MΩ ± 5% (D) 64 kΩ ± 10%

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Q401. JEE Main 2019 (Online) 9th January Morning Slot · MCQ (Single Correct Answer)

A resistance is shown in the figure. Its value and tolerance are given respectively by :

Open solution



(A) 270 Ω, 10 %
(B) 27 kΩ, 10 %
(C) 27 kΩ, 20 %
(D) 270 Ω, 5 %

Q402. JEE Main 2018 (Offline) · MCQ (Single Correct Answer)

In a potentiometer experiment, it is found that no current passes through the galvanometer when the terminals of the cell are connected across 52 cm of the potentiometer wire. If the cell is shunted by a resistance of 5 Ω, a balance is found when the cell is connected across 40 cm of the wire. Find the internal resistance of the cell.

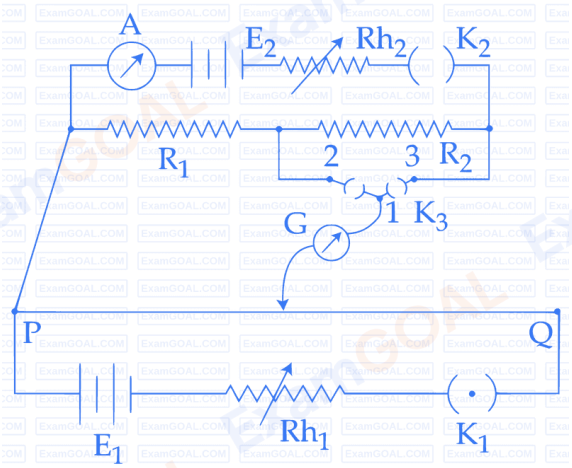
Open solution

(A) 2.5 Ω
(B) 1 Ω
(C) 1.5 Ω
(D) 2 Ω

Q403. JEE Main 2017 (Online) 8th April Morning Slot · MCQ (Single Correct Answer)

A potentiometer PQ is set up to compare two resistances as shown in the figure. The ammeter A in the circuit reads 1.0 A when two way key K_3 is open. The balance point is at a length ℓ_1 cm from P when two way key K_3 is plugged in between 2 and 1, while the balance point is at a length ℓ_2 cm from P when key K_3 is plugged in between 3 and 1. The ratio of two resistances $\frac{R_1}{R_2}$, is found to be :

Open solution



(A) $\frac{l_1}{l_1+l_2}$
(B) $\frac{l_2}{l_2-l_1}$
(C) $\frac{l_1}{l_1-l_2}$
(D) $\frac{l_1}{l_2-l_1}$

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Q404. JEE Main 2014 (Online) 12th April Morning Slot · MCQ (Single Correct Answer)

In the experiment of calibration of voltmeter, a standard cell of e.m.f. 1.1 volt is balanced against 440 cm of potentiometer wire. The potential difference across the ends of resistance is found to balance against 220 cm of the wire. The corresponding reading of voltmeter is 0.5 volt. The error in the reading of voltmeter will be :

Open solution

(A) $-$ 0.15 volt (B) 0.15 volt (C) 0.5 volt (D) -0.05 volt

Q405. AIEEE 2012 (Online) 26th May · MCQ (Single Correct Answer)

In an experiment of potentiometer for measuring the internal resistance of primary cell a balancing length ℓ is obtained on the potentiometer wire when the cell is open circuit. Now the cell is short circuited by a resistance R. If R is to be equal to the internal resistance of the cell the balancing length on the potentiometer wire will be

Open solution

(A) ℓ (B) 2ℓ (C) $\ell/2$ (D) $\ell/4$

Q406. AIEEE 2012 (Online) 12th May · MCQ (Single Correct Answer)

It is preferable to measure the e.m.f. of a cell by potentiometer than by a voltmeter because of the following possible reasons.
(i) In case of potentiometer, no current flows through the cell.
(ii) The length of the potentiometer allows greater precision.
(iii) Measurement by the potentiometer is quicker.
(iv) The sensitivity of the galvanometer, when using a potentiometer is not relevant.
Which of these reasons are correct?

Open solution

(A) (i),(iii),(iv) (B) (i),(iii),(iv)
(C) (i),(ii) (D) (i),(ii),(iii),(iv)

Q407. AIEEE 2005 · MCQ (Single Correct Answer)

In a potentiometer experiment the balancing with a cell is at length $240\ cm$. On shunting the cell with a resistance of 2Ω, the balancing length becomes $120\ cm$. The internal resistance of the cell is

Open solution

(A) 0.5Ω (B) 1Ω (C) 2Ω (D) 4Ω

Q408. AIEEE 2003 · MCQ (Single Correct Answer)

The length of a wire of a potentiometer is $100\ cm$, and the $e.\ m.\ f.$ of its standard cell is E volt. It is employed to measure the $e.m.\ f.$ of a battery whose internal resistance is 0.5Ω. If the balance point is obtained at $l = 30\ cm$ from the positive end, the $e.\ m.\ f.$ of the battery is
where i is the current in the potentiometer wire.

Open solution

(A) $\frac{30E}{100.5}$
(B) $\frac{30E}{(100-0.5)}$
(C) $\frac{30(E-0.5i)}{100}$
(D) $\frac{30E}{100} - 0.5i$, where i is the current in the potentiometer wire

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Answer Key

Numerical

1. **3**	2. **3477**	3. **1**	4. **21**	5. **6**
6. **1**	7. **1080**	8. **4**	9. **5**	10. **2**
11. **2**	12. **1**	13. **2500**	14. **1027**	15. **25**
16. **5**	17. **748**	18. **160**	19. **16**	20. **5**
21. **5**	22. **32**	23. **1**	24. **22**	25. **3**
26. **1**	27. **200**	28. **6**	29. **30**	30. **3**
31. **50**	32. **16**	33. **15**	34. **5**	35. **1**
36. **3**	37. **100**	38. **125**	39. **25**	40. **20**
41. **25**	42. **2**	43. **5**	44. **2**	45. **1.5**
46. **30**	47. **2**	48. **1**	49. **10**	50. **44**
51. **2**	52. **8**	53. **2**	54. **14**	55. **40**
56. **2400**	57. **136**	58. **20**	59. **4**	60. **4**
61. **8**	62. **15**	63. **144**	64. **10**	65. **8**
66. **19**	67. **10**	68. **12**	69. **3**	70. **48**
71. **300**	72. **450**	73. **2**	74. **4**	75. **3840**
76. **3**	77. **6**	78. **9**	79. **20**	80. **1**
81. **45**	82. **50**	83. **15**	84. **10**	85. **4**
86. **48**	87. **70**	88. **4**	89. **2500**	90. **3**
91. **300**	92. **2**	93. **5**	94. **2**	95. **40**
96. **30**	97. **10**			

MCQ (Single Correct Answer)

98. **A**	99. **B**	100. **A**	101. **B**	102. **B**
103. **D**	104. **B**	105. **B**	106. **A**	107. **B**
108. **A**	109. **B**	110. **C**	111. **D**	112. **C**
113. **B**	114. **A**	115. **A**	116. **C**	117. **B**
118. **D**	119. **C**	120. **D**	121. **B**	122. **D**
123. **B**	124. **B**	125. **A**	126. **D**	127. **C**
128. **C**	129. **B**	130. **A**	131. **D**	132. **C**
133. **A**	134. **C**	135. **C**	136. **C**	137. **C**
138. **B**	139. **C**	140. **B**	141. **B**	142. **D**

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143. **C**	144. **B**	145. **B**	146. **C**	147. **D**
148. **D**	149. **B**	150. **C**	151. **B**	152. **B**
153. **C**	154. **D**	155. **C**	156. **C**	157. **A**
158. **C**	159. **C**	160. **D**	161. **D**	162. **C**
163. **A**	164. **A**	165. **C**	166. **B**	167. **D**
168. **B**	169. **A**	170. **A**	171. **D**	172. **C**
173. **C**	174. **C**	175. **C**	176. **A**	177. **A**
178. **B**	179. **A**	180. **B**	181. **B**	182. **B**
183. **C**	184. **D**	185. **D**	186. **D**	187. **A**
188. **C**	189. **B**	190. **A**	191. **B**	192. **B**
193. **C**	194. **A**	195. **B**	196. **A**	197. **A**
198. **A**	199. **D**	200. **B**	201. **A**	202. **A**
203. **B**	204. **C**	205. **C**	206. **A**	207. **D**
208. **B**	209. **B**	210. **C**	211. **B**	212. **C**
213. **D**	214. **C**	215. **D**	216. **A**	217. **D**
218. **A**	219. **C**	220. **B**	221. **C**	222. **A**
223. **C**	224. **A**	225. **C**	226. **B**	227. **B**
228. **A**	229. **B**	230. **C**	231. **D**	232. **D**
233. **D**	234. **C**	235. **A**	236. **D**	237. **C**
238. **C**	239. **B**	240. **C**	241. **A**	242. **C**
243. **B**	244. **C**	245. **B**	246. **B**	247. **C**
248. **B**	249. **C**	250. **D**	251. **C**	252. **C**
253. **B**	254. **D**	255. **B**	256. **A**	257. **D**
258. **D**	259. **C**	260. **D**	261. **B**	262. **A**
263. **A**	264. **D**	265. **B**	266. **C**	267. **A**
268. **A**	269. **D**	270. **D**	271. **B**	272. **B**
273. **C**	274. **C**	275. **D**	276. **A**	277. **A**
278. **A**	279. **A**	280. **C**	281. **A**	282. **A**
283. **B**	284. **C**	285. **A**	286. **D**	287. **B**
288. **D**	289. **C**	290. **D**	291. **C**	292. **D**
293. **A**	294. **A**	295. **A**	296. **C**	297. **D**
298. **A**	299. **A**	300. **C**	301. **B**	302. **C**
303. **C**	304. **A**	305. **B**	306. **C**	307. **C**

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308. **C**	309. **C**	310. **B**	311. **A**	312. **C**
313. **C**	314. **D**	315. **C**	316. **C**	317. **B**
318. **B**	319. **A**	320. **C**	321. **D**	322. **C**
323. **D**	324. **D**	325. **D**	326. **D**	327. **C**
328. **B**	329. **C**	330. **A**	331. **C**	332. **C**
333. **A**	334. **D**	335. **C**	336. **B**	337. **A**
338. **C**	339. **C**	340. **D**	341. **D**	342. **B**
343. **B**	344. **C**	345. **B**	346. **A**	347. **A**
348. **B**	349. **A**	350. **C**	351. **A**	352. **A**
353. **B**	354. **A**	355. **A**	356. **C**	357. **B**
358. **C**	359. **A**	360. **D**	361. **C**	362. **D**
363. **D**	364. **D**	365. **A**	366. **C**	367. **D**
368. **B**	369. **C**	370. **C**	371. **B**	372. **B**
373. **B**	374. **C**			

Out of syllabus

375. **25**	376. **5**	377. **780**	378. **20**	379. **18**
380. **2**	381. **8**	382. **25**	383. **25**	384. **1**
385. **12**	386. **C**	387. **D**	388. **A**	389. **C**
390. **A**	391. **C**	392. **D**	393. **A**	394. **A**
395. **D**	396. **A**	397. **A**	398. **B**	399. **B**
400. **A**	401. **B**	402. **C**	403. **D**	404. **D**
405. **C**	406. **C**	407. **C**	408. **D**	

For numerical answers given as a range, any value within the range is correct. Full solutions: scan the QR code next to each question, or visit questions.examside.com.

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